

A Graph-Based Cluster Dynamics Framework for Irradiated Materials

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Abstract

We present a modular, graph-based cluster-dynamics (CD) framework for the long-term evolution of radiation-induced defect populations in crystalline solids, and instantiate it for the ferritic–martensitic steel EUROFER-97. Two algorithmic primitives carry the formulation: a polarity index $\sigma \in \{-1, +1\}$ that collapses the host-specific taxonomy of loops, voids and bubbles into a four-tuple cluster identifier $(\sigma, n, p, \mathbf{c})$, and a reaction admissibility graph (RAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{\nu}, \mathbf{k})$ that carries every elementary reaction as an edge of exactly one of ten classes, physically forbidden reactions being excluded during graph construction. The master equation is assembled rather than written down: the graph-walking accumulator $dc_a/dt = G_a + \sum_r S_{ar} J_r(\mathbf{c}) - D_a c_a$ is applied edge-by-edge over $\mathcal{E}$, so that Frenkel-pair and helium conservation are left null vectors of the stoichiometric matrix, monitored rather than imposed. The open-source RADCLUSTER code is split along exactly this line—a host-independent core holding the graph, the walker, the state layout, the state-space reductions and the conservation diagnostics, and a host declaration supplying only admissible populations, edges and rate kernels—so that a new material is a declaration and not a rewrite. Two reductions are stated on the abstract graph and inherited by any host: a helium-occupancy collapse of the two-dimensional cavity grid, and a logarithmic size-bin moment scheme whose linear dual-basis (Galerkin) closure is reconstructed, walked with the *exact* per-size kernels and projected back, so that no inter-bin flux and no bin-averaged rate constant appears. For EUROFER-97 under a fission cascade spec-

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trum at 330 °C to 40 dpa, this replaces $\sim 10^5$ discrete size equations by a few hundred stiff ODEs, integrated on the bordered-banded Jacobian with Sherman–Morrison–Woodbury preconditioned GMRES. With no parameter fitted to any measurement, the densities and mean diameters of $\frac{1}{2}\langle 111 \rangle$ loops, $\langle 100 \rangle$ loops and cavities all fall simultaneously within the ranges measured on neutron-irradiated EUROFER-97; loop character converts from $\frac{1}{2}\langle 111 \rangle$ to $\langle 100 \rangle$ sigmoidally in log dose, passing 50 per cent near 3 dpa; and the near-zero swelling of this steel at 330 °C emerges as computed cavity re-dissolution driven by interstitial–vacancy competition rather than as an imposed constraint.

Keywords: cluster dynamics, radiation damage, reaction admissibility graph, ferritic-martensitic steel, helium-vacancy clusters, size-bin moments

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1. Introduction

Structural materials in fission and fusion energy systems must operate under prolonged exposure to neutron radiation that displaces lattice atoms at rates of 10^{-7} to 10^{-5} dpa/s and accumulates damage to total doses of 10–200 dpa. The microstructural changes that result—nucleation and growth of vacancy and self-interstitial atom (SIA) clusters, void and bubble formation, dislocation-loop populations, irradiation-induced segregation, and precipitate coarsening or dissolution—are the principal sources of property degradation, manifesting as hardening, embrittlement, swelling, irradiation creep, and shifts in the ductile-to-brittle transition temperature. Predicting these property changes from the first principles, *over the relevant time and length scales*, and across the full range of temperature, dose, dose rate, and helium-to-dpa ratio is the central quantitative challenge of radiation materials science.

The cluster-dynamics (CD) framework remains the workhorse mesoscopic theory for this challenge, and its lineage is considerably older than its application to irradiation. The population-balance description of clustering originates with Smoluchowski’s coagulation kinetics [1], which already contain the diffusion-limited capture coefficient $4\pi DR$, later recast in the stochastic language still in use [2]; Waite extended the pair-density treatment to the correlated, non-random initial distributions left by irradiation and applied it to damage annealing in germanium [3, 4]; Ham gave the corresponding diffusion-limited capture by extended sinks [5]; and Dienes and Damask first combined production, recombination, and sink loss under irradiation [6]. Void nucleation of quenched-in vacancies followed [7, 8], gas-bubble coalescence was treated in the same spirit [9], and the transient stages of damage accumulation were modeled in the mid-1970s [10]. The capture kernel thus came first; the coupled evolution of *both* defect-cluster families under continuous irradiation came later, with Ghoniem and Cho [11, 12] giving an early systematic irradiation CD model — plausibly the first fully coupled treatment of simultaneous vacancy and interstitial clustering — and the Fokker–Planck description of loop-size evolution following shortly after [13]. CD treats each defect species and size as a separate population whose concentration evolves under a coupled set of master rate equations that balance production in the cascade, diffusion-controlled reactions, thermal emission, and absorption at fixed sinks. Its three defining advantages are direct access to time scales (10^{-6} – 10^8 s) that lie far beyond molecular dynamics [14, 15], explicit res-

olution of the cluster-size distribution and therefore of every quantity that enters macroscopic property models [16, 17]—and modest computational cost relative to spatially resolved kinetic Monte Carlo or atomistic simulation. Together, these features make CD the natural meeting point for atomistic-scale parameterization and engineering-scale prediction [18–20].

Since its inception, CD has been deployed in an unusually broad materials portfolio: void swelling and helium effects on cavity stability in austenitic stainless steels and Fe–Cr–Ni model alloys [12, 21, 22]; SIA-cluster and Cr-precipitate kinetics in ferritic and ferritic–martensitic steels [23–27]; retention of helium and hydrogen-isotopes in plasma-facing tungsten and W–Re alloys [28–30]; evolution and bubble nucleation in nuclear fuels [31]; and the $\langle a \rangle$ -loop and $\langle c \rangle$ -loop populations that underlie irradiation growth. A recent demonstration of how far the framework can be extended is the work of Gao et al. on the Eurofer97 steel with reduced-activation under dual-beam irradiation of He^+ / Fe^{3+} [26, 27]. The several review articles that have shaped the field [20, 32–35] nevertheless, they converge on a common diagnosis: while CD is conceptually mature, the algorithmic framework is fragmented. Existing codes specialize in a single host material, in a particular reaction subset, or in a particular solver philosophy, and there is no widely shared, transparent, and modular formulation that exposes *which* physics has been included, *which* approximation has been invoked, and *which* parameter has been calibrated. This has produced several persistent gaps. The first is a lack of host independence at the level of the algorithmic core: specializing the same machinery for different materials typically requires a non-trivial rewrite of the solver, even though the underlying mathematical structure remains unchanged. The second is the state-space management of helium–vacancy clusters: tracking $\text{He}_\ell V_m$ explicitly across the entire (ℓ, m) grid is prohibitive [18, 36, 37], and reduced descriptions that track only a mean helium load per cavity have been used heuristically [38, 39] without making explicit the limits in which they are exact, the leading correction terms or the conservation laws they preserve. The third is the closure of the cluster-size distribution: Fokker–Planck representations [38, 40, 41] lose accuracy at the discrete-to-continuous transition, grouping or binning closures [16, 20, 42, 43] introduce mass-conservation errors unless the inter-bin flux is explicitly designed, and few formulations rigorously analyze closure accuracy in a way that lets the user trade fidelity for system size with quantified error. The fourth is stiffness and solver design: the CD master equations carry both narrow-banded structure (within a single species) and dense bordering blocks (cross-species

coupling, sink strengths, conservation constraints), generic ODE solvers and standard preconditioners perform poorly on this structure, and few existing CD codes invest in tailored numerical strategies that exploit the bordered-banded form or document explicit conservation-law diagnostics as a built-in part of the solver [44].

This paper pursues two goals, one algorithmic and one physical, and each exists to make the other possible. The algorithmic goal is to close the four gaps above within a single modular framework whose components can be selected, replaced, or specialized for the host material and the irradiation environment without touching the algorithmic core. Host independence is obtained by carrying the kinetics on a reaction admissibility graph, so that changing the material means changing the graph rather than rewriting the solver. The helium–vacancy state space is reduced by two limits derived from the same parent master equations: fast helium equilibration, and a decoupled helium inventory, which carries a single scalar bubble-helium inventory. The cluster-size distribution is closed by a logarithmic size-bin moment method with a linear dual-basis (Galerkin) reconstruction, accompanied by an explicit accuracy and mass-conservation analysis, so that system size can be traded against fidelity with quantified error. Stiffness is met with a Sherman–Morrison–Woodbury preconditioner built for the bordered–banded Jacobian [45–47], and the whole formulation is implemented in the open-source RAD-CLUSTER code.

The physical goal is the test the first must pass. Specialized to the reduced-activation ferritic–martensitic steel EUROFER-97, the model is asked to reproduce what is actually measured in irradiated ferritic–martensitic steels: the number densities and mean sizes of dislocation loops and cavities, and their evolution with dose and irradiation TEMperature against a database of TEM observations compiled from the literature and tabulated in Section S10. Agreement there is not the end but the license: because every channel in the graph is explicit, the converged solution can be interrogated for the mechanisms that produced it. Two are examined here. Cavity nucleation and growth are shown to be controlled by the helium supply and by the vacancy surplus that survives in-cascade clustering, rather than by the point-defect fluxes alone. Loop evolution is shown to be sustained not by net absorption but by the clustered fraction deposited directly in the cascade, so that loops and cavities grow together as a consequence of production bias rather than of a shared critical-radius window. What makes both statements testable, rather than plausible, is the algorithmic discipline of the first goal:

each is a statement about identified terms of the master equations, and each can be switched off and re-run.

RadCluster is a production-spectrum-resolved cluster-dynamics framework that couples asymmetric cascade partitioning, mixed 1D/3D SIA transport, SIA–SIA and vacancy–vacancy coalescence, residual-size SIA–vacancy reactions, cavity capture and emission, helium coupling, and explicit sink accounting, so that loop and cavity populations evolve as size-resolved distributions in a single coupled system rather than as separately nucleated averages. On the physics side, controlled calculations that independently equalise cascade clustering fractions and sink capture efficiencies isolate simultaneous loop and cavity growth caused by production bias rather than by preferential sink capture — a separation that reduced-moment rate theory can argue but not demonstrate. On the numerical side, its reconstruct–evaluate–project bin-moment reduction retains the original size-dependent reaction kernels instead of projecting them onto grouped coefficients, so no constant-property assumption is imposed within a bin and no reaction-specific inter-group operator need be derived for new physics; the reduction conserves projected stoichiometric balances under continuous audit of reconstruction, boundary and solver error, trading the asymptotic work reduction of a strict grouped operator for reaction fidelity in exactly the size-dependent, mixed-dimensional kernels that govern the swelling result.

Section 2.2 defines the state space of the cluster indexed by polarity and the composition vector that distinguishes the host coordinates from the solute coordinates. Section 2.3 defines the reaction admissibility graph and catalogs the elementary edge classes—growth, shrinkage, dissociation, recombination, solute trapping, inter-population transfer, coalescence, source and sink—from which any specialized application selects its subset. Section 2 places the master equation in a code-generation-ready form, comprising stoichiometric matrices and conservation laws. The abstract framework is then instantiated for ferritic-martensitic steels. Section 3 works out the full instantiation of the RAG for EUROFER-97: the reaction admissibility graph, the master equations for SIA clusters, vacancy/helium cavities, and free helium; the Frenkel-pair and helium-conservation laws; the swelling identity; the two helium-occupancy reduction modes; and the explicit mapping to RADCLUSTER code variables [48]. The reductions in the state-space of helium vacancies are derived in Section 4 as systematic limits of the master equations, and the logarithmic reduction of the size-bin moment is developed in Section 4.3 together with both the closures and the accompanying accu-

racy and conservation analysis. The results of the simulation demonstration for EUROFER-97 under prototypical reactor conditions are presented in Section 5.

2. Graph Cluster Dynamics Formulation

2.1. Reaction Network Graphs

Reaction network graphs are the structural language of chemical kinetics. A reaction system is encoded as a graph whose vertices are species and whose edges are elementary reactions, so that the governing rate equations follow from an incidence or stoichiometric matrix, rather than from hand-written balances. The chemical reaction network theory, built on the mass-action formalism of Horn and Jackson [49] and developed into the deficiency theorems of Feinberg [50, 51], established how much of the qualitative dynamics – conservation laws, invariant manifolds, and the existence and uniqueness of steady states – is fixed by network topology alone, independently of the rate constants. Graph-theoretical treatments of reaction mechanisms are reviewed by Temkin *et al.* [52]. Reactions that consume two precursors are not edges of a digraph, so the object is properly a directed hypergraph [53]. The same abstraction underlies the practice of separating a declarative network from the integrator, as in Cantera [54], the kinetic preprocessor KPP [55], SBML [56] and Catalyst.jl [57], and of generating the network itself from reaction-family templates, as in the Reaction Mechanism Generator [58, 59]. The defect population dynamics developed below is a special case of this construction, and rule-based generation of the reaction set is already familiar in radiation damage modeling through object kinetic Monte Carlo [60] and spatially resolved cluster dynamics [61]; what is proposed here is that the graph itself, rather than one host system’s reaction list, be the interface of the framework.

Several features separate the present system from the molecular setting. The rate kernels are state-dependent since the sink strengths and biased absorption depend on the state, so the kinetics is not mass action in the strict sense. And the host lattice provides a signed defect content that has no molecular analog: every vertex carries a polarity $\sigma = \pm 1$, and the resulting classification of the vertex set splits the hypergraph into subgraphs of type vacancy and interstitial coupled only through cross-polarity recombination and annihilation classes. This is what turns a host-specific taxonomy of defects – loops, voids, bubbles – into a two-parameter index, makes Frenkel-pair

conservation a property of the edge classes rather than of a particular kernel calibration, and renders the sparsity structure of the Jacobian a statement about the graph rather than about one material. The detailed correspondence with the reaction-network literature is set out in Section S1 of the Supplementary Material.

2.2. Indexed Cluster State Space

A cluster identifier is a four-tuple

$$\boldsymbol{\tau} = (\sigma, n, p, \mathbf{c}) \in \{-1, +1\} \times \mathbb{Z}_{\geq 1} \times \mathcal{P} \times \mathcal{C}, \quad (1)$$

read as “a cluster of polarity σ , size n , located in the population p , carrying solute composition $\mathbf{c}$.” The polarity index $\sigma \in \{-1, +1\} = \{-1, +1\}$ classifies the cluster by the sign of its net Frenkel-pair content with respect to the host lattice, being -1 for a vacancy-type and +1 for an interstitial-type.

The size $n \in \mathbb{Z}_{\geq 1}$ is the count of host Frenkel-pair sites of the polarity in question carried by the cluster: $n = n_{\text{vac}}$ for $\sigma = -1$ and $n = n_{\text{sia}}$ for $\sigma = +1$. In particular, the monomers are $(\sigma = -1, n = 1) = V$ and $(\sigma = +1, n = 1) = I$. Rate kernels are written as $k_{\sigma\sigma'}(n, n')$ and addressed by polarity, and the state arrays are indexed on a non-negative size axis. The composition vector $\mathbf{c} = (c_1, \dots, c_S) \in \mathbb{Z}_{\geq 0}^S = \mathcal{C}$ records the count of each non-host species $s = 1, \dots, S$ trapped in the cluster. For the EUROFER-97 application, $S = 2$ with $\mathbf{c} = (c_{\text{He}}, c_{\text{H}})$. The composition vector is independent of σ and n . Its admissible range may depend on (σ, n, p) through host-specific saturation rules (e.g. a helium bubble saturates at a He/V ratio set by self-trapping energetics), and these rules enter as edge admissibility constraints on the graph of Section 2.3. The population label $p \in \mathcal{P}$ partitions clusters of the same polarity, size, and composition into spatially or microstructurally distinct populations, between which transport is itself a CD reaction. Typical population sets are

$$\mathcal{P}_- = \{ \text{bulk, GB, disl, ppt} \} \quad (\text{vacancy clusters}), \quad (2)$$

$$\mathcal{P}_+ = \{ \text{bulk-}\langle 111 \rangle, \text{bulk-}\langle 100 \rangle, \text{disl-trapped} \} \quad (\text{bcc SIA clusters}), \quad (3)$$

The set $\mathcal{P}$ is host-specific; they are physically distinct phases of the cluster.

The microstate of the system at time t is a function.

$$\mathbf{s}(t) : \mathcal{V} \rightarrow \mathbb{R}_{\geq 0}, \quad \boldsymbol{\tau} \mapsto C_{\boldsymbol{\tau}}(t), \quad (4)$$

where $C_{\tau}(t)$ is the volumetric concentration of clusters of identifier τ , and $\mathcal{V}$ is the admissible vertex set of the reaction admissibility graph defined in Section 2.3. A particular host material with parameter set Θ specifies $\mathcal{V}$, $\mathcal{P}$, $\mathcal{C}$, and the saturation rules; the algorithmic core operates on $\mathbf{s}$ without further reference to Θ .

Component notation. Let the primitive irradiation-produced species be $\mathcal{E} = \{I, V, \mathbf{g}\}$, where I denotes a self-interstitial atom, V denotes a vacancy, and $\mathbf{g}$ denotes an ordered list of gas species. The gas list is written as $\mathbf{g} = (g_1, g_2, \dots, g_{N_g})$, where N_g is the number of gas species included in the model. Gas atoms in the matrix are represented by reservoir concentrations

$$c_{\mu}^g(t), \quad \mu = 1, \dots, N_g, \quad (5)$$

where μ labels the gas species g_{μ} . These variables represent monomeric gas atoms in the matrix reservoir. They are coupled to vacancy clusters through trapping, emission, re-solution, and gas release from eliminated cavities. The model does not include reactions of the type: $g_{\mu} + g_{\nu} \rightarrow \text{gas cluster}$, so gas-only clusters are excluded.

In the component notation used from here on, the abstract population label p of Eq. (1) is written $\alpha = 1, \dots, N_P^I$ for the interstitial-cluster class $\mathcal{C}_I$ and $\beta = 1, \dots, N_P^V$ for the vacancy-cluster class $\mathcal{C}_V$, with $\mathcal{P}_I^{\alpha} \subset \mathcal{C}_I$ and $\mathcal{P}_V^{\beta} \subset \mathcal{C}_V$ the corresponding populations. Distinct populations may represent mobile SIA clusters, sessile loops, loops aligned or not aligned with a principal stress direction, or clusters attached to a given microstructural environment.

An interstitial-cluster state is written as

$$I_{n,\alpha}, \quad n = 1, \dots, N_I, \quad \alpha = 1, \dots, N_P^I, \quad (6)$$

where n is the number of self-interstitial atoms and α identifies the interstitial-cluster population. Its concentration is $c_{n,\alpha}^I(t)$. A vacancy-cluster, or cavity, state is written as

$$V_{m,\ell,\beta}, \quad m = 1, \dots, N_V, \quad \ell \in \mathcal{G}_m, \quad \beta = 1, \dots, N_P^V. \quad (7)$$

Here m is the number of vacancies in the cavity, β identifies the vacancy-cluster population, and

$$\ell = (\ell_1, \ell_2, \dots, \ell_{N_g}) \in \mathbb{Z}_{\geq 0}^{N_g} \quad (8)$$

is the gas inventory trapped inside the cavity. The component ℓ_μ is the number of atoms of gas species g_μ trapped inside the cavity. The set $\mathcal{G}_m$ denotes the admissible gas occupancies for a cavity containing m vacancies. Its concentration is $c_{m,\ell,\beta}^V(t)$. The notation $V_{m,\ell,\beta}$ denotes a vacancy-cluster or cavity state. Conventional chemical notation is recovered as a special case: with $\mathbf{g} = (\text{He})$, $V_{m,(\ell),\beta}$ denotes $V_m\text{He}_\ell$; with $\mathbf{g} = (\text{He}, \text{T})$, $V_{m,(\ell,p),\beta}$ denotes $V_m\text{He}_\ell\text{T}_p$; and $\ell = \mathbf{0}$ gives the pure vacancy cluster V_m , all in population β .

The full deterministic state vector is therefore

$$\mathbf{c}(t) = \left[\{c_{n,\alpha}^I\}_{n,\alpha}, \{c_{m,\ell,\beta}^V\}_{m,\ell,\beta}, \{c_\mu^g\}_{\mu=1}^{N_g} \right]^\top. \quad (9)$$

The first block contains interstitial-cluster populations, the second block contains vacancy-cluster/cavity populations, and the third block contains matrix gas-reservoir variables.

The generic signed-collision rule is

$$I_{n,\alpha} + V_{m,\ell,\beta} \longrightarrow \begin{cases} I_{n-m,\alpha'} + \mathcal{R}_\ell, & n > m, \\ \mathcal{R}_\ell, & n = m, \\ V_{m-n,\ell,\beta'}, & m > n. \end{cases} \quad (10)$$

Here α' and β' denote the product populations, determined by the specific reaction rule. In the simplest single-population model, $\alpha' = \alpha$ and $\beta' = \beta$ may be suppressed. The operator $\mathcal{R}_\ell$ denotes the release of the gas inventory from an eliminated cavity to the matrix gas reservoir. In the default reservoir model, $\mathcal{R}_\ell = \sum_{\mu=1}^{N_g} \ell_\mu g_\mu$. If $m > n$, the cavity survives and retains its gas inventory ℓ . If $n \geq m$, the supporting vacancy cluster is eliminated and its gas content is returned to the gas reservoir. Equation (10) is the component form of the cross-polarity annihilation class of Section 2.3.

2.3. The Reaction Admissibility Graph (RAG)

The reaction admissibility graph (RAG) is a labeled multidigraph.

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{\nu}, \mathbf{k}), \quad (11)$$

where $\mathcal{V} \subset \{-1, +1\} \times \mathbb{Z}_{\geq 1} \times \mathcal{P} \times \mathcal{C}$ is the (host-specific) vertex set of admissible cluster identifiers; $\mathcal{E} \subset \mathcal{V} \times \mathcal{V} \times \mathbf{N}$ is the edge set, where each edge $e = (\boldsymbol{\tau}_1, \boldsymbol{\tau}_2, \nu_e)$ carries a stoichiometric tuple $\nu_e \in \mathbf{N}$ that records all reactants and products of the elementary reaction; and $\mathbf{k} : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a

rate kernel $k_e(t, \mathbf{s}, \Theta)$ to each edge. The graph is host-specific in $(\mathcal{V}, \mathcal{E}, \boldsymbol{\nu})$ and calibration-specific in $\mathbf{k}$; the algorithmic core is the graph walker. Here, “graph walker” refers to the algorithmic traversal of the reaction admissibility graph. The ODE right-hand side is assembled by looping over admissible reaction edges $r \in \mathcal{E}$, evaluating the corresponding reaction flux $J_r(\mathbf{C})$, and accumulating its stoichiometric contribution $S_{ar}J_r$ to each affected cluster concentration. Here a labels one state variable of the model, that is, a single cluster population or gas-species reservoir, and the stoichiometric coefficient S_{ar} is the net number of objects of state a produced by one occurrence of reaction r , counted negatively when the state is consumed.

Every edge in $\mathcal{G}$ falls into exactly one of the following classes, identified by the pattern of polarity and population at its endpoints and by the form of its stoichiometric tuple: (1) Growth: $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n + 1, p, \mathbf{c})$ by absorption of a σ -monomer. Both endpoints share polarity, population, and composition. (2) Shrinkage: $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n - 1, p, \mathbf{c})$ by absorption of a $-\sigma$ -monomer (impingement of the opposite polarity). (3) Dissociation: $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n - 1, p, \mathbf{c})$ by thermal emission of a σ -monomer; the released monomer is added to the σ -monomer pool. The rate kernel involves a binding energy. (4) Recombination: $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n - 1, p, \mathbf{c})$ paired with $(-\sigma, 1, p', \mathbf{0}) \rightarrow \emptyset$. A cluster of polarity σ and size n absorbs a single $(-\sigma)$ -monomer; the monomer is annihilated rather than added to any pool. (5) Annihilation (cross-polarity binary hyperedge): $(\sigma, n, p, \mathbf{c})$ and $(-\sigma, n', p', \mathbf{c}')$ jointly $\rightarrow (\sigma, |n - n'|, p_{\text{surv}}, \mathbf{c}_{\text{surv}})$, where

$$\sigma = \text{sgn}(n - n'), \quad \mathbf{c}_{\text{surv}} = \begin{cases} \mathbf{c} & n > n', \\ \mathbf{c}' & n < n', \\ \mathbf{0} & n = n'. \end{cases} \quad (12)$$

When $n = n'$ both precursor vertices are removed ($\rightarrow \emptyset$); when $n \neq n'$ the larger cluster survives at a reduced size with the polarity of the larger reactant and the composition of the surviving fragment. This is the opposite polarity analog of *coalescence*: it is a *hyperedge* with two precursor vertices and a *quadratic* rate kernel. (6) Inter-population (cross-population edge): $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n, p', \mathbf{c})$ with $p \neq p'$. This edge represents segregation, sweep, or capture by a moving sink; it changes only the population coordinate. (7) Solute trapping/detrapping (intra-vertex edge): $(\sigma, n, p, \mathbf{c}) \rightarrow (\sigma, n, p, \mathbf{c} \pm \mathbf{e}_s)$, where $\mathbf{e}_s$ is the unit composition vector for species s . This edge changes only the composition coordinate. (8) Coalescence (binary growth):

$(\sigma, n, p, \mathbf{c})$ and $(\sigma, n', p, \mathbf{c}')$ together $\rightarrow (\sigma, n + n', p, \mathbf{c} + \mathbf{c}')$. This is a hyperedge in the strict sense; in the implementation, it is represented as a multi-source edge with a quadratic rate kernel. (9) Source (cascade-injection edge): $\emptyset \rightarrow (\sigma, n, p, \mathbf{c})$ with a rate set by the displacement-cascade spectrum. (10) Sink (fixed sink edge): $(\sigma, n, p, \mathbf{c}) \rightarrow \emptyset$ in a fixed sink (free surface, grain boundary acting as perfect sink or dislocation).

Admissibility rules. A vertex $(\sigma, n, p, \mathbf{c})$ is admissible if and only if (i) the polarity is consistent with the population (e.g., a “vacancy-loop” population p admits $\sigma = -1$ only); (ii) the composition lies in the host-specific admissible region $\mathcal{C}_{\sigma, n, p} \subseteq \mathcal{C}$ (e.g., a helium–vacancy cluster respects the maximum self-trapping ratio); and (iii) the size is at or above the minimum size for the population (monomers always exist in the bulk population; loops exist only for $n \geq n_{\text{loop}}^{\min}$). An edge is admissible if and only if its endpoints are admissible and its stoichiometric tuple respects the host conservation laws.

2.4. RAG Implementation

Cluster identifiers $(\sigma, n, p, \mathbf{c})$ are species or Petri-net places; unary and binary edges are directed hyperedges with one or two tails, or transitions; stoichiometric tuples $\boldsymbol{\nu}_e$ are columns of a stoichiometric matrix $\mathbf{S}$; and the Frenkel-pair and helium conservation laws of Section S4 are the left null vectors of $\mathbf{S}$, the P-invariants of the net. The complete dictionary, together with the sense in which the set of generated vertex and the state-dependent kernels place the system outside the classical deficiency theorems, is given in Section S1.

The framework is realized in the open-source RADCLUSTER code, split along exactly the line drawn in Section 2.3: a host-independent core that knows the graph but no physics and a host declaration that knows the physics but no algorithm. Adding a new host material amounts to specifying the admissible population set $\mathcal{P}_\sigma$, the admissible vertex set $\mathcal{V}$, the admissible edge set $\mathcal{E}$, and the rate-kernel library $\mathbf{k}$, while the solver, the preconditioner, the state-space reductions, and the conservation diagnostics are inherited unchanged. The state-space reduction theorems are themselves graph-theoretic; both are stated and proved on the abstract graph and inherited by any host. The analogy to object-oriented programming is precise: the abstract graph $\mathcal{G}$, together with its solver, preconditioner, and reduction theorems, acts as an abstract base class that defines the interface and implements all methods; and running a simulation for a particular irradiation scenario is instantiation.

2.5. The Master Equation as a “Graph Walker”

The concentration rate of change of a state X_a is assembled by a graph traversal that we call the reaction-graph *walker*. Visit each admissible edge $e \in \mathcal{E}$, read its stoichiometric update vector $\boldsymbol{\nu}_e$ and rate law k_e , and deposits the associated gains-loss contributions into the master equation. Writing $c_a(t)$ for the concentration of state X_a , the explicit variables are

$$c_a \in \{c_{n,\alpha}^I, c_{m,\ell,\beta}^V, c_\mu^g\}. \quad (13)$$

The mean-field master equation is

$$\boxed{\frac{dc_a}{dt} = G_a + \sum_r S_{ar} J_r(\mathbf{c}) - D_a c_a.} \quad (14)$$

Here G_a is the direct production rate of state a from collision cascades, $J_r(\mathbf{c})$ is the flux of reaction r , and $D_a c_a$ is a first-order loss to unresolved fixed sinks.

The reaction flux J_r depends on the reaction type. For a unary transition, $X_b \longrightarrow \text{products}$, $J_r(\mathbf{c}) = \Gamma_r c_b$, where Γ_r is a first-order rate coefficient. For a binary collision, $X_b + X_c \longrightarrow \text{products}$, $J_r(\mathbf{c}) = K_{bc}^{(r)} c_b c_c$, where $K_{bc}^{(r)}$ is the appropriate collision or capture rate. For a direct source process, $\emptyset \longrightarrow X_a$, $J_r = G_r$. The generalized model is built from the admissible process classes listed in the first two columns of Table 1: irradiation production, interstitial–interstitial and vacancy–vacancy coalescence, interstitial–vacancy collisions (Eq. (10)), gas trapping by cavities, gas emission and resolution, trap mutation, thermal vacancy and thermal SIA emission, population transfer, and loss to unresolved sinks. The remaining columns of that table record the instance each class takes in the EUROFER-97 host of Section 3, so that the abstract classes and their concrete realization are read from a single object.

3. The Ferritic–Martensitic Steel System

The abstract framework of Section 2 specializes to reduced-activation EUROFER-97 ferritic–martensitic steel in three steps: the host admissibility graph $\mathcal{G}^{\text{EUR}}$ is declared through its vertex set, active populations, and gas list; its eight edge classes P1–P8 are enumerated together with their closed-form rate kernels; and the graph-walker accumulator is applied to each

population. Previous CD calculations demonstrated that one-dimensional interstitial-cluster motion changes black-dot density, loop size, and loop size distributions in low-temperature ferritic iron. RadCluster extends this treatment to a coupled cavity–loop–helium system for ferritic/martensitic steels, combining cascade production bias, mixed 1D/3D $1/2\langle 111 \rangle$ -loop transport, loop coalescence and character conversion, residual-size SIA–vacancy reactions, and simultaneous cavity evolution.

Gas list and composition vector. For EUROFER-97 under neutron irradiation, the only resolved trapped solute is helium: $N_g = 1$, $g_1 = \text{He}$, $\ell = (\ell)$, so the gas reservoir $c_1^g(t) \equiv c_h(t)$ and the composition vector reduce to the occupancy of scalar helium $\ell \in \mathbb{Z}_{\geq 0}$.

Active populations. EUROFER-97 carries one population per polarity: $\mathcal{P}_{-}^{\text{EUR}} = \{\text{bulk}\}$, $N_P^V = 1$, $\mathcal{P}_{+}^{\text{EUR}} = \{\text{bulk}\}$, $N_P^I = 1$. The single SIA population covers both 3D-mobile clusters ($n < 4$) and $\frac{1}{2}\langle 111 \rangle$ prismatic loops ($n \geq 4$); the distinction is encoded in size-dependent rate kernels, not a separate population label. Grain boundaries, the dislocation network, and MX/M₂₃C₆ precipitates are treated as fixed unresolved sinks. With one population per polarity the population labels are suppressed:

$$c_{n,\alpha}^I(t) \rightarrow c_n(t), \quad c_{m,(\ell),\beta}^V(t) \rightarrow c_{m,\ell}(t), \quad c_1^g(t) \rightarrow c_h(t). \quad (15)$$

Admissible vertex set.

$$\mathcal{V}_{-}^{\text{EUR}} = \{ (-1, m, \text{bulk}, (\ell)) : 1 \leq m \leq N_V, 0 \leq \ell \leq \ell_{\max}(m) \}, \quad (16)$$

$$\mathcal{V}_{+}^{\text{EUR}} = \{ (+1, n, \text{bulk}, \mathbf{0}) : 1 \leq n \leq N_I \}, \quad (17)$$

$$\mathcal{V}_0^{\text{EUR}} = \{ g_1 = \text{He} \}, \quad (18)$$

comprising size-resolved SIA clusters, size-resolved voids ($\ell = 0$) and helium bubbles ($\ell \geq 1$), and the single matrix He reservoir. The maximum helium content obeys the saturation rule

$$\ell_{\max}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor, \quad \alpha_{\text{He}} \in [1.5, 2.0]. \quad (19)$$

Three primary populations. Equations (16)–(18) yield three populations evolved by the master equations: (1) **SIA clusters** $c_n(t)$, $n = 1, \dots, N_I$: loops with $\frac{1}{2}\langle 111 \rangle$ Burgers vector for $n \geq 4$, and 3D-mobile clusters for $n = 1, 2, 3$, carrying no helium; (2) **vacancy clusters** / **cavities** $c_{m,\ell}(t)$, $m = 1, \dots, N_V$,

$\ell = 0, \dots, \ell_{\max}(m)$: spherical cavities, pure voids when $\ell = 0$ and helium bubbles when $\ell \geq 1$; (3) **free helium** $c_h(t)$, the single gas-reservoir variable. The signed lattice-defect contents are

$$q(I_n) = +n, \quad q(V_m \text{He}_\ell) = -m, \quad q(\text{He}) = 0, \quad (20)$$

as required by the conservation law of Section S4.

Process classes P1–P8. The admissible edges of $\mathcal{G}^{\text{EUR}}$ partition into eight process classes, each an instance of one of the abstract edge classes of Section 2.3: vacancy–SIA recombination (P1) and impingement of a 1D-gliding SIA cluster on a cavity (P6) are cross-polarity edges; absorption of a monomer by a cavity (P2) or by a loop (P3) are growth/shrinkage edges; absorption at a fixed sink (P4) is a sink edge; thermal monomer emission (P5) is a dissociation edge; and trap mutation (P7) and radiation re-solution (P8) act on the helium occupancy ℓ as the gas-pressure-driven and solute-detrapping edges. Because only one population is carried per polarity, the inter-population transfer edge is suppressed and defect production is treated as a source term. Each class, the abstract process class of Section 2.3 that it instantiates, and its rate kernel are listed in Table 1.

Table 1: Abstract process classes (columns 1–2) and the edge classes they generate in the EUROFER-97 admissibility graph (columns 3–4; mono-defect, $i_{\text{mobile}} = v_{\text{mobile}} = 1$, one population per polarity, $g_1 = \text{He}$). The host declaration suppresses the population labels α, β and fixes the gas index μ , so each abstract class collapses onto one or more concrete classes P1–P8. Processes that connect adjacent nodes on a size ladder appear as separate growth and shrinkage rows. Sums over repeated indices are implied.

Abstract class	General reaction and flux	P	EUROFER-97 reaction and flux J_r
Irradiation production	$\emptyset \longrightarrow X_a$ $G_a = G_a^{\text{FP}} + G_a^{\text{cascade}} + G_a^{\text{gas}} + G_a^{\text{trans}}$	—	cascade sources $G_n, G_{m,\ell}, G_{\text{He}}$
Interstitial–interstitial coalescence	$I_{n,\alpha} + I_{n',\alpha'} \longrightarrow I_{n+n',\alpha''}$ $J_{n\alpha,n'\alpha'}^{II} = K_{n\alpha,n'\alpha'}^{II} c_{n,\alpha}^I c_{n',\alpha'}^I$	P3	$I_1 + I_n \rightarrow I_{n+1}$ $\mathcal{K}_{i,n}^{\text{loop}} c_1 c_n$
Vacancy–vacancy coalescence	$V_{m,\ell,\beta} + V_{m',\ell',\beta'} \longrightarrow V_{m+m',\ell+\ell',\beta''}$	P2v	$V_1 + V_m \rightarrow V_{m+1}$ $\mathcal{K}_{v,m}^{\text{cav}} c_{1,0} c_{m,\ell}$

continued on next page

clusters are 3D mobile for $1 \leq m \leq v_{\text{mobile}}$; larger cavities grow or shrink only through single-monomer absorption or emission.

Reaction Rate Kernels. Every binary edge of $\mathcal{G}^{\text{EUR}}$ carries a diffusion-limited kernel, written in atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$, $\Omega = a^3/2$) through $\omega_\alpha = D_\alpha/a^2$ and $\xi(s) = (3s/8\pi)^{1/3}$, so that a cluster of s defects has radius $r_s = \xi(s)a$. The five cases below, in order of increasing generality, are derived in Section S2 of the Supplementary Material and collected in Table 2.

(i) *3D migration to a fixed sink*; The steady-state problem is Laplace’s equation, so the kernel depends on the shape of the sink through a single length and its electrostatic capacitance C : $\mathcal{K}^{\text{vol}} = 4\pi C D_\alpha$ [1, 3, 62]. A cavity of m vacancies has $C = r_m$, a loop of n SIAs is a disk of radius $r_n = a[n/(\pi\sqrt{3})]^{1/2}$ with $C = \pi r_n/\ln(8r_n/r_c)$ [63], and hence

$$\mathcal{K}_{\alpha,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_\alpha, \quad A_{\text{sph}} = (192\pi^2)^{1/3} \simeq 12.37, \quad (21)$$

$$\mathcal{K}_{\alpha,n}^{\text{loop}} = A_{\text{loop}} Z_\alpha^{\text{loop}} n^{1/2} \omega_\alpha, \quad A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \simeq 10.77, \quad (22)$$

with Z_α^{loop} carrying the loop geometry factor and the elastic bias, $Z_i^{\text{loop}} > 1 = Z_v^{\text{loop}}$ being the only species-specific quantity [64, 65].

(ii) *Two mobile clusters.* In the relative coordinate the two random walks add, so the pair diffuses with $D_A + D_B$ and reacts on the contact surface swept by the two bodies; for spheres that surface is a sphere of radius $r_s + r_{s'}$, so additivity of radii and diffusivities is exact and the three binary channels — SIA–SIA and V–V coalescence, and V–I annihilation ($V_m + I_n \rightarrow V_{m-n}$, I_{n-m} , or $\emptyset$) — share one kernel,

$$\mathcal{K}_{s,s'}^{ab} = 8\pi (\xi_s + \xi_{s'}) (\omega_s^{\text{eff}} + \omega_{s'}^{\text{eff}}), \quad a, b \in \{i, v\}, \quad (23)$$

giving $\mathcal{K}_{n,n'}^{ii}$, $\mathcal{K}_{m,m'}^{vv}$ and $\mathcal{K}_{m,n}^{vi}$; here $\omega_\alpha^{\text{eff}}$ is the three-dimensionally equivalent reaction frequency, equal to ω_α for a 3D migrator and to $\omega_n^{1D}/3$ for a 1D glider, the factor $\frac{1}{3}$ being the fraction of space explored at any instant. Both reactants must be mobile ($n, n' \leq i_{\text{mobile}}$, $m, m' \leq v_{\text{mobile}}$), like-polarity channels obey $n' \leq n$ and $m' \leq m$ to avoid double counting, and He–He coalescence is excluded (binding $\lesssim 0.05$ eV). For a mobile disc capacitances are not additive, so $C \simeq r_m + \pi r_n/\ln(8r_n/r_c)$ is an interpolation. Anisotropic diffusion reduces to the same form by an affine rescaling with $\bar{D} = (D_x D_y D_z)^{1/3}$ [66], extended to mixed mobilities and arbitrary D_A/D_B — with characteristic exponents 0, $-1/2$ and $-1/3$ for 3D–3D, 3D–1D and 1D–1D pairs — in [67–69].

(iii) *One-dimensional migration.* SIA clusters with $n \geq 4$ glide along $\langle 111 \rangle$. A 1D walker samples a line rather than a volume, so no binary rate constant exists: absorption is set by the whole sink field through $k_{(1),m}^2 = 6\kappa_m\kappa$, with $\kappa_m = \sigma_m N_m$, $\sigma_m = \pi r_m^2$ and $\kappa = \sum_j \kappa_j$ — quadratic in sink density and scaling as r_m^4 , against $k_{(3),m}^2 = 4\pi r_m N_m$ in three dimensions [70, 71] — so that the kernel $6D_n\sigma_m\kappa$ is not a property of the reacting pair alone.

(iv) *Mixed 1D/3D migration.* Direction changes after a mean straight-line length ℓ_n interpolate between the limits along the Trinkaus–Heinisch–Barashev–Golubov–Singh master curve [72, 73]: $k_{n,m}^2 = y_{n,m}k_{(1),m}^2$, $y = \frac{1}{2}[1 + (1 + 4x^2)^{1/2}]$, with x a single dimensionless variable built from ℓ_n , $k_{(1),m}^2$ and $k_{(3),m}^2$. (v) *SIA cluster capture at a cavity.* Case (i) and the mixed-mobility reduction of (iv) combine into the kernel that enters the master equations for SIA absorption at, and SIA-induced shrinkage of, cavities,

$$\mathcal{K}_{n,m}^{ic} = \begin{cases} A_{\text{sph}} m^{1/3} \omega_i^{\text{eff}}, & 1 \leq n \leq 3 \text{ (3D, P2)}, \\ \frac{A_{\text{sph}} m^{1/3} \omega_n^{\text{eff}}}{[1 + B_{\text{rot}} \hat{L}^2 \xi(m)^{-2}]^{1/2}}, & 4 \leq n \leq i_{\text{mobile}} \text{ (1D/3D, P6)}, \\ 0, & n > i_{\text{mobile}} \text{ (sessile)} \end{cases} \quad (24)$$

with $\hat{L} = \ell_n/a$ and $B_{\text{rot}} = 2/9$. The middle branch is the near-3D reduction of the master curve, Eq. (S28) of the Supplementary Material: it returns the Smoluchowski value as $\ell_n \rightarrow 0$ and decays as ℓ_n^{-1} once the glide segment exceeds the cavity radius r_m . The thermal emission of vacancies, helium and Frenkel pairs by trap mutation is fixed by detailed balance with the corresponding capture rate and the binding energy E_b^α . The geometric prefactors are collected in Table S12, and the sink parameters, energetics, and mobility cutoffs in Table S13.

In Table 2, $\mathcal{K}$ is in units of atomic-fraction, $\omega_\alpha = D_\alpha/a^2$, $\xi(s) = (3s/8\pi)^{1/3}$, $r_s = \xi(s)a$, $\sigma_m = \pi r_m^2$, $\kappa = \sum_j \sigma_j N_j$; C is the capacitance entering $\mathcal{K}^{\text{vol}} = 4\pi C\mathcal{D}$, and $\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$ with $\Omega = a^3/2$, so that $A_{\text{sph}} = 8\pi\xi(1) = (192\pi^2)^{1/3} \simeq 12.37$. The loop capacitance is the asymptotic slender-torus result, $r_n \gg r_c$; the constant inside the logarithm depends on the torus convention adopted for the core radius r_c . In the dislocation-network row ρ_d is the physical line density (m^{-2}) and Z_α the capture efficiency carrying the elastic size-interaction bias only, with no core correction. The sphere–sphere capture length is exact within the continuum absorbing-sphere (Smoluchowski) model,

Table 2: Diffusion-limited reaction kernels of the EUROFER-97 admissibility graph.

Case (mobility)	Kernel [Hz]	Capture length C	Scaling	Ref.
3D mobile + fixed cavity	$A_{\text{sph}} m^{1/3} \omega_\alpha$	r_m	$m^{1/3} N_m$	[1, 3]
3D mobile + fixed loop	$A_{\text{loop}} Z_\alpha^{\text{loop}} n^{1/2} \omega_\alpha$	$\pi r_n / \ln(8r_n/r_c)$	$n^{1/2} N_n$	[63, 64]
3D mobile + dislocation network	$Z_\alpha \rho_d a^2 \omega_\alpha$	— (line sink)	ρ_d	[64]
Two 3D mobile spheres	$8\pi(\xi_s + \xi_{s'}) (\omega_s^{\text{eff}} + \omega_{s'}^{\text{eff}})$	$r_s + r_{s'}$ (continuum)	$s^{1/3} + s'^{1/3}$	[3, 66]
V–I recombination ($m=n=1$)	$4\sqrt{3}\pi(\omega_i + \omega_v) \simeq 21.8 \omega_i$	$\sqrt{3}a/2$	—	[3]
Mobile sphere + mobile loop	$(8\pi C/a)(\omega_s^{\text{eff}} + \omega_{s'}^{\text{eff}})$	r_m + $\pi r_n / \ln(8r_n/r_c)$ (interp.)	—	[67, 69]
1D/fixed sinks	$\mathcal{K}_{n \rightarrow m}^{(1)} = 6D_n \sigma_m N_m \kappa$; $k_{(1),m}^2 = 6\sigma_m N_m \kappa$	— (line sampling)	$m^{4/3} N_m^2$	[70, 71]
1DR/fixed sinks	$\mathcal{K}_{n \rightarrow m}^{(1R)} = D_n y_{n,m} k_{(1),m}^2$; $y = \frac{1}{2}[1 + \sqrt{1 + 4x^2}]$	= — (master curve)	$k_{(1)}^2 \rightarrow k_{(3)}^2$	[72]
1DR/immobile	$\mathcal{K}_{n \rightarrow m}^{(1R)} = D_n y_{n,m} k_{(1),n,m}^2$; $k_{(3),n,m}^2 = 4\pi R_{n,m} N_m$	= $R_{n,m} = r_n + r_m$	$N_m^2 \rightarrow N_m$	[70, 72]

whereas the sphere–loop entry is an interpolation between the two limiting capacitances and not the exact capacitance of the combined geometry. In the three 1D rows, $k_{(1),m}^2$ and $k_{(3),m}^2$ are the one- and three-dimensional sink strengths of family m and ℓ_n is the mean straight-line glide length of cluster n between Burgers-vector rotations, $\ell_n^2 = d_j^2 n_{\text{ch}}$ with $d_j = \sqrt{3}a/2$ the glide jump distance and n_{ch} the mean number of jumps between rotations; $y \equiv y_{n,m} = \frac{1}{2}[1 + (1 + 4x_{n,m}^2)^{1/2}]$ is the Trinkaus–Heinisch master-curve factor that interpolates the mixed 1D/3D sink strength, $k_{n,m}^2 = y_{n,m} k_{(1),m}^2$, with $x \equiv x_{n,m} = [\ell_n^2 k_{(1),m}^2 / 12 + k_{(1),m}^4 / k_{(3),m}^4]^{-1/2}$, so that $x \rightarrow 0$ gives $y \rightarrow 1$ and the pure 1D limit while $x \gg 1$ gives $y \rightarrow x$ and the 3D limit; and N denotes a number density. Every entry is a loss frequency in hertz: for the 3D channels the tabulated coefficient is the atomic-fraction rate constant $\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$ that multiplies the partner’s atomic fraction, while the 1D entries are written directly as $\mathcal{K} = D k^2$. In the 1DR/immobile row m is an immobile

cluster family of number density N_m , so that the kernel is quadratic in N_m in the 1D limit and linear in the 3D limit, $\mathcal{K} \rightarrow 4\pi R_{n,m} D_n N_m$. Derivations in Section S2 of the Supplementary Material. The graph admits no reaction between two mutually 1D-mobile clusters. Those kernels are derived in Sec. S2 and collected in Table S3 of the Supplementary Material, and they are excluded here on two grounds. First, the geometric 1D–1D results assume straight flights maintained over the whole encounter, a picture established for pure iron; in EUROFER-97 the Cr, W, C and N solutes and their traps randomize the glide direction far more frequently, so a mutually 1D-mobile pair crosses over to the three-dimensional kinetics already tabulated above rather than sampling a shared line. Second, the 1D–1D channels are of higher order in cluster concentration and are numerically negligible at the mobile-cluster densities reached here. Writing each channel as an equivalent sink strength $\mathcal{K}/D$, the dislocation network contributes $Z_i \rho_d \simeq 10^{14} \text{ m}^{-2}$, while the non-collinear 1D–1D channel contributes $2\pi R \alpha N_B \simeq 1.3 \times 10^{12} \text{ m}^{-2}$ at $N_B = 10^{21} \text{ m}^{-3}$ with $R = 1 \text{ nm}$ and $\alpha \simeq 0.2$, about one percent of it, and the collinear channel contributes $6\pi^2 R^4 N_A N_B \simeq 6 \times 10^7 \text{ m}^{-2}$, seven orders of magnitude below the network. The non-collinear term would rival the fixed sinks only for $N_B \gtrsim 8 \times 10^{22} \text{ m}^{-3}$, a mobile-cluster atomic fraction near 10^{-6} , which these simulations never approach.

Master Equation Assembly. The abstract master equation 14 is algorithmically assembled by looping over every admissible edge $r \in \mathcal{E}^{\text{EUR}}$, computing the flux J_r , and accumulating the signed contribution $S_{ar} J_r$ into each affected population a . Table 1 lists every edge class P1–P8 with its reaction and its flux; the stoichiometric contributions these make to the three EUROFER-97 population equations $\dot{c}_n$, $\dot{c}_{m,\ell}$ and $\dot{c}_h$ are collected in the boxed equations below. Where a process connects adjacent nodes on a ladder (e.g. cavity growth/shrinkage), gain and loss appear as separate rows, the net effect being the tridiagonal ladder operator visible in those equations. Figure 1 shows the EUROFER-97 admissibility graph, with vertices arranged by size within the $\frac{1}{2}\langle 111 \rangle$ loop, $\langle 100 \rangle$ loop, and vacancy/cavity populations, the cascade source and unresolved sinks at either end, and the unary, source, and sink edge classes color-coded. Binary reactions, whose arc count grows as $O(n_{\text{max}}^2)$, are tabulated instead in Fig. 2.

Reading Table 1 column by column yields the three ODEs for SIA clusters, vacancy–helium clusters and free helium; every term in the master equations below corresponds one-to-one to a row and column of that table. Define the

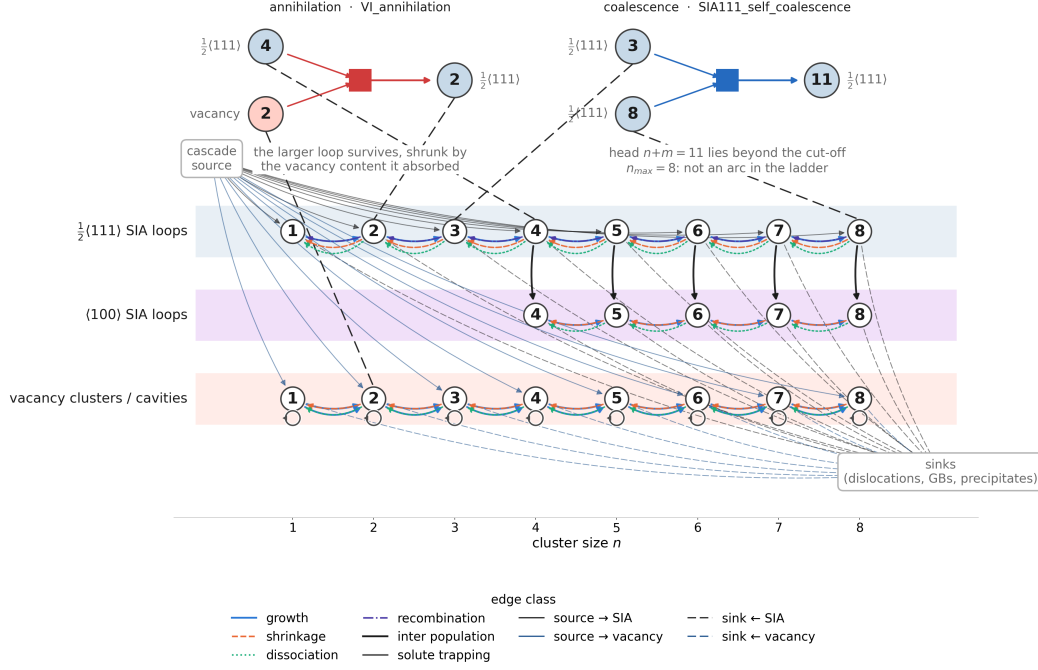


Figure 1: EUROFER-97 reaction admissibility graph: cluster vertices by size in three populations, with source, sink, and unary edge classes.

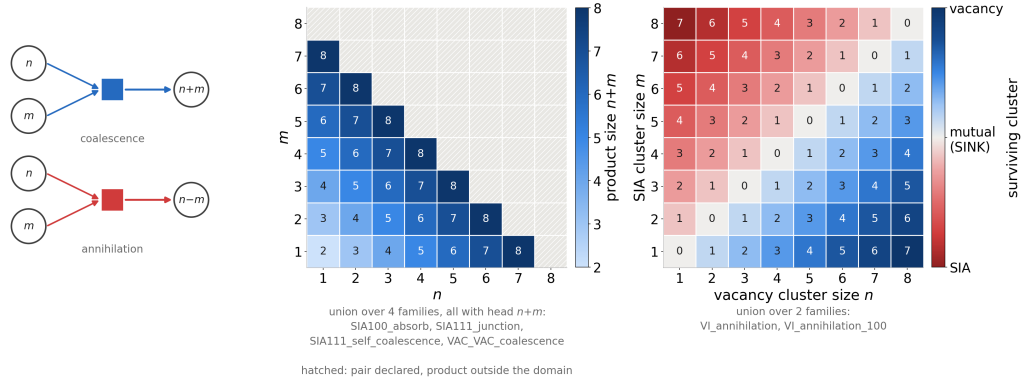


Figure 2: Binary hyperedges: coalescence product $n+m$ and surviving annihilation cluster over all admissible reactant pairs.

total cavity concentration at vacancy size m and the combined fixed-sink

rate:

$$\bar{c}_m(t) \equiv \sum_{\ell=0}^{\ell_{\max}(m)} c_{m,\ell}(t), \quad \mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p. \quad (25)$$

The equations below are written in the general (coalescence) form; the mono-defect simplification is recovered by setting $i_{\text{mobile}} = v_{\text{mobile}} = 1$, for which the coalescence kernels reduce as given in Section S2 of the Supplementary Material.

Defect Class 1: SIA clusters c_n .

$$\begin{aligned} \frac{dc_n}{dt} = & \underbrace{G_n}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} \mathcal{K}_{n', n-n'}^{ii} c_{n'} c_{n-n'}}_{\text{SIA-SIA coalescence gain } (I_{n'} + I_{n-n'} \rightarrow I_n)} \\ & - \underbrace{c_n \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n, n'}^{ii} c_{n'}}_{\text{SIA-SIA coalescence loss } (I_n + I_{n'} \rightarrow I_{n+n'})} + \underbrace{\varepsilon_i(n+1) c_{n+1} - \varepsilon_i(n) c_n}_{\text{thermal emission (SIA from loops, P5)} \delta_{n \geq 2}} \\ & + \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n+m'}^{vi} c_{m',0} c_{n+m'}}_{\text{V-I annihilation gain } (V_{m'} + I_{n+m'} \rightarrow I_n)} - \underbrace{\sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m', n}^{vi} c_{m',0} c_n}_{\text{V-I annihilation loss } (V_{m'} + I_n \rightarrow I_{n-m'})} \\ & - \underbrace{\sum_{m=1}^{N_V} \mathcal{K}_{n,m}^{ic} c_n \bar{c}_m}_{\text{SIA cluster } \rightarrow \text{cavity absorption (P2/P6)}} - \underbrace{\mathcal{D}_i(n) c_n}_{\text{fixed-sink loss (mobile clusters only, P4)}} \\ & + \underbrace{\delta_{n=1} \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\max}(m)} \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap-mutation SIA source at } n=1 \text{ (P7)}} \end{aligned} \quad (26)$$

where $\mathcal{K}_{n,m}^{ic}$ is the SIA-cluster capture kernel at cavities, Eq. (24).

Defect Class 2: Vacancy clusters $c_{m,\ell}$. Voids ($\ell = 0$) and bubbles ($\ell \geq 1$) share a unified (m, ℓ) grid. Spherical cavities are unbiased ($Z_i^{\text{cav}} = Z_v^{\text{cav}} = 1$).

The v-v coalescence terms are active only for $m \leq v_{\text{mobile}}$; SIA-induced shrinkage spans all mobile SIA sizes through $\mathcal{K}_{n,m}^{ic}$.

$$\begin{aligned}
\frac{dc_{m,\ell}}{dt} = & \underbrace{G_{m,\ell}}_{\text{production}} + \underbrace{\frac{1}{2} \sum_{m'=1}^{\min(m-1, v_{\text{mobile}})} \mathcal{K}_{m',m-m'}^{vv} c_{m',0} c_{m-m',\ell} \delta_{m \leq 2 v_{\text{mobile}}}}_{\text{V-V coalescence gain } (V_{m'}+V_{m-m'} \rightarrow V_m)} \\
& - \underbrace{c_{m,\ell} \sum_{m'=1}^{v_{\text{mobile}}} \mathcal{K}_{m,m'}^{vv} c_{m',0}}_{\text{V-V coalescence loss } (V_m+V_{m'} \rightarrow V_{m+m'})} + \underbrace{\varepsilon_v(m+1, \ell) c_{m+1,\ell} - \varepsilon_v(m, \ell) c_{m,\ell}}_{\text{thermal emission (vacancy from cavities, P5)}} \\
& + \underbrace{\sum_{n=1}^{\min(i_{\text{mobile}}, N_V - m)} \mathcal{K}_{n,m+n}^{ic} c_n c_{m+n,\ell}}_{\text{SIA-induced shrinkage gain } (I_n+V_{m+n} \rightarrow V_m)} - \underbrace{\sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{ic} c_n c_{m,\ell}}_{\text{SIA-induced shrinkage loss } (I_n+V_m \rightarrow V_{m-n})} \\
& + \underbrace{A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell-1} \delta_{\ell \geq 1} - A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h c_{m,\ell} \delta_{\ell < \ell_{\text{max}}}}_{\text{He absorption by cavities (P2, He-side)}} \\
& + \underbrace{\varepsilon_h(m, \ell+1) c_{m,\ell+1} - \varepsilon_h(m, \ell) c_{m,\ell} \delta_{\ell \geq 1}}_{\text{He emission from cavities (P5, He-side)}} + \underbrace{\Gamma_{\text{TM}}(m-1, \ell) c_{m-1,\ell} - \Gamma_{\text{TM}}(m, \ell) c_{m,\ell}}_{\text{trap mutation (P7)}} \\
& + \underbrace{b_0 \dot{\phi}(\ell+1) c_{m,\ell+1} - b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}} - \underbrace{\mathcal{D}_v c_{m,\ell} \delta_{m \leq v_{\text{mobile}}}}_{\text{fixed-sink loss (mobile cavities only, P4)}}
\end{aligned} \tag{27}$$

Free helium c_h . He-He coalescence is excluded, and helium interacts with the vacancy-cluster grid only through mono-He trapping and emission.

$$\begin{aligned}
\frac{dc_h}{dt} = & \underbrace{G_{\text{He}}}_{\text{production}} - \underbrace{\sum_{m=1}^{N_V} A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \sum_{\ell=0}^{\ell_{\text{max}}(m)-1} c_{m,\ell}}_{\text{He absorption by cavities (P2)}} \\
& - \underbrace{\mathcal{D}_h c_h}_{\text{fixed-sink loss (P4)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \varepsilon_h(m, \ell) c_{m,\ell}}_{\text{He emission (P5)}} + \underbrace{\sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} b_0 \dot{\phi} \ell c_{m,\ell}}_{\text{radiation re-solution (P8)}}
\end{aligned} \tag{28}$$

Mass Conservation Constraints. The master equations (26)–(28) satisfy two exact conservation identities, monitored as diagnostics rather than imposed as constraints. The first is the Frenkel-pair balance: cascades deposit SIA and vacancy atoms in a balanced ratio, every mutual $I_n + V_m$ reaction destroys $\min(n, m)$ of each so that the destruction sums cancel, and only mobile clusters reach fixed sinks. With the inventories

$$\begin{aligned}
S_I(t) &\equiv \sum_{n=1}^{N_I} n c_n(t), & S(t) &\equiv \sum_{m=1}^{N_V} m \bar{c}_m(t), \\
\Delta J_{\alpha}^{\text{fix}}(t) &\equiv \int_0^t \sum_s s \mathcal{D}_{\alpha}(s) c_s(t') dt',
\end{aligned} \tag{29}$$

the balance reads

$$S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t) = S(0) - S_I(0) = \text{const.}, \tag{30}$$

where S_I is the SIA content held in clusters and loops, S the vacancy content of all cavities, and $\Delta J_{\alpha}^{\text{fix}}$ the cumulative atomic content of species $\alpha \in \{i, v\}$ delivered to the fixed sinks (dislocations, grain boundaries, precipitates) through the loss coefficient $\mathcal{D}_{\alpha}(s)$, summed over the entire mobile range rather than over the monomers alone. For a defect-free initial condition this is the swelling identity: once loops saturate, swelling grows at the rate at which the biased dislocation channel ($Z_i^d \neq Z_v^d$) drains the surplus SIA inventory faster than the vacancy inventory, the unbiased grain-boundary and precipitate channels contributing equally to both sums.

The second identity is helium conservation. Trap mutation, radiation re-solution and thermal emission only redistribute helium between the free and

trapped pools, so their contributions cancel pairwise and the total inventory $S_{\text{He}} = c_h + \sum_m \sum_\ell \ell c_{m,\ell}$ changes only through free helium captured at fixed sinks and helium carried there by mobile cavities,

$$\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - \mathcal{D}_h c_h - \mathcal{D}_v \sum_{m=1}^{m_{\text{max}}^v} \sum_{\ell} \ell c_{m,\ell}, \quad (31)$$

with c_h the free-helium concentration, $c_{m,\ell}$ the concentration of cavities holding m vacancies and ℓ helium atoms, G_{He} the helium generation rate, and m_{max}^v the mobility cutoff of cavities. Helium is therefore *not* conserved against $\int G_{\text{He}} dt'$ alone, and the fixed-sink loss must be tracked explicitly, though it is numerically small.

The cumulative fixed-sink fluxes are evolved as auxiliary ODEs appended to the state vector, so the residuals share the integrator's error control. Such identities are rarely tested in the cluster-dynamics literature; derivations and explicit residuals are given in Section S4 of the Supplementary Material.

Conditions for Cavity and Loop Growth. Equations (26) and (27) decide at each node whether a cluster advances or retreats along its ladder. In a ferritic–martensitic steel, that decision collapses to a single number, the ratio of the mobile vacancy flux to the mobile SIA flux,

$$\Phi_I = \sum_{n=1}^{i_{\text{mobile}}} n D_n c_n, \quad \Phi_V = \sum_{m=1}^{v_{\text{mobile}}} m D_m c_{m,0}, \quad x \equiv \Phi_V / \Phi_I, \quad (32)$$

measured against two thresholds. Because bcc iron is diffusionally cubic, every D_α is a scalar and the capture efficiencies carry no orientation factor, so the entire species asymmetry of the model is elastic and sits at two sinks only, $Z_i^d > 1 = Z_v^d$ at the dislocation network and $Z_i^{\text{loop}} > 1 = Z_v^{\text{loop}}$ at the loops; cavities, lath boundaries and precipitates are unbiased. Requiring the net capture rate, less thermal emission, to be positive gives one threshold per family,

$$\underbrace{x < Z_i^{\text{loop}}}_{\text{loops grow}} \quad \underbrace{x > 1 + \varepsilon_{m,\ell}}_{\text{cavity } (m,\ell) \text{ grows}}, \quad (33)$$

where $\varepsilon_{m,\ell} = D_v c_v^{\text{th}}(m,\ell) / \Phi_I$ is the thermal emission flux of that cavity reduced by Φ_I . Loops and cavities are of opposite character and do not share a threshold, so they grow together only inside the window $1 + \varepsilon_{m,\ell} < x < Z_i^{\text{loop}}$

— the critical-radius statement in flux-ratio form — which the elastic bias alone opens, thermal emission closes from below, and helium widens by lowering $\varepsilon_{m,\ell}$.

Whether x falls inside that window is decided by the production bias, since only the unclustered part of each species migrates freely. With f_i^{cl} and f_v^{cl} the in-cascade clustered fractions and k_I^2/k_V^2 the sink-strength ratio, the quasi-steady balance gives $x \simeq \eta k_I^2/k_V^2$ with $\eta = (1 - f_v^{\text{cl}})/(1 - f_i^{\text{cl}})$, so that $x < Z_i^{\text{loop}}$ requires $f_i^{\text{cl}} - f_v^{\text{cl}} \lesssim 0.04$. Iron’s cascade statistics miss this by an order of magnitude — $f_i^{\text{cl}} - f_v^{\text{cl}} \simeq 0.43$ under a fission spectrum and 0.45 under a fusion one, hence $\eta \simeq 2.0\text{--}2.3$ — placing x above the window. Production bias thus preferentially partitions SIAs into clusters and leaves an excess of freely migrating vacancies that promotes cavity growth. Because the interstitial capture bias of sessile loops is typically only a few percent, monomer fluxes alone may cause these loops to shrink. Simultaneous cavity and interstitial-loop growth therefore generally requires an additional interstitial supply, most plausibly through the capture of glissile SIA clusters, although direct cascade production and other non-monomer mechanisms may also contribute [74–79]. In EUROFER-97 that supply is the clustered fraction deposited directly as loops in the cascade, and the same production asymmetry leaves the vacancy surplus that drives swelling. Co-growth here is a production-bias phenomenon rather than a shared-window one. The thresholds, the flux ratio, the cascade-statistics estimate, and the loop budget that follows from them are derived in Section S5 of the Supplementary Material.

4. State Space Reduction

4.1. Phase Space Reduction Strategy

The dimension of the state space is fixed by what must be resolved experimentally. Transmission electron microscopy of ferritic–martensitic steels irradiated to 50–100 dpa in fission and fusion-relevant spectra shows $\frac{1}{2}\langle 111 \rangle$ SIA loops of mean diameter 10–30 nm at 350–500 °C, the largest approaching 50 nm; a 30 nm loop holds $n \approx 1.3 \times 10^4$ SIAs and a 50 nm loop $n \approx 3.7 \times 10^4$, so the SIA ladder must extend to $N_I \sim 10^3\text{--}10^4$. Cavities span a comparable range: helium bubbles stabilize at 1–3 nm ($m \approx 45\text{--}1200$) at 350–450 °C, and near the swelling peak (480–550 °C) a subset grows to 5–10 nm ($m \approx 5 \times 10^3\text{--}4 \times 10^4$), giving $N_V \sim 10^3\text{--}10^4$. The ladder lengths are therefore set by the observable microstructure rather than chosen.

The full state vector has three blocks:

$$n_{\text{eq}}^{\text{full}} = \underbrace{N_I}_{\text{SIA ladder}} + \underbrace{n_{VH}^{\text{full}}}_{\text{gas-vacancy grid}} + \underbrace{1}_{c_h}. \quad (34)$$

The SIA block is a one-dimensional ladder of N_I equations and scales benignly. The gas-vacancy block does not: every vacancy size m carries a helium sub-ladder of depth $\ell_{\text{max}}(m) = \lfloor \alpha_{\text{He}} m^{2/3} \rfloor$ (Eq. 19), so

$$n_{VH}^{\text{full}} = \sum_{m=1}^{N_V} [\ell_{\text{max}}(m) + 1] \approx N_V \left(1 + \frac{3}{5} \alpha_{\text{He}} N_V^{2/3} \right) \sim N_V^{5/3}, \quad (35)$$

which reaches 10^4 – 10^5 for $N_V = 10^3$ – 10^4 and brings the unreduced system to coupled ODEs coupled with $n_{\text{eq}}^{\text{full}} \sim 10^4$ – 2×10^5 .

Helium is not the only gas the cavities must accommodate. The transmutation reactions that supply 10–15 appm He/dpa under a fusion spectrum supply three to four times as much hydrogen, which is trapped in the same cavities, lowers the vacancy emission barrier and at high loading adds to the internal pressure. Resolving it explicitly promotes the cavity state from (m, ℓ) to $(m, \ell_{\text{He}}, \ell_{\text{H}})$. Each additional gas species raises the exponent of N_V by $2/3$. Two species give $N_V^{7/3}$: states of order 10^6 at $N_V = 10^3$ and approaching 10^9 at $N_V = 10^4$. A gridded gas axis is thus marginal for one species and untenable for two.

Cost compounds size. The right-hand side is severely stiff, so an implicit BDF method is mandatory and an $n_{\text{eq}} \times n_{\text{eq}}$ Newton system is solved at every step; dense LU costs $O(n_{\text{eq}}^3)$, or $\sim 10^{15}$ operations per solve at $n_{\text{eq}} = 10^5$. Even with the sparse Jacobian with bordered bands exploited in RADCLUSTER, each right-hand side evaluation sweeps all (m, ℓ) pairs at $O(n_{VH}^{\text{full}})$ cost. Because the dominant cost is superlinear in system size, a factor-of-ten reduction returns two orders of magnitude in run time: reduction is a prerequisite for practical simulation, not an optional refinement.

Two independent reductions are therefore required, and both are available in RADCLUSTER: the first removes the gas dimensions from the cavity state (Sec. 4.2), and the second shortens the size ladders themselves (Sec. 4.3). Each is exact in a stated limit or systematically improvable, each preserves the conservation identities of Sec. S4, and the two make up.

4.2. Gas-Vacancy Space Reduction

Two reductions collapse the two-dimensional cavity state (m, ℓ) onto the vacancy axis alone, each exact in a stated limit. They differ in which rate is

fast: the delivery of gas to the cavity population or its redistribution among cavities relative to cavity growth.

(1) *Decoupled gas inventory.* When the gas generation rate is low, most cavities carry $\ell \ll m$ and the gas weakly perturbs the kinetics of the void rather than driving the nucleation, so the load per-cluster $\bar{\ell}(m)$ does not carry dynamical information of its own. The trapped inventory is followed as a single scalar ℓ_{tot} and redistributed algebraically over cavity sizes in proportion to the capture rate $A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$, the steady-state loading at which trapping and re-resolution balance. The vacancy equation is unchanged apart from a gas-corrected binding energy in its emission rate, and the gas-vacancy block shrinks to the equations $N_V + 1$. The approximation is controlled by the smallness of $\bar{\ell}/m$ and degrades once the cavities become gas-dominated.

(2) *Fast gas equilibration.* Helium in iron migrates with $E_m^h \approx 0.06$ eV [80], a jump rate exceeding the vacancy jump rate by $\sim 10^6$ at 400 °C. When the gas reaction frequency dominates, $\omega_h^{\text{eff}} \gg \omega_v^{\text{eff}}$, gas redistributes within and between cavities faster than a cavity grows or shrinks, and at each vacancy size the loading equilibrates to a mean value $\bar{\ell}(m, t)$. The distribution is then carried as two one-dimensional arrays, the total cavity concentration $\bar{c}_m$ and the trapped inventory Q_m , giving $2N_V$ equations. The evolution of $\bar{\ell}$ contains a local kinetic part — trapping from the free pool, thermal emission, and athermal re-resolution, the last driving $\bar{\ell}$ toward zero — and an advection correction for cavities entering size class m with a loading different from $\bar{\ell}(m)$.

The derivations — the ℓ -summation of the cavity master equation, the quotient-rule form of $d\bar{\ell}/dt$ with its advection corrections, and the algebraic redistribution rule with its conservation check — are given in Sec. S6 of the Supplementary Material.

4.3. ODE System Size Reduction

Coarse-graining (grouping) of the cluster size-distribution function (SDF) has a long lineage [7, 10, 16, 42, 81–83]. Geometric grouping [7] broadens the SDF and drifts the mean size; carrying a zeroth moment and a slope per group removes that mass leakage [77] but cannot treat cluster-cluster coalescence. Later work matched a discrete core to a Fokker–Planck continuum [84], generalized grouping to a basis-function framework for arbitrary reaction networks [85], and coupled deterministic dynamics to stochastic sampling of the sparse large-size tail [86].

RADCLUSTER [48] uses a hybrid discrete plus bin-moment reconstruct–project scheme. No analytic grouped coefficients are pre-computed: every right-hand-side evaluation reconstructs the per-size SDF, applies the *exact* size-dependent master equation, and reprojects onto the bin moments, so the grouped system is consistent with the discrete model by construction. Conservation is *monitored* through the Frenkel-pair and gas accounting integrals rather than enforced structurally, and only the state-vector dimension, not the per-step arithmetic, is reduced.

The smallest sizes are resolved one ODE per size (*discrete range*) and larger sizes are aggregated into logarithmically spaced bins (*moment range*). The intra-bin closure used throughout this work is the linear dual basis defined below: exactly $P = 2$ moments per bin, the zeroth and the first, with no higher-order shape function. The SIA and vacancy populations are reduced identically, and the gas state follows Sec. 4.2. The ODE count drops from $O(I + V)$ to

$$N_{\text{eq}} = \underbrace{i_{\text{discrete}} + 2 I_{\text{bin}}}_{\text{SIA block}} + \underbrace{v_{\text{discrete}} + 2 V_{\text{bin}}}_{\text{vacancy block}} + n_{\text{He}} + 5, \quad (36)$$

the trailing 5 counting the cumulative-loss accounting integrals (Sec. S4). Typically 20–60 bins suffice for $I, V = 10^4\text{--}10^5$ with $r \in [1.5, 2]$.

The SIA axis is split into a discrete prefix $\{1, \dots, i_{\text{discrete}}\}$ followed by I_{bin} geometrically spaced bins covering $\{i_{\text{discrete}}+1, \dots, I\}$; the cutoff is a numerical parameter, independent of the physical mobility cutoff n_{max}^i , which it may exceed. The edges $\{s_k\}_{k=0}^{I_{\text{bin}}}$ satisfy

$$s_0 = i_{\text{discrete}} + 1, \quad s_{I_{\text{bin}}} \leq I + 1, \quad (37)$$

and follow the recursion

$$s_{k+1} = \max(\lfloor r s_k \rfloor, s_k + 1), \quad r > 1, \quad (38)$$

which guarantees at least one integer per bin even when $r s_k$ floors back to s_k , with the ratio set by the target bin count,

$$r = \max\left((I/i_{\text{discrete}})^{1/I_{\text{bin}}}, 1.01\right). \quad (39)$$

Bin k spans $\mathcal{B}_k = \{n \in \mathbb{Z} : s_k \leq n < s_{k+1}\}$, has width $\Delta_k = s_{k+1} - s_k$ and geometric midpoint

$$\hat{s}_k = \sqrt{s_k (s_{k+1} - 1)}. \quad (40)$$

The vacancy axis is partitioned identically with parameters v_{discrete} , V_{bin} , edges $\{t_k\}_{k=0}^{V_{\text{bin}}}$ and ratio $r_v = (V/v_{\text{discrete}})^{1/V_{\text{bin}}}$.

The tracked unknowns in bin k are the two moments of the linear closure,

$$\mu_k^{(0)}(t) \equiv \sum_{n \in \mathcal{B}_k} c_n(t), \quad \mu_k^{(1)}(t) \equiv \sum_{n \in \mathcal{B}_k} n c_n(t). \quad (41)$$

The mean cluster size in bin k is $\langle n \rangle_k = \mu_k^{(1)}/\mu_k^{(0)}$, the total SIA content is $S_I = \sum_{n=1}^{i_{\text{discrete}}} n c_n + \sum_{k=1}^{I_{\text{bin}}} \mu_k^{(1)}$, and the vacancy moments $\nu_k^{(p)} = \sum_{m \in \mathcal{B}_k^v} m^p \bar{c}_m$, $p = 0, 1$, are defined analogously.

Intra-bin closure: the linear dual basis.. Tracking only two moments per bin leaves the per-size concentrations inside $\mathcal{B}_k$ undetermined, so a *closure* is required: a rule that rebuilds a profile c_n , $n \in \mathcal{B}_k$, from $(\mu_k^{(0)}, \mu_k^{(1)})$ alone. The profile is taken to be linear in n and expanded in the basis *dual* to the two moment functionals,

$$c_n \simeq \phi_{k,0}(n) \mu_k^{(0)} + \phi_{k,1}(n) \mu_k^{(1)}, \quad n \in \mathcal{B}_k, \quad (42)$$

where $\phi_{k,0}$ and $\phi_{k,1}$ are the unique linear functions on the integer set $\mathcal{B}_k$ obeying the discrete biorthogonality conditions

$$\sum_{n \in \mathcal{B}_k} n^p \phi_{k,q}(n) = \delta_{pq}, \quad p, q \in \{0, 1\}, \quad (43)$$

so that each basis function returns its own moment and annihilates the other. Setting $\phi_{k,q}(n) = \alpha_q + \beta_q n$ and solving (43) gives the closed form used in the code,

$$\phi_{k,0}(n) = \frac{S_2^{(k)} - S_1^{(k)} n}{\mathcal{N}_k}, \quad \phi_{k,1}(n) = \frac{\Delta_k n - S_1^{(k)}}{\mathcal{N}_k}, \quad (44)$$

with the bin geometry sums $S_1^{(k)} = \sum_{n \in \mathcal{B}_k} n$, $S_2^{(k)} = \sum_{n \in \mathcal{B}_k} n^2$ and $\mathcal{N}_k = \Delta_k S_2^{(k)} - (S_1^{(k)})^2 > 0$, evaluated on the *integer* sizes of the bin and computed once at setup. Equation (42) reproduces $\mu_k^{(0)}$ and $\mu_k^{(1)}$ exactly by construction, and negative reconstructed values are clamped to zero. The pair $(\phi_{k,0}, \phi_{k,1})$ is the discrete counterpart of the linear Galerkin shape functions on a bin and reduces to the piecewise-linear finite-element hat functions in the continuum limit $\Delta_k \rightarrow \infty$, hence the name. The construction of the basis, its Galerkin setting, and the truncation-error analysis are given in Sec. S7 of the Supplementary Material.

Rather than re-deriving the bin-integrated rates symbolically, which would introduce inter-bin fluxes and bin-averaged rate constants, the moments are evolved by three steps executed at every right-hand-side call:

1. *Reconstruct* c_n^{rec} on the full grid $n = 1, \dots, I$ from the discrete unknowns $\{c_n\}_{n \leq i_{\text{discrete}}}$ and the linear closure ansatz $c_n[\mu_k^{(0)}, \mu_k^{(1)}]$ on $\mathcal{B}_k$ [Eq. (42)]; likewise for the vacancy axis.
2. *Evaluate* the per-size master equations (26), (27), (28) *at every* n *and every* m from c_n^{rec} and $\bar{c}_m^{\text{rec}}$. Every reaction channel is evaluated *without bin-averaging the rate constants*; bin averaging enters only through the reconstructed concentrations.
3. *Project* the per-size rates onto the tracked moments,

$$\frac{d\mu_k^{(p)}}{dt} = \sum_{n \in \mathcal{B}_k} n^p \left(\frac{dc_n}{dt} \right)^{\text{full}}, \quad p = 0, 1, \quad (45)$$

and analogously for the vacancy moments; sizes $n \leq i_{\text{discrete}}$ are copied verbatim into dy/dt .

Because the per-size rates are exact, all conservation properties of the master equations (Eqs. 26–28) are inherited automatically: no inter-bin flux and no bin-averaged rate constants appear, and the closure in step (1) is the only approximation. The fixed-sink term keeps its size dependence, $\sum_n \mathcal{D}_i(n) c_n$, since $\mathcal{D}_i(n) \propto \omega_n^{\text{eff}}(n)$ varies with n through the 1D-glide diffusivity.

The vacancy block follows the same procedure with $n \rightarrow m$ and the rate constants of (27): V–V coalescence, thermal vacancy emission $\varepsilon_v(m, \ell)$ with its gas-pressure correction, SIA-induced shrinkage for all mobile $n = 1, \dots, n_{\text{max}}^i$, and the fixed-sink loss $\sum_{m \leq m_{\text{max}}^v} \mathcal{D}_v \bar{c}_m$. Gas capture leaves the vacancy size class m unchanged and enters only the equations dQ/dt : a decoupled inventory evolves the single scalar Q_{tot} along with the moments, while fast equilibration carries one Q_k per vacancy bin, making $\bar{\ell}_k = Q_k/\nu_k^{(0)}$ constant within bin k .

Because the projection sums the *exact* per-size rates, the Frenkel-pair, gas-balance and swelling identities survive unmodified, and closure error enters only through the reconstructed c_n of Eq. (42): the per-bin rate-constant error is about 12% at $r = 2$ and 1.5% at $r = 1.5$, making $r \approx 1.5$ –2 the best accuracy/cost compromise. The truncation-error analysis is given in Sec. S7 of the Supplementary Material.

5. Results

All calculations reported in this section were performed with RADCLUSTER on the EUROFER97 host declaration of Section 3, under a fission cascade spectrum at a displacement rate $G = 10^{-7}$ dpa s⁻¹ and an irradiation temperature of 330 °C, and were carried to 40 dpa. The interstitial and vacancy axes span sizes $I = 80\,000$ and $V = 20\,000$, and the size axis is carried in the hybrid bin-moment representation of Section 4.3 with linear dual-basis closure; the integration uses GMRES with the Woodbury preconditioner. The energetics of the material and other model input parameters are those listed in Section S9 of the Supplementary Material.

The section is organized in four parts. Section 5.1 sets the calculated microstructure against the EUROFER97 neutron-irradiation database of Section S10, comparing six independent observables. Section 5.2 examines the mechanism that partitions the interstitial loop population between its two Burgers-vector families and contrasts the treatment adopted here with the alternatives in the literature. Section 5.3 follows the underlying defect kinetics through dose, and Section 5.4 summarizes the numerical validation of the state space reduction.

5.1. Comparison with experimental data

The database. The measurements used here are collected in Section S10 of the Supplementary Material, which tabulates dislocation loops, cavities, helium bubbles and loop character for reduced-activation ferritic-martensitic steels and for model iron alloys. The entries are reproduced as the source papers report them so that no interpretation is imposed; the short source keys are expanded in Table S21. The data is for EUROFER97 under neutron irradiation at 300–350 °C, which is the simulated condition. Those measurements span 0.30–34.0 and $2.24\text{--}12.0 \times 10^{21}$ m⁻³ in $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ loop density, 3.3–35 and 2.8–48 nm in the corresponding mean diameters, and $0.32\text{--}6.30 \times 10^{21}$ m⁻³ and 1.6–16 nm for cavities. The spread is wide and reflects genuine differences in irradiation temperature within the quoted window, in dose, in heat and heat treatment, and — especially for the loop families — in the diffraction conditions and the visibility cutoff of the individual TEM analyzes [87].

Six independent observables. Figure 3 carries the calculation over five decades of dose against that database. The comparison rests on six quantities that are

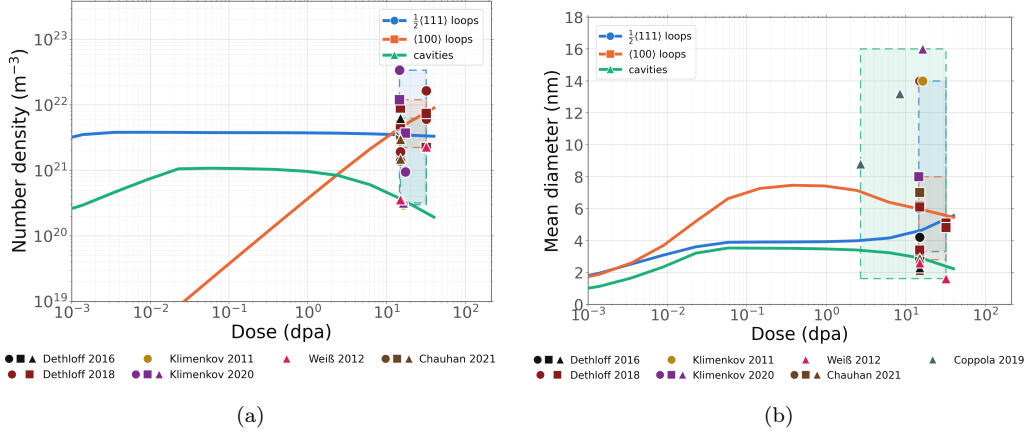


Figure 3: Calculated dose evolution of (a) the number densities and (b) the mean diameters of the $\frac{1}{2}\langle 111 \rangle$ loop, $\langle 100 \rangle$ loop and cavity populations in EUROFER97 at 330 °C under a fission cascade spectrum, with the EUROFER97 neutron-irradiation measurements of Section S10 overlaid. Dashed rectangles mark the measured range of each observable over the dose interval in which it was measured.

independently measured and independently predicted — the number density and the mean diameter of each of the three populations. Table 3 scores them at 15.72 dpa, within the dose interval where all three populations have been characterized.

Table 3: Calculated observables at 15.72 dpa against the range measured on neutron-irradiated EUROFER97 at 300–350 °C. Loop densities are in 10²¹ m⁻³, the cavity density in 10²⁰ m⁻³, diameters in nm.

	N_{111}	d_{111}	N_{100}	d_{100}	N_{cav}	d_{cav}
Calculated	3.50	4.64	4.72	5.94	3.61	2.89
Measured range	0.30–34.0	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16

Every one of the six lies inside the measured band, and the two that are best constrained experimentally — the $\langle 100 \rangle$ density and mean diameter, for which the loop families were resolved separately by diffraction contrast — sit near the middle of their ranges rather than at an edge. The bands are wide, so falling inside them is a weak test taken one observable at a time; what is not weak is that the same parameter set places all six simultaneously, with the correct ordering among them ($d_{100} > d_{111} > d_{cav}$ and $N_{100} > N_{111} \gg N_{cav}$), and with the correct sense of the dose dependence of each.

The trajectories themselves contain more information than the endpoint values. The density of $\frac{1}{2}\langle 111 \rangle$ loops (Fig. 3a) is established below 10^{-3} dpa and is then flat to within a factor of two over four further decades: these are the glissile loops, whose density saturates as soon as the sink field they themselves constitute balances their production. by contrast, the density of $\langle 100 \rangle$ loops increases steeply and continuously from $\sim 2 \times 10^{-2}$ dpa, crosses the cavity density near 2 dpa and the density of $\frac{1}{2}\langle 111 \rangle$ loops near 10 dpa, and is still rising at 40 dpa. The two loop families therefore exchange dominance within the dose window that the experiments sample, which is precisely the behavior the database encodes for ferritic–martensitic steels under both neutron and ion irradiation, where the dominant Burgers vector reported in a given study switches from $\frac{a}{2}\langle 111 \rangle$ to $a\langle 100 \rangle$ as dose increases [88–90]. Section 5.2 discusses the responsible mechanism.

The cavity population behaves differently. Its density peaks near 5×10^{-2} dpa at $\approx 1.1 \times 10^{21} \text{ m}^{-3}$ and then *declines* monotonically, and its mean diameter (Fig. 3b) peaks at ≈ 3.5 nm over the same interval before falling back to ≈ 2.2 nm at 40 dpa. The cavities do not coarsen at this temperature; they are progressively re-dissolved. That is the correct outcome for EUROFER97 at 330 °C, where the measured swelling is below 0.01–0.1 per cent even at 32 dpa [91, 92], and it follows here from the vacancy balance examined in Section 5.3 rather than from any imposed suppression of cavity growth. The mean $\langle 100 \rangle$ diameter shows the same non-monotonic shape, peaking near 7.5 nm at ≈ 0.4 dpa and relaxing to ≈ 5.5 nm by 40 dpa, while the $\frac{1}{2}\langle 111 \rangle$ diameter holds near 3.9 nm through 1 dpa and then climbs to ≈ 5.2 nm. The two SANS determinations of Coppola *et al.* at 2.5 and 8 dpa [93], which are the only measurements in the compilation below 14 dpa that resolve a mean size, lie above the calculated cavity trajectory; SANS and TEM weight the size distribution differently, and the discrepancy should not be read as a straightforward disagreement.

What the comparison does not establish should also be said. The calculation was run at one temperature, and the strongest experimental dependence in the database is on the irradiation temperature, not the dose. A quantitative test of the temperature dependence requires a temperature series and an uncertainty quantification of the underlying DFT and MD database, both of which are in preparation and are beyond the scope of this paper.

5.2. Dislocation loop mechanisms

Interstitial loops in bcc iron and its alloys form with two Burgers vectors, the glissile $\frac{1}{2}\langle 111 \rangle$ and the sessile $\langle 100 \rangle$, and the partition between them controls both the hardening contribution of the loop population and its ability to transport interstitials to sinks.

Energetics. The driving force follows the prismatic-loop formation free energy of Dudarev *et al.* [94],

$$E_l^X(n, T) = P_X(n) \left[\hat{F}_X(T) \ln \frac{4R_X^*(n)}{e\delta} + F_\delta^X + F_c^X \right], \quad X \in \{\tfrac{1}{2}\langle 111 \rangle, \langle 100 \rangle\}, \quad (46)$$

in which $P_X(n)$ is the perimeter of the loop, $R_X^*(n)$ its effective radius, δ the core cutoff, and $\hat{F}_X$ the line-tension prefactor. The temperature dependence enters through the magnetic softening of the $\langle 100 \rangle$ prefactor,

$$\hat{F}_{100}(T) = \hat{F}_{100}^0 \left(1 - \frac{T}{T_c} \right)^{1/4}, \quad T_c = 1185 \text{ K}, \quad (47)$$

while $\hat{F}_{111}$ is held temperature-independent. The single free constant $\hat{F}_{100}^0$ is fixed by one calibration condition, $\Delta F(n_{\text{ref}}, T^*) = 0$ with $\Delta F = E_l^{111} - E_l^{100}$, which defines a crossover size $n^*(T)$ below which $\langle 100 \rangle$ is the thermodynamically favored character.

Three channels transfer content between the families. All three are edges of the reaction admissibility graph, and their rate constants are collected in Section S3 of the Supplementary Material. The first is a direct single-loop rotation,

$$\Gamma_{\text{uni}}(n) = \nu_0 e^{-E_a(n)/k_B T} \max \left[0, 1 - e^{-\Delta F(n, T)/k_B T} \right], \quad E_a(n) = E_{a0} + \gamma_a \frac{P_{111}(n)}{b_{111}}, \quad (48)$$

in which $\nu_0 = 10^{13} \text{ s}^{-1}$ is the attempt frequency, $E_{a0} = 2.1 \text{ eV}$ the barrier for a direct rotation, $\gamma_a = 0.02 \text{ eV}$ the additional cost per perimeter segment b_{111} , and $\Delta F(n, T) = E_l^{111} - E_l^{100}$ the driving force of Eqs. (46)–(47). Since $P_{111} \propto \sqrt{n}$, the barrier grows with the loop line length, $E_a(n) = 2.10 + 0.065\sqrt{n} \text{ eV}$. The thermodynamic factor vanishes when the two characters are degenerate and is clipped at zero beyond the crossover size $n^*(T)$, so the channel is one-way: it converts $\frac{1}{2}\langle 111 \rangle$ into $\langle 100 \rangle$ and never the reverse. At 330°C the driving force is positive only for $n \leq 23$, so the channel operates in a

narrow window of small loops. It is turned off below the $\langle 100 \rangle$ loop-onset floor $n_{\text{loop}}^{\min} = 4$, where there is no $\langle 100 \rangle$ loop to receive the content.

The second is the junction reaction of Marian *et al.* [95], in which two $\frac{1}{2}\langle 111 \rangle$ loops of comparable size react through the metastable $\frac{1}{2}\langle 110 \rangle$ configuration to produce a $\langle 100 \rangle$ product, $\frac{1}{2}\langle 111 \rangle_n + \frac{1}{2}\langle 111 \rangle_{n'} \rightarrow \langle 100 \rangle_{n+n'}$, with branching efficiency

$$\begin{aligned} \varphi_{\text{junc}}(n, n') &= P_{\text{succ}}(T) \varphi_{\text{max}} e^{-\ln^2(n/n')/2\sigma_s^2} \Theta(\min(n, n') \geq n_j^{\min}), \\ P_{\text{succ}}(T) &= \frac{e^{-\Delta H_2/k_B T}}{e^{-\Delta H_2/k_B T} + e^{-\Delta H_{\text{rev}}/k_B T}} = \left[1 + e^{(\Delta H_2 - \Delta H_{\text{rev}})/k_B T}\right]^{-1}, \end{aligned} \quad (49)$$

where $\varphi_{\text{max}} = 0.1$ is the peak efficiency, $\sigma_s = 0.35$ the width of the size-comparability window in $\ln(n/n')$, Θ the Heaviside step, $n_j^{\min}$ the smallest partner able to sustain a junction segment, and $\Delta H_2 = 0.44$ eV and $\Delta H_{\text{rev}} = 0.30$ eV the forward ($\frac{1}{2}\langle 110 \rangle \rightarrow \langle 100 \rangle$) and reverse ($\frac{1}{2}\langle 110 \rangle \rightarrow \frac{1}{2}\langle 111 \rangle$) barriers of the two-step path. The Gaussian in $\ln(n/n')$ peaks at $n = n'$, so this channel is by construction a same-size reaction. P_{succ} depends only on the difference, equals 0.063 at 330 °C. Because φ_{junc} is a branching fraction rather than an encounter rate, it *partitions* the $\frac{1}{2}\langle 111 \rangle$ collision kernel instead of adding to it: a fraction φ_{junc} of every collision is deposited in the $\langle 100 \rangle$ population and the complementary fraction $1 - \varphi_{\text{junc}}$ remains in $\frac{1}{2}\langle 111 \rangle$, so the total collision rate is conserved by construction and no reaction is counted twice.

The collision kernel itself is not that of a three-dimensional random walk. $\frac{1}{2}\langle 111 \rangle$ loops glide along their Burgers vector and sample new directions only when that vector is reoriented, so the encounter rate carries the mixed 1D/3D correlation factor derived in Section S2,

$$\mathcal{K}_{n,n'}^{ii} \longrightarrow \frac{\mathcal{K}_{n,n'}^{ii}}{[1 + B_{\text{rot}}(\ell/R_{nn'})^2]^{1/2}}, \quad B_{\text{rot}} = \frac{2}{9}, \quad (50)$$

with $R_{nn'}$ the pair capture radius and ℓ the straight-line glide length between reorientations. Which regime is physical depends on the host. In high-purity iron the glide is long-range and genuinely one-dimensional [96]. In the alloys considered here, carbon and nitrogen trap the gliding cluster strongly and the substitutional solutes act as weaker obstacles, giving $\ell \approx 22$ nm for an effective obstacle fraction of 2.8×10^{-3} [97–99]. The resulting suppression is moderate and strongly size dependent: 19–33 for the junction, where both

partners are small, and falling from 15 to 2 across the $\langle 100 \rangle$ size range in the absorption channel, so large sessile loops are far less affected by one-dimensional correlation than the small mobile clusters they absorb.

The third channel is a size-asymmetric absorption gate: a sessile $\langle 100 \rangle$ loop of hundreds of atoms captures a mobile $\frac{1}{2}\langle 111 \rangle$ cluster and the product inherits the character of the host, $\langle 100 \rangle_m + \frac{1}{2}\langle 111 \rangle_n \rightarrow \langle 100 \rangle_{m+n}$, at the rate

$$\mathcal{K}_{m,n}^{\text{abs}} = P_{\text{succ}}^{\text{abs}}(T) \frac{8\pi}{\Omega^{2/3}} \frac{(\xi_m + \xi_n) D_n^{\text{1D}}}{[1 + B_{\text{rot}}(\ell/R_{mn})^2]^{1/2}}, \quad \xi_k = \left(\frac{3k}{8\pi}\right)^{1/3}, \quad m \geq n_{\text{loop}}^{\text{min}}, \quad n \leq i_{\text{mobile}}, \quad (51)$$

which is the cross-character form of the binary kernel (S10) with the diffusivity sum collapsed to the single mobile partner, since the $\langle 100 \rangle$ host is sessile. The incorporated segment is negligible against the host and is templated by its existing $\langle 100 \rangle$ character, so reversion through $\frac{1}{2}\langle 110 \rangle$ is far less competitive than in the symmetric junction and the forward barrier is correspondingly lower, $\Delta H_2^{\text{abs}} = 0.36$ eV. The absorption gate therefore carries its own success probability,

$$P_{\text{succ}}^{\text{abs}}(T) = A_{\text{abs}} \left[1 + e^{(\Delta H_2^{\text{abs}} - \Delta H_{\text{rev}})/k_B T} \right]^{-1}, \quad A_{\text{abs}} = 2, \quad (52)$$

equal to 0.48 at 330 °C, some seven times the junction value, so the asymmetric route dominates. This is the kinetic expression of the in-situ observation that a loop can change its Burgers vector on interacting with another loop, with the resultant character following the larger of the two [100].

Loop incorporation into the network. The loop populations are also depleted by the pre-existing dislocation network. The transfer is not tied to the loop's own diffusivity — were it so, it would vanish for exactly the large sessile loops that dominate incorporation — but to the rate at which the network sweeps or climbs past a stationary loop,

$$\Lambda_n^{\text{net}} = v_{\text{net}} \rho_{\text{net}} w_c P_{\ell d}(n), \quad P_{\ell d}(n) = \frac{1}{2} \left[1 + \tanh \frac{R_{\text{int}}(n) - s_{\ell d}(n)}{b_c} \right], \quad (53)$$

in which $R_{\text{int}}(n) = \chi d_n^{(c)}$ is the elastic interaction range of a loop of diameter $d_n^{(c)} = 2(n\Omega/\pi b_c)^{1/2}$, $s_{\ell d}(n) = \max[b_c, \rho_{\text{net}}^{-1/2} - \frac{1}{2}d_n^{(c)}]$ the edge-to-edge gap to the nearest network line, w_c a capture width of order b_c , and v_{net} the irradiation-driven climb velocity, built from the same biased point-defect

fluxes that set the fixed-sink strength. The switch $P_{\ell d}$ vanishes while the loop sits far from the nearest line and approaches unity once its elastic zone reaches the mean network spacing. With ρ_{net} held at its steady-state value here, the tanh has a width of one Burgers vector and therefore acts as a sharp size threshold, switching the channel on near $n \approx 13$ and leaving it on for every larger loop. It applies to both characters for $n \geq n_{\text{inc}}$ and enters as a diagonal loss $-\Lambda_n^{\text{net}} c_n$ added to the fixed-sink strength rather than as a separate edge; the interstitials it carries away are routed into the same fixed-sink ledger, so the Frenkel-pair diagnostic is unchanged in form. Because Λ_n^{net} carries no factor of the loop diffusivity and grows with loop size through $P_{\ell d}$, it supplies the size-dependent loss the loop densities need in order to saturate. It is developed in Section S2 of the Supplementary Material.

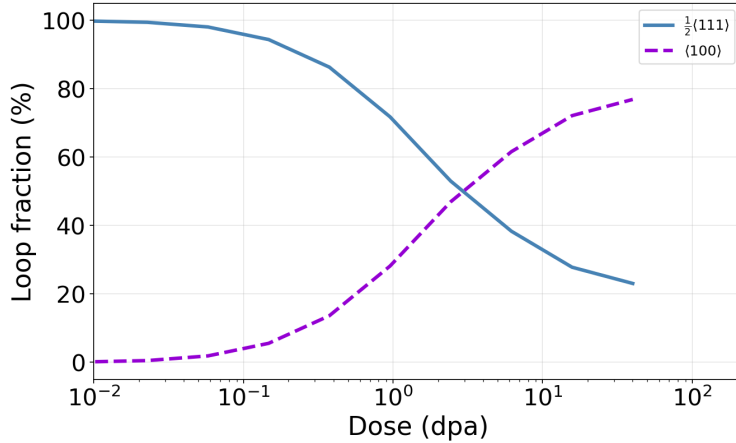


Figure 4: Partition of the interstitial loop population between the $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ families as a function of dose, at 330 °C under a fission cascade spectrum. The character crossover occurs near 3 dpa; at 40 dpa the $\langle 100 \rangle$ family holds about three quarters of the population.

Figure 4 shows the resulting partition. Below 5×10^{-2} dpa the population is essentially pure $\frac{1}{2}\langle 111 \rangle$: loops are being produced directly in cascades and have not yet accumulated to the densities at which they interact. The $\langle 100 \rangle$ share then grows through more than two decades of dose, passes 50 per cent near 3 dpa, and reaches about 77 per cent at 40 dpa. Two features of this curve matter. It is sigmoidal in log dose rather than step-like, which is the signature of an interaction-driven rather than a thermodynamic transition — a purely energetic crossover at fixed temperature would predict no dose dependence at all. And the crossover dose falls squarely in the range over

which the database records the reported dominant Burgers vector switching, near 4 dpa for HT9 under Kr^+ ions and between 15 and 35 dpa for T91 and HT9 under BOR-60 neutrons (Tables S14 and S15) [88, 89].

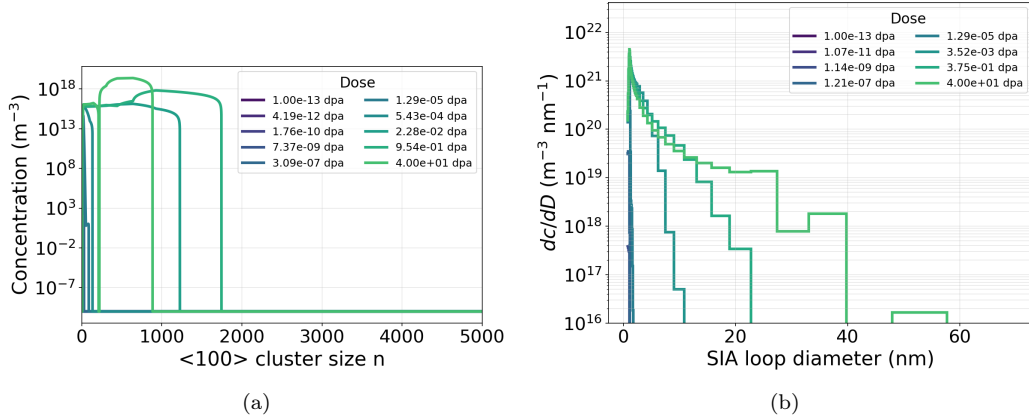


Figure 5: (a) Evolution of the $\langle 100 \rangle$ cluster size distribution with dose. (b) Evolution of the interstitial loop size distribution expressed as a function of loop diameter, the form in which TEM reports it; the wide steps above ≈ 15 nm are the logarithmic bins of the reduced representation rather than structure in the distribution.

The size distributions in Figure 5 show how the partition is achieved. The $\langle 100 \rangle$ population (Fig. 5a) first appears at small sizes, advances its upper edge to $n \approx 1750$ by ≈ 1 dpa, and then, between 1 and 40 dpa, *gains* a factor of four in amplitude while its upper edge retreats to $n \approx 900$. Conversion is feeding new $\langle 100 \rangle$ loops at small and intermediate sizes faster than the existing ones grow. This is the microscopic content of the non-monotonic d_{100} curve in Figure 3b: the mean $\langle 100 \rangle$ diameter falls at high dose not because loops shrink but because the population is continuously refreshed from below. The diameter-resolved interstitial distribution (Fig. 5b) is monotonically decreasing at every dose, peaked at 1.5–2 nm, with a tail that extends to ≈ 40 nm at 40 dpa. A steeply falling distribution with most of the number density below the TEM visibility limit and a long resolved tail is what the EUROFER97 examinations report at these conditions [87, 101], and it explains why quoted mean loop diameters depend so strongly on the analysis cutoff.

The closest published treatment of this problem is that of Gao, Gaganidze and Aktaa, who parameterize the loop formation free energy to 1100 K from anisotropic elasticity with Varshni-fitted elastic constants [102] and

then use it in a cluster dynamics model of the relative $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ populations in α -Fe [103]. Their free energy has the same functional form as Equation (46) — since $P_X \propto \sqrt{N}$ and $\ln R^* = \frac{1}{2} \ln N + \text{const}$, their $E_{\text{tot}}(N, T) = \epsilon_0(T)\sqrt{N} \ln N + \epsilon_1(T)\sqrt{N}$ is a reparameterization of it — so the two models differ in calibration and in kinetics rather than in physical content. Two differences in the energetics are structural. Their fitted coefficients for $\frac{1}{2}\langle 111 \rangle$ carry significant linear and cubic temperature terms, while holding $\hat{F}_{111}$ fixed here attributes the entire crossover to $\langle 100 \rangle$ softening; and they resolve the $\langle 11\bar{2} \rangle$ and $\langle 1\bar{1}0 \rangle$ side orientations of the loop $\frac{1}{2}\langle 111 \rangle$, whose crossover temperatures differ by $\sim 80^\circ\text{C}$ at 100 interstitials, which is comparable to the effect being modeled. Table 4 compares the crossover temperature $T^*(n)$ obtained here with the value reported.

Table 4: Character crossover temperature T^* , at which $\Delta F(n, T^*) = 0$, as a function of loop size, and the resulting $\langle 100 \rangle$ -favored size window at 450°C . The shipped calibration of Equation (47) uses $\hat{F}_{100}^0 = 0.4799 \text{ eV}/\text{\AA}$ fixed at $(T^*, n_{\text{ref}}) = (450^\circ\text{C}, 50)$.

Calibration target	$n = 20$	$n = 50$	$n = 100$	$\langle 100 \rangle$ window
$(450^\circ\text{C}, n_{\text{ref}}=50)$, as used here	-13°C	450°C	556°C	$n \leq 49$
$(455^\circ\text{C}, n_{\text{ref}}=100)$	—	319°C	455°C	$n \leq 96$
$(375^\circ\text{C}, n_{\text{ref}}=100)$	—	215°C	375°C	$n \leq 183$
Gao <i>et al.</i> [102]	—	—	$375\text{--}455^\circ\text{C}$	—

Both treatments agree in direction — small loops favor $\langle 100 \rangle$ and the crossover temperature rises with size — but the calibration adopted here sits above theirs at $n = 100$, and moving the calibration target to $(455^\circ\text{C}, n_{\text{ref}} = 100)$ would bring the size dependence onto their curve and roughly double the $\langle 100 \rangle$ -favored window at 450°C .

The treatment of reaction kinetics is different here. Gao *et al.* test both loop–loop routes and reach an asymmetric verdict: the symmetric junction reaction is negligible in their simulations, because two $\frac{1}{2}\langle 111 \rangle$ loops of nearly identical size with overlapping glide cylinders rarely meet, whereas the asymmetric $\frac{1}{2}\langle 111 \rangle + \langle 100 \rangle$ reaction is, in their words, the origin of interstitial transfer between loops of different type [103]. Channels (49) and the absorption gate map exactly onto their negligible and dominant routes. That correspondence also identifies the weakest element of the present parameterization: φ_{junc} is peaked at $n = n'$ and additionally gated by $\min(n, n') \geq n_j^{\text{min}}$, so it maximizes precisely the encounter they find improbable. The unary

channel Γ_{uni} is more exposed still. It is calibrated rather than derived, and Gao *et al.* state explicitly that the population rearrangement proceeds *not* by a single loop transferring its orientation but through mutual loop interactions, consistent with heavy-ion experiments that see no direct transformation. Equation (48) is therefore best understood as a coarse-grained effective channel standing in for unresolved processes, and is labelled as such here rather than advanced as a mechanism.

One difference is not a matter of parameterization. In the model of Gao *et al.*, $\langle 100 \rangle$ loops nucleate mainly through the collapse of C15 Laves-phase clusters [104], which also act as an interstitial buffer that suppresses the $\langle 100 \rangle$ density; RADCLUSTER carries no C15 population, so every $\langle 100 \rangle$ loop in Figure 4 arises by conversion. These are genuinely different nucleation hypotheses, not a missing detail, and the two are discriminated by the dose dependence rather than by the temperature dependence: a thermodynamic crossover alone predicts none, and the sigmoidal rise of Figure 4 is produced entirely by interaction terms. Two further differences are worth recording. Gao *et al.* adopt annular reaction cross-sections on the grounds that a spherical reaction volume becomes invalid for loops larger than a few nanometers, where RADCLUSTER uses the $A_{\text{sph}}m^{1/3}$ and $A_{\text{loop}}n^{1/2}$ capture-radius kernels of Section S2 throughout; and they use an irradiation-driven trap-mediated loop diffusivity [105], against the thermal $D_n^{1D} \propto n^{-s}e^{-E_m/k_B T}$ with rotational correlation adopted here. Conversely, their model excludes vacancy-type loops and reports that voids barely form, whereas the coupled cavity-loop-helium system solved here covers a regime theirs does not.

5.3. Defect cluster evolution

The free point defect concentrations (Fig. 6a) reach quasi-steady state extremely early: the mono-interstitial by 10^{-9} dpa at $\approx 2 \times 10^{18} \text{ m}^{-3}$, the mono-vacancy by 10^{-7} dpa at $\approx 3.6 \times 10^{20} \text{ m}^{-3}$. The ratio $C_{v1}/C_{i1} \approx 180$ at saturation is determined by the mobility disparity, the faster species being held at the lower concentration by the same sink field. Then both remain essentially constant for ten decades of dose and decrease only beyond ≈ 1 dpa, at which point the sink strength of the accumulated microstructure has grown enough to draw them down. The transient is short because interstitial clusters are produced directly in cascades and migrate with very low activation energies; so, the sink field establishes itself long before an appreciable dose has accumulated.

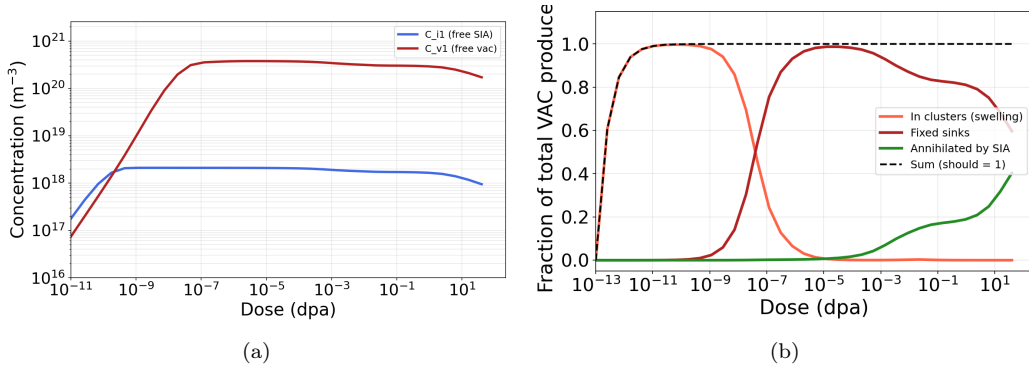


Figure 6: (a) Free mono-interstitial and mono-vacancy concentrations against dose. (b) Partition of all vacancies produced among retention in clusters, absorption at fixed sinks, and annihilation by interstitials. The dashed sum is the Frenkel-pair conservation check of Section S4 and holds at unity throughout.

Figure 6b partitions every vacancy produced among three fates. In the initial transient, below 10^{-9} dpa, essentially all are held in clusters, simply because nothing else has yet formed to take them. Between 10^{-9} and 10^{-5} dpa the fixed sinks — the dislocation network, grain boundaries and precipitate interfaces — take over completely, the two curves crossing near 4×10^{-8} dpa. From 10^{-4} dpa onwards, annihilation by interstitials grows steadily, from a few per cent to about 40 per cent of all vacancies produced by 40 dpa, at the expense of the fixed-sink channel. Vacancy retention in cavities remains negligible at every dose. This is the origin of the declining cavity population noted in Section 5.1: as the sessile $\langle 100 \rangle$ loop population builds up and the interstitial flux to cavities rises, the net vacancy flux into cavities turns negative and the cavities are re-dissolved. The near-zero swelling of EURO-FER97 at 330 °C is thus a computed consequence of the interstitial–vacancy competition, not an input. The dashed curve in Figure 6b is the Frenkel-pair conservation identity of Section S4 evaluated along the trajectory; it holds at unity to graphical accuracy over thirteen decades of dose, which is the strongest available check that the stiff integration has not leaked mass.

Figure 7a resolves the class of the interstitial axis by class. The discrete sizes fill in sequentially through the transient, each rising later and saturating near 10^{20} m^{-3} by 10^{-4} dpa, while the binned classes switch on abruptly near 10^{-3} dpa and settle at $2\text{--}3 \times 10^{20} \text{ m}^{-3}$. Beyond 10^{-2} dpa nothing grows in the figure: the interstitial cluster population is established before 10^{-2} dpa and then only redistributes among sizes. The size distributions of Figure 5

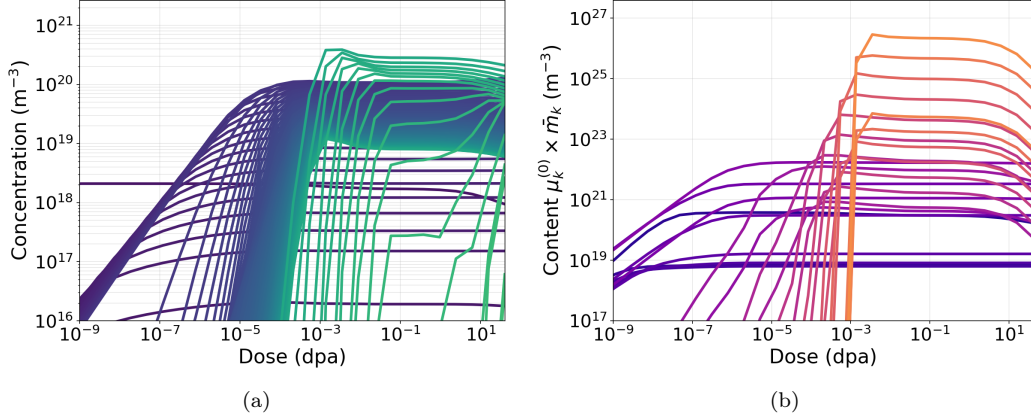


Figure 7: (a) Concentration of each interstitial size class against dose, discrete sizes and logarithmic bins shown together. (b) Vacancy content $\mu_k^{(0)} \times \bar{m}_k$ carried by each size bin against dose, colored by bin index from the smallest (dark) to the largest (light).

are moving on a fixed total, which is why the loop densities of Figure 3a saturate while the loop *character* continues to evolve for three more decades of dose.

The vacancy axis (Fig. 7b) tells a different story. Plotting the content $\mu_k^{(0)} \bar{m}_k$ of each bin rather than its number density weights the display toward the large cavities that dominate swelling. The small bins saturate below 10^{-6} dpa; successively larger bins switch on later and carry more, the largest turning on sharply near 10^{-3} dpa at $\approx 2.5 \times 10^{26} \text{ m}^{-3}$. All bins then plateau from 10^{-2} to a few dpa, and beyond ≈ 5 dpa they decline, the largest bins losing roughly a decade of content by 40 dpa. The loss is concentrated where the swelling would be, which is the same statement as the falling cavity density and mean size of Figure 3, seen now as a redistribution of vacancy content out of the large-size tail.

Figure 8 shows the cavity distribution in the diameter-resolved form. Two features distinguish it from its interstitial counterpart. It is peaked rather than monotonically decreasing, and the maximum moves from about 2.5 nm at 3.5×10^{-3} dpa to 4.5–5 nm at 0.375 dpa. Small vacancy clusters must survive thermal emission and interstitial impingement long enough to reach a size at which growth is favorable, which depletes the small-size end and builds a peak. And the whole distribution then decays. Between 0.375 and 40 dpa the peak amplitude falls by a factor of about 17 and the 6–7.6 nm bins fall by two orders of magnitude, so the population is not coarsening in the Ostwald

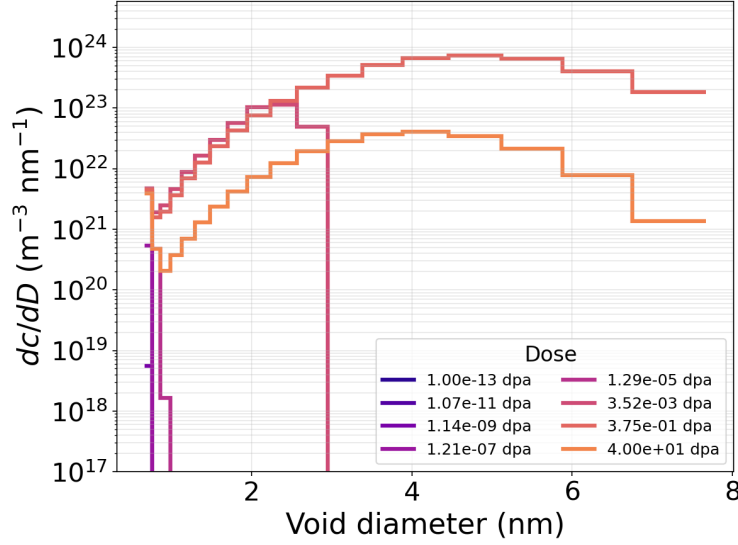


Figure 8: Evolution of the cavity size distribution expressed against cavity diameter. The distribution is broad and peaked, unlike the monotonically decreasing interstitial loop distribution of Figure 5b, and its amplitude falls by more than an order of magnitude between 0.375 and 40 dpa.

sense — in which the large end would grow at the expense of the small — but shrinking at every size. Within a coupled cavity–loop–helium treatment, this is the expected low-temperature response of a ferritic–martensitic steel under fission-spectrum irradiation, where the helium inventory is too small to stabilize cavities against the interstitial flux.

5.4. Numerical validation

Every result above was obtained with the hybrid bin-moment reduction of Section 4.3, which carries sizes up to i_{discrete} individually and partitions the remainder into logarithmically spaced bins holding only the zeroth and first moments, the intra-bin distribution being reconstructed from the linear dual-basis closure. The economy is large — a few hundred equations in place of 10^4 – 10^5 — and it is worth nothing unless the reduced solution reproduces the discrete one. Section S8 of the Supplementary Material reports the full validation; its results are summarized here because they bound the confidence that can be placed in Sections 5.1–5.3. The six observables scored throughout are those of Table 3.

The reduction is internally converged.. Refining the discrete core on the production domain, with the bin count adjusted to hold the bin ratio approximately constant, changes nothing of consequence: as i_{discrete} falls from 1600 to 400 to 100 the system shrinks from 2048 to 562 to 189 equations while d_{100} spans only 5.868–6.001 nm and N_{111} only $3.449\text{--}3.512 \times 10^{21} \text{ m}^{-3}$. Taking the 189-equation configuration as reference and varying one parameter at a time, two orders of magnitude in relative integration tolerance move no observable in any reported digit, and reducing the bin count from 18 to 10 displaces them by a few per cent at most (Table S7). Replacing the dynamic helium sub-system by its quasi-steady-state form reproduces all six observables to every digit, but the reference calculation holds a total helium concentration of only $1.18 \times 10^{13} \text{ m}^{-3}$ under fission, so that agreement establishes the safety of the reduction at fission helium levels and says nothing about fusion generation rates.

The reduction reproduces the exact discrete solution.. Verification proper requires the exact solution of the same equations, which is affordable only on a domain small enough for the fully discrete system to be integrated. On $I = V = 4000$ at 2 dpa, with the $\frac{1}{2}\langle 111 \rangle$ coarsening channel disabled in both arms so that the two solvers integrate identically the same equations, the linear closure holds d_{100} to within 1 per cent from 830 equations down to 244 — a thirtyfold reduction in cost for a one-per cent error — and N_{100} to 2 per cent, degrading only to 4.4 per cent at the coarsest rung of 108 equations (Table S8). The exceptions are informative. N_{111} degrades by 44 per cent at that coarsest rung, because the $\frac{1}{2}\langle 111 \rangle$ population is peaked at small sizes and, once i_{discrete} falls to 25, that peak lies inside the binned region where the closure represents it rather than resolving it. The cavity observables err in the opposite sense, low by some 13 per cent on the finer grids and crossing to +9.2 per cent on the coarsest. Convergence along this ladder is not monotonic, and the reason is that the rungs vary two parameters at once.

One axis governs the residual error.. Separating the two axes settles it. Varying only the bin count at a fixed discrete core, five of the six observables converge quickly: by thirteen bins, at 158 equations against 2006 for the exact arm, N_{100} , d_{100} , N_{cav} and d_{cav} all lie within 0.3 per cent of the exact solution and N_{111} has plateaued at +1.0 per cent (Table S9). The sixth does not converge: d_{111} remains between -5 and -6.6 per cent throughout and

does not improve with refinement. Sweeping both parameters independently (Table S10) shows why — the error falls with the extent of the discrete core and is nearly flat in the bin count, reaching better than one per cent at $i_{\text{discrete}} = 800$ at every bin count. The residual is the ordinary closure error, and it tracks the fraction of $\frac{1}{2}\langle 111 \rangle$ content lying above the discrete core: raising i_{discrete} through 25, 50, 100, 200, 400 and 800 leaves 96.6, 91.2, 83.4, 70.6, 47.7 and 8.0 per cent of the content binned and returns a monotone approach to the exact d_{111} from -8.6 to $+0.1$ per cent. Three alternative explanations — a difference between the two code paths, a domain-edge artifact, and slow accumulation — were excluded by direct measurement.

What this bounds.. The consequence for the preceding sections is specific rather than general. The $\langle 100 \rangle$ observables, which carry the loop-character argument of Section 5.2, are the ones the reduction represents best, to within one per cent in mean diameter and two per cent in density. The cavity observables are next, converging to a few tenths of a per cent once the bin count is adequate. The $\frac{1}{2}\langle 111 \rangle$ mean diameter is the one quantity carrying a systematic residual of five to six per cent at the discrete-core extent used in production, which is small against the factor-of-ten spread of the measurements in Table 3 but should be remembered when that observable is used quantitatively. None of these errors is of the size that would change any conclusion drawn above.

6. Summary & Conclusions

This paper has developed a general, modular, graph-based cluster-dynamics (CD) framework for the long-term evolution of radiation-induced defect populations in crystalline solids, and has instantiated it for the ferritic–martensitic steel EUROFER-97 under a fission cascade spectrum at 330 °C carried to 40 dpa. The development proceeds in a unified notation from the reaction admissibility graph and its graph-walking master equation (Section 2), through the EUROFER-97 host declaration and its exact conservation laws (Section 3), the helium-occupancy and size-bin moment state-space reductions (Sections 4 and 4.3), to the calculations of Section 5 and their verification against the exact discrete solution.

The reaction graph as the interface. The first contribution is a pair of algorithmic primitives that make a host-independent CD formulation possible. The *polarity index* $\sigma \in \{-1, +1\}$ separates the cluster state space into

vacancy- and SIA-type sub-spaces without hard-coding a taxonomy of loops, voids and bubbles, so that a cluster is the four-tuple $(\sigma, n, p, \mathbf{c})$ rather than a named object. The *reaction admissibility graph* (RAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \boldsymbol{\nu}, \mathbf{k})$ then encodes every elementary reaction—growth, shrinkage, dissociation, recombination, cross-polarity annihilation, inter-population transfer, solute trapping/detrapping, coalescence, source and sink—as an edge belonging to exactly one of ten classes, each carrying a fixed stoichiometric contract. Reactions that crystallography, Burgers-vector compatibility, migration dimensionality or energetics forbid are excluded when the graph is built, not suppressed when it is solved. What remains is the host-independent graph-walking accumulator $dc_a/dt = G_a + \sum_r S_{ar} J_r(\mathbf{c}) - D_a c_a$, applied edge-by-edge over $\mathcal{E}$: the right-hand side is assembled, never written down. Because the sign pattern of S_{ar} is fixed by the edge class rather than by a kernel calibration, the Frenkel-pair and helium conservation laws of Section S4 are the left null vectors of the stoichiometric matrix, and they are carried as auxiliary accounting integrals under the integrator’s error control—monitored as diagnostics rather than imposed as constraints.

The algorithmic structure of RADCLUSTER. The second contribution is that this abstraction is not a presentational device but the actual decomposition of the code. RADCLUSTER is split along exactly the line the graph draws. A host-independent core defines the cluster identifiers, the ten edge classes and their stoichiometric contracts, the RAG container and its admissibility logic, the graph walker, the *state layout* that maps RAG vertices bijectively onto a flat ODE state vector, the state-space reductions, and the conservation diagnostics; it contains no material parameter. A host declaration supplies the admissible population set $\mathcal{P}_\sigma$, the vertex and edge sets $\mathcal{V}$ and $\mathcal{E}$, and the rate-kernel library $\mathbf{k}$; it contains no algorithm. The EUROFER-97 declaration is a single module that names one bulk SIA and one bulk vacancy population with helium as the only resolved gas, maps each of its process classes onto one of the ten abstract edge classes, registers the size-resolved kernels as precomputed arrays, and assembles the state layout—never an integrator. The analogy to object-oriented programming is exact: the abstract graph, together with its solver, preconditioner and reduction theorems, is a base class that implements every method; a host material is a declaration; and an irradiation scenario is an instantiation. Two practical consequences follow. First, the reduction theorems are proved once on the abstract graph and inherited by any host, rather than re-derived per material. Second, the

production path keeps the graph *implicit*—an edge is a size-parameterized family applied vectorized over n , and no vertex is instantiated in the inner loop—while the same declaration can be materialized explicitly, truncated at a chosen size, for the structural queries (connectivity, reachability from the monomer pools, orphaned vertices) that an implicit graph does not expose. The figure of the graph and the equations that are solved therefore come from one object, and cannot drift apart.

Reductions owned by the core, not by the host. The third contribution is a pair of state-space reductions declared on the abstract graph. The gas–vacancy grid, whose unreduced size scales as $N_V^{5/3}$ for one gas species and $N_V^{7/3}$ for two, is collapsed onto the vacancy axis in either of two limits—a mean-field loading $\bar{\ell}(m)$ when gas redistribution outruns cavity growth, or a single decoupled inventory Q_{tot} when the gas is a weak perturbation—each exact in a stated limit and each preserving the helium moment exactly. The size ladders themselves are then shortened by the hybrid discrete plus logarithmic bin-moment scheme, in which two moments per bin are evolved by *reconstruct–evaluate–project* at every right-hand-side call: the per-size distribution is rebuilt from the linear dual-basis (Galerkin) closure, the *exact* per-size master equations are evaluated at every size, and the resulting rates are projected back onto the moments. No grouped coefficient is pre-computed, so no inter-bin flux has to be made consistent and no bin-averaged rate constant enters; the closure is the only approximation, and every conservation identity survives it unmodified. Only the dimension of the state vector is reduced, not the per-step arithmetic, which is what makes the reduction safe to audit. The resulting Jacobian is bordered–banded—narrow-banded in the size index within each species and bordered through the cross-species couplings—and is inverted by Sherman–Morrison–Woodbury preconditioned GMRES, which separates the dense border updates from the banded core.

What the EUROFER-97 calculations show. The production calculation carries $I = 80\,000$ and $V = 20\,000$ size classes in a few hundred equations, and no parameter in it was adjusted to any measurement. Four results follow.

(i) *Six unfitted observables land simultaneously.* The number density and the mean diameter of each of the three populations— $\frac{1}{2}\langle 111 \rangle$ loops, $\langle 100 \rangle$ loops and cavities—are independently measured and independently predicted, and all six lie inside the ranges reported for neutron-irradiated EUROFER-97 at 300–350 °C (Table 3), with the correct ordering among

them and the correct sense of the dose dependence of each. The measured bands are wide, so no single observable is a strong test; what is not weak is that one unfitted parameter set places all six at once.

(ii) *Loop character is an interaction-driven, not a thermodynamic, transition.* Carrying the two Burgers-vector families as separate size-resolved populations on the graph, with conversion as explicit edges, produces a $\langle 100 \rangle$ share that rises sigmoidally in log dose, passes 50 per cent near 3 dpa and reaches about 77 per cent at 40 dpa. A purely energetic crossover at fixed temperature would predict no dose dependence at all, and the computed crossover falls in the range over which the experimental database records the reported dominant Burgers vector switching. The same conversion explains the non-monotonic mean $\langle 100 \rangle$ diameter: the family is continuously refreshed from below rather than shrinking. This is also where the present parameterization is most exposed, since the symmetric junction channel is peaked precisely at the encounter that independent simulations find improbable, and the unary rotation channel is a coarse-grained effective process rather than a mechanism.

(iii) *Near-zero swelling is an outcome, not an input.* The vacancy balance partitions every vacancy produced among retention in clusters, absorption at fixed sinks and annihilation by interstitials. Fixed sinks dominate from 10^{-9} to 10^{-5} dpa, annihilation then grows to about 40 per cent of all vacancies produced by 40 dpa, and retention in cavities remains negligible throughout. As the sessile $\langle 100 \rangle$ population builds and the interstitial flux to cavities rises, the net vacancy flux into cavities turns negative and the cavity population is progressively re-dissolved—peaked in density near 5×10^{-2} dpa and declining thereafter, and shrinking at every size rather than coarsening in the Ostwald sense. The measured near-zero swelling of EUROFER-97 at 330 °C is thus a computed consequence of the interstitial–vacancy competition. Along the same trajectory the Frenkel-pair identity holds at unity to graphical accuracy over thirteen decades of dose, which is the strongest available check that the stiff integration has not leaked mass.

(iv) *Distribution shape, not mean size, carries the comparison.* The interstitial distribution is monotonically decreasing at every dose, peaked at 1.5–2 nm with a tail reaching ≈ 40 nm, so most of the number density lies below the TEM visibility limit and quoted mean diameters depend strongly on the analysis cutoff. The cavity distribution is peaked instead, and its amplitude falls by more than an order of magnitude between 0.375 and 40 dpa. Neither population is representable by a single mean size.

What the verification bounds. The economy of the reduction is worth nothing unless the reduced solution reproduces the discrete one, and the claim made here is specific rather than general. Refining the reduction’s own resolution changes nothing of consequence—the system shrinks by an order of magnitude, from 2048 to 189 equations, while d_{100} moves by 2 per cent—and two decades of integration tolerance move no observable in any reported digit, so the tolerance is not a free parameter concealed within the results. Against the exact discrete solution of the same equations, the linear closure holds d_{100} to within 1 per cent from 830 equations down to 244, a thirtyfold reduction in cost for a one-per cent error, and N_{100} to 2 per cent; five of the six observables converge to within 0.3 per cent by thirteen bins, at 158 equations against 2006 for the exact arm. The sixth does not: the $\frac{1}{2}\langle 111 \rangle$ mean diameter carries a systematic residual of five to six per cent, which a two-dimensional sweep localizes on the extent of the discrete core rather than on the bin count, and which falls below one per cent once $i_{\text{discrete}} = 800$. That residual is small against the factor-of-ten spread of the measurements, and none of these errors is of a size that would change a conclusion drawn above—but it is the one quantity to be remembered when this framework is used quantitatively.

Outlook. Several directions are immediately open, and the graph structure is what makes each of them an extension rather than a rewrite. Additional species—hydrogen and other transmutation products, $\langle a \rangle$ and $\langle c \rangle$ loops in hcp hosts, carbide or α' precipitates—enter by extending the vertex set and the admissibility filter, leaving the walker, the reductions and the solver untouched. The bin-moment closure generalizes to multi-dimensional cluster spaces, size together with composition or gas content, once the projection is extended along each axis, which is the natural route to coupled solute segregation and precipitation. The bordered-banded structure is preserved under spatial discretization, so embedding the same kernel in a first-wall or fuel-pin transport calculation, or coupling it to dislocation dynamics or phase-field models, requires no new solver. Because the reaction graph and its rate kernels are separable, data-driven surrogates—learned binding energies or coarse-grained rate constants, and active-learning loops over atomistic calibration—can replace individual edge labels without disturbing the conservation laws or the numerical core. The most immediate need is experimental rather than algorithmic: the calculations reported here were run at a single temperature, whereas the strongest dependence in the EUROFER-

97 database is on irradiation temperature. A temperature series, together with an uncertainty quantification of the underlying DFT and MD parameter set, is in preparation and is the test that will decide how far the agreement reported here extends.

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Declarations

- **Ethics Declaration:** The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Competing interests:** The author declares that he has no known competing financial interests or personal relationships that could appear to influence the work reported in this document.
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- **Declaration of generative AI-assisted technologies :** During the preparation of this work, the author(s) used Anthropic Claude to research published literature, assist in coding and prepare figures. After using this service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Supplementary Material

A Graph-Based Cluster Dynamics Framework for Irradiated Materials

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This Supplementary Material collects the derivations and implementation detail supporting the main text. Section S1 documents the reaction admissibility graph and its realization in the RADCLUSTER code; Section S2 derives the complete set of reaction rate constants for ferritic–martensitic steels; Section S3 derives the three dislocation-loop reaction kernels that transfer interstitial content between the $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ characters; Section S5 derives the growth thresholds for cavities and loops; Section S4 derives the Frenkel-pair and helium conservation identities together with the swelling identity and its numerical diagnostics; Section S6 derives the two helium-occupancy reductions; and Section S7 derives the closures of the size-bin moment equations and their truncation errors; and Section S8 reports the numerical validation of that reduction against the exact discrete solution.

S1. RAG Implementation Details

This appendix collects the supporting detail behind Section 2.4: the precise correspondence between the reaction admissibility graph and the established reaction-network formalisms of chemical kinetics, and the software realization of the graph in the RADCLUSTER code.

Section 2.3 states the correspondence between the RAG and the directed reaction hypergraph of chemical kinetics; this appendix makes it precise. What the correspondence buys is the machinery of those fields – structural conservation laws, incidence-matrix sparsity, graph coloring, and code generation from a declarative network – imported here rather than re-derived. Table S1 gives the dictionary.

Read in the notation of chemical reaction network theory, that of Horn and Jackson [49] and Feinberg [51], the master equation of Section 2.5 is the

Table S1: Correspondence between the reaction admissibility graph of Section 2.3 and the standard objects of chemical reaction network theory, directed hypergraph theory, and Petri nets.

This work	CRN / hypergraph	Petri net
Cluster identifier $\tau = (\sigma, n, p, \mathbf{c}) \in \mathcal{V}$	Species (node of the species–reaction hypergraph)	Place
Unary edge $e \in \mathcal{E}$ (growth, shrinkage, dissociation, inter-population, solute trapping, sink)	Directed hyperedge with one tail	Transition with one input arc
Binary edge (coalescence, annihilation)	Directed hyperedge with two tails	Transition with two input arcs
Stoichiometric tuple $\boldsymbol{\nu}_e$	Column of the stoichiometric matrix $\mathbf{S}$	Arc-weight (incidence) matrix
Rate kernel $k_e(t, \mathbf{s}, \Theta)$	Rate law; mass action when k_e is constant	Transition firing rate
Frenkel-pair and helium conservation laws	Left null vectors of $\mathbf{S}$ (conserved moieties)	Place invariants (P-invariants)
Admissibility rules on $(\sigma, n, p, \mathbf{c})$	Restriction of the species set	Restriction of the place set

mass-action-type system $\dot{\mathbf{s}} = \mathbf{S}\mathbf{J}(\mathbf{s})$ on the species set $\mathcal{V}$, with $\mathbf{S}$ assembled column-by-column from the stoichiometric tuples $\boldsymbol{\nu}_e$. The two properties noted in Section 2.3 – a vertex set that is generated rather than enumerated, and state-dependent kernels $k_e = k_e(\mathbf{s})$ – place the system outside the reach of the classical deficiency theorems, and we state this explicitly rather than claim more than the formalism delivers. What *is* inherited from CRNT is structural and is used throughout: the conservation laws quoted in Section S4 are exactly the left null space of $\mathbf{S}$, so Frenkel-pair and helium conservation hold by construction of the edge classes rather than as a property of any particular kernel calibration. Because the ten edge classes of Section 2.3 fix the sign pattern of $\mathbf{S}$ once and for all, a host material cannot declare a network that violates them.

The same network read as a hypergraph or as a Petri net supplies the other half of the dictionary. The bipartite place/transition encoding of the

reaction hypergraph [53, 106, 107] is what makes the incidence structure, and hence the Jacobian sparsity pattern discussed below, a purely combinatorial object. In that reading the Frenkel-pair and helium balances are P-invariants of the net.

The declarative reading of the network carries over just as directly. The reaction-family templates of the Reaction Mechanism Generator [58, 59] are the analogue of the edge families introduced below: in both, the network is constructed by applying templates to a growing species set rather than being written out. Rule-based generation of the reaction set is likewise standard in object kinetic Monte Carlo [60] and in spatially resolved cluster dynamics [61], though there the network is generally written for one host system rather than exposed as the framework’s interface.

What none of these formalisms carries is a polarity grading, because in molecular kinetics there is no analogue of the signed lattice-defect content $q = \sigma n$ that the host lattice supplies here. It is this grading that makes the bordered-banded Jacobian sparsity a theorem about the graph rather than an observation about one material.

That formalism is realized in the open-source RADCLUSTER code.¹ The remainder of this section documents how the graph is represented, how it is walked, which external libraries carry which part of the computation, and why the graph is *not* built with a general-purpose graph library. All statements below refer to release RADCLUSTER 2.1.

The code is split along exactly the line drawn in Section 2.3: a host-independent core that knows the graph but no physics, and a host declaration that knows the physics but no algorithm (Table S2). The core layer is small by design – approximately 1400 lines of Python across seven modules – because its only responsibilities are the vertex algebra, the stoichiometric contract of the ten edge classes, the index map onto a flat ODE state vector, and the traversal. Adding a host material means adding one declaration module; no core file is touched. The EUROFER-97 declaration of Section 3 consists of three populations (**bulk**- $\langle 111 \rangle$ and **bulk**- $\langle 100 \rangle$ SIA loops, and bulk vacancy-type cavities), one resolved gas species, and 24 edge families drawn from the ten abstract classes.

Within that core the graph itself is never built. A vertex of $\mathcal{V}$ is never instantiated as an object, and an edge of $\mathcal{E}$ is never stored; the unit of repre-

¹<https://github.com/Ghoniem-Org/RadCluster>

Table S2: Software realization of the objects of Section 2.3. Layer 1 is host-independent; Layer 2 is the EUROFER-97 declaration; Layer 3 is the production integrator.

Object	Module	Role
<i>Layer 1 — host-independent core</i>		
$\tau = (\sigma, n, p, \mathbf{c})$	core/cluster_identifier.py	Polarity, population and composition algebra of the vertex set
Edge classes, ν	core/edge_classes.py	The ten classes and their stoichiometric contracts
$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \nu, \mathbf{k})$	core/rag.py	Graph container: population registry, edge families, kernel library, structural validation
Index map	core/state_layout.py	Blocked packing of $\mathcal{V}$ into the flat ODE vector
Graph walker	core/graph_walker.py	Edge-by-edge accumulation of dc_a/dt
Reductions	core/reductions/	Bin-moment and helium-occupancy projections
<i>Layer 2 — host declaration</i>		
$\mathcal{G}_{\text{EUROFER}}$	materials/eurofer97/	3 populations, 24 edge families, size-resolved kernels
<i>Layer 3 — production solver (C++17)</i>		
Integrator	cpp_utils/core/solver.cpp	CVODE BDF driver, window modes
Preconditioners	cpp_utils/core/rhs_dispatch.cpp	Jacobi and Sherman–Morrison–Woodbury
Sparsity/coloring	cpp_utils/core/sparse_jacobian.cpp	CSC pattern, CPR coloring, colored finite differences

sentation is the *edge family*: one record carrying an edge class, a population (or population pair), and a key into the rate-kernel library, standing for the entire one-parameter family of elementary reactions that the class generates along the size axis. A single growth family is $O(N)$ scalar reactions; a single coalescence or annihilation family is $O(N^2)$. The graph is therefore specified in 24 records and generated on demand.

This is a deliberate choice against the obvious alternative, and the reason is quantitative. A general-purpose graph library – NetworkX [108], igraph [109], the Boost Graph Library [110], or graph-tool [111] – stores an explicit incidence structure and traverses it by pointer chasing. At the size

cut-offs used in Section 5 ($N_{\text{SIA}}, N_{\text{vac}} \sim 10^4$), the binary edge families alone generate in excess of 10^8 elementary reactions; materializing them would consume tens of gigabytes, and the traversal would then be repeated at every right-hand-side evaluation of a stiff BDF integration spanning eight decades in dose. The cost is not the graph but the *re-traversal* of the graph.

What the implicit representation buys is that the inner loop is a contiguous array operation. Because every edge family is parameterized by n and its kernel is a precomputed array indexed on the same size axis, each family contributes to the residual through a small number of shifted vector operations over the corresponding block of $\mathbf{s}$; the traversal never leaves NumPy [112] in the Python reference path, and never leaves a contiguous C++ array in the production path. The walker itself is a dispatch table from edge class to accumulator: it reads the class, never the physics, which is precisely the modularity claim of Section 2.3 made executable.

The trade-off is that structural graph queries – connectivity, reachable sets, orphaned vertices, cycle structure – are not available for free, as they would be from a graph library. For those diagnostics the code exports the declared graph, at any chosen size cut-off, into an explicit **NetworkX** multi-digraph [108]; this export is used to verify that the declared EUROFER-97 network is connected, that every non-source vertex is reachable from the monomer pools, and to draw Fig. 1. It is a diagnostic path only and is never entered during integration.

Beyond the inner loop, the incidence structure of the RAG is the sparsity pattern of the master equation’s Jacobian, and this is the point at which the graph formulation pays for itself numerically. A conservative compressed-sparse-column pattern is assembled directly from the edge classes – each class contributes a fixed stencil in (size, population) space – and is therefore known before any kernel is evaluated. Columns are then grouped by greedy Curtis–Powell–Reid coloring [113], so that structurally orthogonal columns are perturbed simultaneously and the entire Jacobian is recovered in a number of residual evaluations equal to the number of colors rather than to the number of unknowns. The reduction is the classical one analyzed by Coleman and Moré [114] and surveyed by Gebremedhin *et al.* [115]; here the graph coloring problem is posed on the same object that defines the physics.

The colored Jacobian feeds a production integrator built on CVODE from SUNDIALS 7.1 [116, 117], run in variable-order variable-step BDF mode. Four linear-solver paths are available and are selected by problem size and physics option:

1. **Sparse direct.** `SUNLinSol_KLU`, i.e. KLU from SuiteSparse [118] with approximate-minimum-degree ordering [119], applied to the colored finite-difference Jacobian described above. This is the default for the fully discrete size-resolved systems.
2. **Preconditioned Krylov.** `SUNLinSol_SPGMR`, i.e. restarted GMRES [120], with a preconditioner that exploits the bordered-banded structure of the Jacobian: the banded block is factored with the LAPACK routines `dgbtrf/dgbtrs` and the low-rank border is folded in by the Sherman–Morrison–Woodbury identity [121] using `dgetrf/dgetrs`. A diagonal (Jacobi) preconditioner is retained as a fallback. This is the default for the bin-moment systems, whose reduced blocks are dense.
3. **Banded and dense.** `SUNLinSol_Band` and `SUNLinSol_Dense`, used for verification on small systems.

The sliding-window solver mode parallelizes the residual sweep over the two polarity layers with OpenMP. The Python reference layer uses NumPy [112] and SciPy [122]; SciPy’s sparse module is used for the diagnostic assembly of $\mathbf{S}$ and for the null-space computation that verifies the conservation laws of Section S4 numerically. The build is CMake-based (C++17) and locates SUNDIALS, SuiteSparse/KLU, LAPACK/BLAS and OpenMP at configure time; each optional dependency degrades to a documented fallback if absent.

Finally, because the Python walker and the C++ residual are two independent realizations of the same edge-class contract, they can be checked against each other. The Python walker is maintained as the executable specification: it is the reference against which the production residual, the bin-moment reduction, and the helium-occupancy reductions are differentially tested at fixed states. The repository carries the corresponding harnesses, which additionally assert that `rag.validate()` succeeds (every declared edge has a registered kernel), that the signed defect content $q = \sigma n$ is conserved edge-by-edge to round-off for every intra-host class, and that the helium inventory closes. Because the sign pattern of $\mathbf{S}$ is fixed by the edge-class table rather than by any host, these checks are host-independent and are inherited by any future material declaration.

S2. Reaction Rates

This section derives the rate constants for all reactions between defect clusters, expressed in units of atomic-fraction (s^{-1}) and in terms of reaction

frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The cluster notation (m, ℓ) and the concentration units (at/at) follow the conventions established in the diffusion treatment of the cluster-dynamics formulation. The cases are ordered by increasing generality: a single mobile species absorbed by a fixed sink (S2), two mobile clusters (S2), purely one-dimensional migration (S2), the mixed 1D/3D regime (S2), and thermal emission by detailed balance (S2).

Common framework: capacitance of the absorbing body. All diffusion-limited kernels below follow from one steady-state problem. If a perfectly absorbing, immobile body is embedded in a medium of far-field concentration C_∞ and the mobile species diffuses isotropically with D_α , the concentration obeys Laplace's equation with $C = 0$ on the absorber. The absorbed current is therefore fixed by a single geometric length, the electrostatic capacitance $\mathcal{C}$ of a conductor of the same shape (in Gaussian units, a length) [62]:

$$\mathcal{K}^{\text{vol}} = 4\pi \mathcal{C} D_\alpha, \quad \mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega, \quad \Omega = a^3/2 \text{ (bcc)}. \quad (\text{S1})$$

Every geometry-specific result in this section is Eq. (S1) evaluated for the appropriate $\mathcal{C}$: $\mathcal{C} = r$ for a sphere, $\mathcal{C} = 2r/\pi$ for a flat disc, and $\mathcal{C} = \pi r_n / \ln(8r_n/r_c)$ for a torus of major radius r_n and core radius r_c . Only the one-dimensional cases of Sections S2–S2 fall outside this framework, because a 1D walker does not solve a three-dimensional Laplace problem.

Mobile point defect absorbed by a fixed sink. For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant [1] is $\mathcal{K}^{\text{vol}} = 4\pi r_m (D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m) a$, and $\Omega = a^3/2$:

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_\alpha \quad [\text{s}^{-1}]. \quad (\text{S2})$$

When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}_{AB}^{3D} = 8\pi \xi_{\text{rec}} (\omega_A + \omega_B), \quad (\text{S3})$$

with $\xi_{\text{rec}} = \sqrt{3}/2 \approx 0.866$ for nearest-neighbor capture in bcc.

Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 4\sqrt{3} \pi (\omega_i + \omega_v) \approx 21.8 \omega_i, \quad (\text{S4})$$

since $\omega_i \gg \omega_v$ at all relevant temperatures.

Point defect absorption by a spherical cluster (void or bubble) of m vacancies is Eq. (S2) written with the size exponent made explicit:

$$\mathcal{K}_{\alpha,m}^{\text{void}} = A_{\text{sph}} m^{1/3} \omega_{\alpha}, \quad A_{\text{sph}} = 8\pi \left(\frac{3}{8\pi} \right)^{1/3} = (192\pi^2)^{1/3} \approx 12.37. \quad (\text{S5})$$

This is the recommended code-ready form: a single precomputed constant multiplied by $m^{1/3}$ and the appropriate reaction frequency. The prefactor follows from $\Omega = a^3/2$; adopting $\Omega = a^3$ instead would lower it by $4^{1/3} = 1.587$ to $(48\pi^2)^{1/3}$, so the atomic-volume convention must be applied consistently to A_{sph} , A_{loop} and $\xi(m)$ alike.

Point defect absorption by a dislocation loop of n SIAs requires species-specific rate constants because only SIAs experience a non-trivial elastic bias ($Z_i^{\text{loop}} > 1$), while vacancies are unbiased ($Z_v^{\text{loop}} = 1$):

$$\mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i, \quad (\text{S6})$$

$$\mathcal{K}_{v,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} \omega_v, \quad (\text{S7})$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{\text{loop}} \approx 1.02\text{--}1.10$ (Table S13). SIA absorption grows the loop by one; vacancy absorption shrinks it by one. The prefactor is pure geometry. A perfect $\frac{1}{2}\langle 111 \rangle$ loop has Burgers vector $b = \sqrt{3}a/2$, so each SIA occupies an area $\Omega/b = a^2/\sqrt{3}$ of the habit plane and $\pi r_n^2 = n\Omega/b$ gives

$$r_n = a \sqrt{\frac{n}{\pi\sqrt{3}}} = 0.4287 a \sqrt{n}, \quad \frac{8\pi r_n}{a} = 8\sqrt{\pi/\sqrt{3}} \sqrt{n} = A_{\text{loop}} \sqrt{n}. \quad (\text{S8})$$

Equations (S6)–(S7) therefore correspond to $\mathcal{C} = r_n$ in Eq. (S1), i.e. to a sphere of the loop radius. The exact toroidal solution of Seeger and Gösele [63] gives instead

$$\mathcal{C}_{\text{loop}} = \frac{\pi r_n}{\ln(8r_n/r_c)} \simeq (0.5\text{--}0.7) r_n \quad (r_n/r_c \sim 10\text{--}10^2), \quad (\text{S9})$$

bracketed below by the flat-disc value $2r_n/\pi \approx 0.637 r_n$ [62]; for $\langle 100 \rangle$ loops ($b = a$) the geometric prefactor becomes $8\sqrt{\pi/2} \approx 10.03$ instead of 10.78. The efficiency Z_{α}^{loop} used in the runs therefore carries both the geometric factor $\mathcal{C}_{\text{loop}}/r_n$ and the elastic bias [64, 65].

Point defect absorption rate by the dislocation network (first-order sink): $-Z_{\alpha} \rho_d a^2 \omega_{\alpha} c_{\alpha}$, where ρ_d (m^{-2}) is the dislocation line density, and no n_{at} conversion is needed because this term is first-order in c_{α} .

Two mobile clusters. When both partners move, transform to the relative coordinate $\mathbf{r} = \mathbf{r}_A - \mathbf{r}_B$. Two independent isotropic random walks add, so $\mathbf{r}$ diffuses with $D_A + D_B$, and reaction occurs when $\mathbf{r}$ reaches the contact surface swept by the two bodies. For two spheres that surface is again a sphere, of radius $r_A + r_B$, so both the radius and the diffusivity additivity in Eq. (S3) are exact [3]. Written for the polarity-labelled channels of the main text,

$$\mathcal{K}_{s,s'}^{ab} = 8\pi (\xi_s + \xi_{s'}) (\omega_s^{\text{eff}} + \omega_{s'}^{\text{eff}}), \quad a, b \in \{i, v\}, \quad (\text{S10})$$

which reduces to Eq. (S2) when one partner is immobile ($\omega^{\text{eff}} \rightarrow 0$) and to Eq. (S4) for the monomer–monomer channel. Throughout, $\omega_\alpha^{\text{eff}}$ denotes the three-dimensionally equivalent reaction frequency of that reactant: $\omega_\alpha^{\text{eff}} = \omega_\alpha$ for a three-dimensional migrator, and $\omega_n^{\text{eff}} = \omega_n^{1D}/3$ for a one-dimensionally gliding cluster, Eq. (S20) below.

Two qualifications matter for defect clusters. First, additivity of capture radii is a theorem only for spheres: in general $\mathcal{K}^{\text{vol}} = 4\pi(D_A + D_B)\mathcal{C}[\mathcal{A} \oplus (-\mathcal{B})]$, the capacitance of the Minkowski sum of the two bodies, and capacitance is not additive. For a mobile cavity meeting a mobile loop we therefore use

$$\mathcal{K}_{m,n}^{\text{cav-loop}} = \frac{8\pi}{a} \left[r_m + \frac{\pi r_n}{\ln(8r_n/r_c)} \right] (\omega_m^{\text{eff}} + \omega_n^{\text{eff}}), \quad (\text{S11})$$

which is exact in either single-mobile limit and is an interpolation in between. Second, cluster diffusion is generally anisotropic. Gösele and Seeger [66] showed that an anisotropic diffusion tensor can be removed by the affine rescaling $x' = x\sqrt{\bar{D}/D_x}$ (and cyclically), which maps the anisotropic Laplacian onto the isotropic one at the price of deforming the reaction surface into an ellipsoid; to leading order the isotropic kernel is recovered with the geometric mean $\bar{D} = (D_x D_y D_z)^{1/3}$. Adjanor [67, 68] and Jourdan and Adjanor [69] have carried this analysis to arbitrary ratios D_A/D_B and to mutually 1D-mobile partners, and report the characteristic scalings

$$\mathcal{K} \propto (D_A/D_B)^\delta, \quad \delta = \begin{cases} 0, & \text{3D–3D,} \\ -1/2, & \text{3D–1D,} \\ -1/3, & \text{1D–1D,} \end{cases} \quad (\text{S12})$$

together with the result that the mutually mobile 1D–1D regime persists down to diffusivity ratios of order 10^{-6} , so that treating the slower partner as a fixed sink is a much stronger approximation than is usually assumed.

For two 1D-mobile partners the loss is cubic rather than quadratic in concentration [67],

$$\left. \frac{\partial c_A}{\partial t} \right|_{\parallel} = -6\pi^2 R^4 (c_A^2 c_B D_B + c_B^2 c_A D_A). \quad (\text{S13})$$

Equation (S13) is not used in the EUROFER-97 runs, where only one polarity is 1D mobile, but it bounds the error of treating sessile loops as fixed sinks.

One-dimensional migration. SIA clusters of size $n \geq 4$ migrate by 1D glide along $\langle 111 \rangle$ with the size-dependent SIA-cluster reaction frequency:

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad s \approx 0.5\text{--}1.0, \quad E_m^{1D} \lesssim 0.05 \text{ eV}. \quad (\text{S14})$$

A one-dimensional walker samples a line rather than a volume, so its absorption is not a binary process: it is controlled by the entire sink field and must be written as a sink strength rather than as a pair rate constant. For sink family m ,

$$\left. \frac{dc_n}{dt} \right|_m = -D_n^{\text{eff}} k_{n,m}^2 c_n, \quad (\text{S15})$$

where D_n^{eff} is the effective diffusivity used in the rate equation and $k_{n,m}^2$ is the sink strength for absorption of the mobile cluster by sink family m . In the pure 1D limit [70, 71, 73],

$$k_{(1),m}^2 = 6\kappa_m \kappa, \quad \kappa_m = \sigma_m N_m, \quad \kappa = \sum_j \kappa_j, \quad (\text{S16})$$

where N_m is the number density of sinks of type m , σ_m is their geometrical absorption cross section, and κ^{-1} is the mean 1D diffusion range in the sink field. For a monodisperse population of spherical sinks with radius r_m ,

$$\sigma_m = \pi r_m^2, \quad k_{(1),m}^2 = 6\pi^2 r_m^4 N_m^2. \quad (\text{S17})$$

Thus, since $r_m \propto m^{1/3}$, the pure 1D sink strength scales as $m^{4/3} N_m^2$, in contrast to the ordinary 3D sink strength,

$$k_{(3),m}^2 = 4\pi r_m N_m, \quad (\text{S18})$$

which scales as $m^{1/3} N_m$. Their ratio, $k_{(1),m}^2 / k_{(3),m}^2 = \frac{3}{2} \pi r_m^3 N_m = \frac{9}{8} \Phi_m$ with Φ_m the sink volume fraction, shows that in a dilute microstructure purely

1D transport is far less efficient at finding sinks than 3D transport at the same jump frequency. The cross section σ_m is the only geometry-dependent quantity: for toroidal loops the orientation-averaged projected area is $\sigma_n = \pi^2 r_c r_n$ [123], and Eq. (S16) applies unchanged.

Mixed 1D/3D migration: the master curve. Real SIA clusters change their Burgers-vector direction at a finite rate, so neither limit applies. Let ℓ_n denote the mean straight-line migration length of cluster n between changes of its 1D migration direction. With $d_j = \sqrt{3}a/2$ the glide jump distance and n_{ch} the mean number of jumps between rotations [123],

$$\ell_n^2 = d_j^2 n_{\text{ch}} = 2D_n^{\text{lo}} \tau_{\text{rot},n}, \quad (\text{S19})$$

where D_n^{lo} is the longitudinal 1D diffusivity and $\tau_{\text{rot},n}$ is the mean time between direction changes. Rotations are assumed instantaneous and to a randomly chosen equivalent variant; with v glide variants only a fraction $f_v = (v-1)/v$ of rotation events change the direction [67].

It is worth stating explicitly what an effective diffusivity can and cannot do here. Long-range transport is a 3D random walk of step ℓ_n taken every $\tau_{\text{rot},n}$, so $\langle r^2 \rangle = 6D_n^{\text{3D,eff}} t$ gives

$$D_n^{\text{3D,eff}} = \frac{\ell_n^2}{6 \tau_{\text{rot},n}} = \frac{D_n^{\text{lo}}}{3}, \quad \omega_n^{\text{3D,eff}} = \frac{\omega_n^{\text{1D}}}{3}, \quad (\text{S20})$$

independent of ℓ_n : at any instant the cluster explores one of three spatial directions. Inserting Eq. (S20) into Eq. (S2) would therefore give an absorption rate that is independent of the direction-change frequency, which is wrong: the 1D and 3D sink strengths differ not by a constant but in their scaling with sink density (N_m^2 against N_m). No rescaling of D can reproduce that, and the transition must be handled at the level of the sink strength.

The transition from correlated 1D kinetics to 3D kinetics is represented by the Trinkaus–Heinisch–Barashev–Golubov–Singh master-curve interpolation [72, 73]. The effective sink strength is written as

$$k_{n,m}^2 = y_{n,m} k_{(1),m}^2, \quad (\text{S21})$$

with

$$y_{n,m} = \frac{1}{2} \left[1 + (1 + 4x_{n,m}^2)^{1/2} \right], \quad (\text{S22})$$

and

$$x_{n,m} = \left[\frac{\ell_n^2 k_{(1),m}^2}{12} + \frac{k_{(1),m}^4}{k_{(3),m}^4} \right]^{-1/2}. \quad (\text{S23})$$

This form recovers the correct limiting behaviors. When $\ell_n \gg \kappa^{-1}$, direction changes are rare, $x_{n,m} \rightarrow 0$, $y_{n,m} \rightarrow 1$, and therefore

$$k_{n,m}^2 \rightarrow k_{(1),m}^2. \quad (\text{S24})$$

When direction changes are very frequent, $\ell_n \ll r_m$, the second term in Eq. (S23) dominates, $x_{n,m} \rightarrow k_{(3),m}^2/k_{(1),m}^2 \gg 1$, $y_{n,m} \rightarrow x_{n,m}$, and the interpolation gives

$$k_{n,m}^2 \rightarrow k_{(3),m}^2. \quad (\text{S25})$$

The loss term may therefore be written compactly as

$$\left. \frac{dc_n}{dt} \right|_m = -D_n^{\text{eff}} y_{n,m} k_{(1),m}^2 c_n, \quad (\text{S26})$$

or, dividing by N_m to recover a kernel in the form used by the graph walker,

$$\mathcal{K}_{n,m}^{\text{mixed}} = \frac{D_n^{\text{eff}} k_{n,m}^2}{N_m} = 6 y_{n,m} D_n^{\text{eff}} \sigma_m \kappa. \quad (\text{S27})$$

In the near-3D regime, $y_{n,m} \simeq x_{n,m}$, and Eqs. (S17)–(S18) reduce Eq. (S27) for a single spherical sink family to the closed form

$$\mathcal{K}_{n,m}^{\text{mixed}} \simeq \frac{A_{\text{sph}} m^{1/3} \omega_n^{\text{eff}}}{[1 + B_{\text{rot}} (\ell_n/r_m)^2]^{1/2}}, \quad B_{\text{rot}} = \frac{2}{9}, \quad (\text{S28})$$

which returns the Smoluchowski value as $\ell_n \rightarrow 0$ and decays as ℓ_n^{-1} once the straight-line segment exceeds the sink radius. Written with $\hat{L} = \ell_n/a$ the denominator is $[1 + B_{\text{rot}} \hat{L}^2 \xi(m)^{-2}]^{1/2}$, i.e. the size dependence enters as $m^{-2/3}$; this is the form evaluated by the code and quoted as Eq. (24) of the main text. The pure 1D limit is quadratic in sink density and scales as r_m^4 , whereas the 3D limit is first order in sink density and scales as r_m . This difference makes the absorption of one-dimensionally migrating clusters especially sensitive to sink size, sink density, and the direction-change mean free path.

Reactions between two 1D-mobile clusters. Let clusters A and B glide along $\mathbf{e}_A$ and $\mathbf{e}_B$ with diffusivities D_A , D_B and mutual capture distance $R = R_A + R_B$. Three geometries are distinct.

Collinear glide lines. When both species glide on the same line family, the partner population also sets the spacing between interceptions. Applying the marked interception result symmetrically then gives the cubic loss frequency quoted in Eq. (S13),

$$\mathcal{K}_{A|B}^{\parallel} = 6\pi^2 R^4 (D_B N_A N_B + D_A N_B^2). \quad (\text{S29})$$

Non-parallel intersecting lines. For two lines that intersect at an angle θ , the displacements s_A, s_B give $r^2 = s_A^2 + s_B^2 - 2s_A s_B \cos \theta$, so the encounter is a two-dimensional first-passage problem in (s_A, s_B) . The long-time solution of the 2D absorbing-disc problem is logarithmically slow,

$$\mathcal{K}_{A|B}^{\times}(t) = 2\pi D R \alpha(t) N_B, \quad \alpha(t) \simeq \frac{4}{\ln(4Dt/\pi R^2)}, \quad (\text{S30})$$

with D the relevant pair diffusivity. Unlike the collinear channel this contribution is only quadratic in concentration. For $D_A \neq D_B$, or for non-orthogonal axes, the problem maps to diffusion toward an ellipse and a diffusion-tensor transformation is required; replacing D by $D_A + D_B$ is not generally valid.

Skew, non-coplanar lines. Two non-parallel lines separated by a minimum distance h meet the capture condition only over a reduced radius,

$$R_{\text{eff}}(h) = \begin{cases} (R^2 - h^2)^{1/2}, & h < R, \\ 0, & h \geq R, \end{cases} \quad (\text{S31})$$

and the intersecting-line result applies with $R \rightarrow R_{\text{eff}}$,

$$\mathcal{K}_{A|B}^{\text{skew}}(t, h) = 2\pi D R_{\text{eff}}(h) \alpha[t; R_{\text{eff}}(h)] N_B. \quad (\text{S32})$$

Because α also depends on the radius, the lattice average must be taken over the product, $\overline{R_{\text{eff}}\alpha} = \sum_h P(h) R_{\text{eff}}(h) \alpha[t; R_{\text{eff}}(h)]$, and not over R_{eff} alone.

Orientation average. With v equally populated unoriented glide variants the non-collinear fraction is $f_v = (v - 1)/v$, equal to 3/4 for the four $\langle 111 \rangle$ axes of bcc iron. Separating intersecting from skew pairs by the conditional probability $p_\times$,

$$\mathcal{K}_{A|B}^{1D-1D} = (1 - f_v) \mathcal{K}_{A|B}^{\parallel} + f_v \left[p_\times \mathcal{K}_{A|B}^\times + (1 - p_\times) \langle \mathcal{K}_{A|B}^{\text{skew}} \rangle \right]. \quad (\text{S33})$$

In a dilute population the approximately quadratic non-collinear terms normally exceed the cubic collinear one, so quoting Eq. (S29) alone understates the 1D–1D channel.

Rotating gliders: immobile and mutually mobile partners. Section S2 treats a rotating glider absorbed by the fixed microstructural sink field. Two further cases occur in the EUROFER-97 graph: a rotating glider reacting with an immobile *cluster* family, and two rotating gliders reacting with each other. Both follow from the master curve (S22) applied to the pair of limiting sink strengths of the case at hand, so that each kernel is exact in both limits and interpolates between them.

Rotating glider and immobile cluster (1DR/immobile). Let the immobile family m have number density N_m , capture radius $R_{n,m} = r_n + r_m$ and cross section $\sigma_{n,m} = \pi R_{n,m}^2$. The walker terminates on that family with the line-interception sink strength

$$k_{(1),n,m}^2 = 6 \sigma_{n,m} N_m \Sigma_n, \quad \Sigma_n = \sum_j \sigma_{n,j} N_j, \quad (\text{S34})$$

in which the immobile family both terminates the walk and contributes to the screening sum Σ_n . When rotations are frequent the same family is sampled as a three-dimensional point sink,

$$k_{(3),n,m}^2 = 4\pi R_{n,m} N_m, \quad (\text{S35})$$

first order rather than quadratic in N_m . Inserting Eqs. (S34) and (S35) into Eq. (S23) fixes $x_{n,m}$, and

$$\mathcal{K}_{n \rightarrow m}^{(1R)} = D_n y_{n,m} k_{(1),n,m}^2 \longrightarrow \begin{cases} 6 D_n \sigma_{n,m} N_m \Sigma_n, & \ell_n \text{ long,} \\ 4\pi R_{n,m} D_n N_m, & \ell_n \text{ short.} \end{cases} \quad (\text{S36})$$

Formally this is Sec. S2 with a different target, but the distinction matters in practice: N_m is a state variable, so the kernel is quadratic in an evolving population at one end of the crossover and linear at the other.

Two rotating gliders (1DR/1DR). Let A and B both rotate, with flight lengths ℓ_A, ℓ_B . In the relative coordinate the two limits are known. Rare rotations return the purely 1D pair kinetics $\mathcal{K}_{A|B}^{1D-1D}$ of Eq. (S33); frequent rotations make both trajectories isotropic on the encounter scale, and the pair obeys Smoluchowski kinetics with summed diffusivities,

$$\mathcal{K}_{A|B}^{(3)} = 4\pi R (D_A + D_B) N_B. \quad (\text{S37})$$

Defining the pair sink strengths

$$k_{AB}^2 \equiv \frac{\mathcal{K}_{A|B}^{1D-1D}}{D_A + D_B}, \quad k_{(3),AB}^2 \equiv 4\pi R N_B, \quad (\text{S38})$$

and adding the variances of the two independent walkers, $\ell_{AB}^2 = \ell_A^2 + \ell_B^2$, the master curve reads

$$\mathcal{K}_{A|B}^{(1R|1R)} = y_{AB} \mathcal{K}_{A|B}^{1D-1D}, \quad y_{AB} = \frac{1}{2} \left[1 + (1 + 4x_{AB}^2)^{1/2} \right], \quad (\text{S39})$$

$$x_{AB} = \left[\frac{\ell_{AB}^2 k_{AB}^2}{12} + \frac{k_{AB}^4}{k_{(3),AB}^4} \right]^{-1/2}. \quad (\text{S40})$$

For $x_{AB} \rightarrow 0$, $y_{AB} \rightarrow 1$ and the purely 1D pair result is recovered; for $x_{AB} \gg 1$, $y_{AB} \rightarrow x_{AB} \rightarrow k_{(3),AB}^2/k_{AB}^2$ and $\mathcal{K}^{(1R|1R)} \rightarrow (D_A + D_B)k_{(3),AB}^2$, which is Eq. (S37). Since $\mathcal{K}^{1D-1D}$ already mixes a cubic collinear term with approximately quadratic non-collinear ones, $\mathcal{K}^{(1R|1R)}$ is an instantaneous frequency evaluated from the current populations and not a binary rate constant. Equations (S39)–(S40) are constructed to be exact in both limits, in the same spirit as Eq. (S22), rather than derived independently from a first-passage treatment of the mutually rotating pair; they are used here as a controlled closure.

Thermal emission by detailed balance. Thermal emission of species α from a cluster of size n is related to the capture rate by detailed balance. The daughter cluster has size $(n-1)$, so the capture rate that enters the detailed-balance relation uses $(n-1)$:

$$\alpha_\alpha(n) = A_{\text{sph}} (n-1)^{1/3} \omega_\alpha \exp\left(-\frac{E_b^\alpha(n)}{k_B T}\right) \quad [\text{s}^{-1}], \quad (\text{S41})$$

Table S3: Reaction kernels for two mutually 1D-mobile clusters, in hertz. $R = R_A + R_B$ is the mutual capture distance, $\alpha(t) \simeq 4/\ln(4Dt/\pi R^2)$ the slowly varying logarithmic factor of two-dimensional encounter kinetics, h the minimum separation of two skew glide lines, $f_v = (v-1)/v = 3/4$ the non-collinear fraction for the four unoriented $\langle 111 \rangle$ variants of bcc iron, and $p_\times$ the conditional probability that a non-collinear pair intersects rather than skews; N denotes a number density and ℓ a flight length between rotations. None of these channels is active in the EUROFER-97 calculations, for the reasons given after Table 2 of the main text.

Case	Kernel [Hz]	Capture length	Scaling	Eq.
1D/1D-collinear	$\mathcal{K}_{A B}^\parallel = 6\pi^2 R^4 \times (D_B N_A N_B + D_A N_B^2)$	$R = R_A + R_B$	cubic in c	(S29)
1D/1D-intersecting	$\mathcal{K}_{A B}^\times = 2\pi D R \alpha(t) N_B$	R	quadratic in c	(S30)
1D/1D-skew	$\mathcal{K}_{A B}^{\text{skew}} = 2\pi D R_{\text{eff}} \alpha N_B$, $R_{\text{eff}} = (R^2 - h^2)^{1/2}$	$R_{\text{eff}}(h)$	quadratic in c	(S32)
1D/1D-averaged	$\mathcal{K}_{A B}^{1\text{D}-1\text{D}} = (1-f_v)\mathcal{K}^\parallel + f_v[p_\times \mathcal{K}^\times + (1-p_\times)\langle \mathcal{K}^{\text{skew}} \rangle]$	—	mixed	(S33)
1DR/1DR	$\mathcal{K}_{A B}^{(1R 1R)} = y_{AB} \mathcal{K}_{A B}^{1\text{D}-1\text{D}}$, $\ell_{AB}^2 = \ell_A^2 + \ell_B^2$	R	$\rightarrow$ 3D limit	(S39)

where $E_b^\alpha(n)$ is the binding energy of the last defect of type α to the cluster (from detailed-balance dissociation). The emission enters the master equation as a first-order process:

$$\left. \frac{dc_n}{dt} \right|_{\text{emission}} = +\alpha_\alpha(n) c_n - \alpha_\alpha(n+1) c_{n+1}. \quad (\text{S42})$$

For a dislocation loop emitting an SIA, replace $A_{\text{sph}}(n-1)^{1/3}$ with $A_{\text{loop}}(n-1)^{1/2}$:

$$\alpha_i^{\text{loop}}(n) = A_{\text{loop}}(n-1)^{1/2} \omega_i \exp\left(-\frac{E_b^i(n)}{k_B T}\right). \quad (\text{S43})$$

For $\text{He}_\ell \text{V}_m$ bubbles, three emission channels exist (detailed-balance for the He–bubble system):

Vacancy emission:

$$\varepsilon_v(m, \ell) = A_{\text{sph}}(m-1)^{1/3} \omega_v \exp\left(-\frac{E_b^v(m, \ell)}{k_B T}\right). \quad (\text{S44})$$

Helium emission:

$$\varepsilon_h(m, \ell) = A_{\text{sph}} m^{1/3} \omega_h \exp\left(-\frac{E_B^{\text{He}}(m, \ell)}{k_B T}\right). \quad (\text{S45})$$

He emission uses $m^{1/3}$ (not $(m-1)^{1/3}$) because removal of a He atom does not change the vacancy count; the capture radius for detailed balance is that of the daughter $\text{He}_{\ell-1}\text{V}_m$, which retains the same m . The binding energy E_B^{He} is from the He–vacancy detailed-balance relation summarised in this section.

Trap mutation (Frenkel pair emission):

$$\Gamma_{\text{TM}}(m, \ell) = \nu_0 \exp\left(-\frac{E_{\text{TM}}(m, \ell)}{k_B T}\right), \quad (\text{S46})$$

producing $\text{He}_{\ell}\text{V}_{m+1} + \text{SIA}$. Table S13 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions (Section S2).

Mono-defect reduction of the coalescence kernels. Under the mono-defect simplification ($i_{\text{mobile}} = v_{\text{mobile}} = 1$), the coalescence rate constants reduce to

$$\mathcal{K}_{n,1}^{ii}|_{n'=1} = \mathcal{K}_{i,n}^{\text{loop}} = A_{\text{loop}} n^{1/2} Z_i^{\text{loop}} \omega_i^{\text{eff}}, \quad (\text{S47})$$

$$\mathcal{K}_{m,1}^{vv}|_{m'=1} = \mathcal{K}_{v,m}^{\text{cav}} = A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}, \quad (\text{S48})$$

$$\mathcal{K}_{m=1,n=1}^{vi} = \mathcal{K}_{iv} = 4\sqrt{3} \pi (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}). \quad (\text{S49})$$

Loop incorporation into the dislocation network. Loops are also removed from the explicit inventory by the pre-existing dislocation network. Loop–loop encounters are already carried by the coalescence kernel (S10) and its junction branch (S73), so this channel models only the loop–network transfer and adds no loop–loop term, which would otherwise be double counted.

For a circular prismatic loop of n interstitials of character c , either $\frac{1}{2}\langle 111 \rangle$ or $\langle 100 \rangle$, with Burgers-vector magnitude b_c , the platelet area is $n\Omega/b_c$ and the diameter is

$$d_n^{(c)} = 2\left(\frac{n\Omega}{\pi b_c}\right)^{1/2}, \quad (\text{S50})$$

the per-character convention also used by the TEM-diameter post-processing. With a network line density ρ_{net} , the mean loop–network spacing and the edge-to-edge gap to the nearest line are

$$L_{\ell d} = \rho_{\text{net}}^{-1/2}, \quad s_{\ell d}(n) = \max\left[b_c, L_{\ell d} - \frac{1}{2}d_n^{(c)}\right], \quad (\text{S51})$$

clamped from below to exclude unphysical negative gaps. The elastic interaction range of the loop is

$$R_{\text{int}}(n) = \chi d_n^{(c)} + \xi b_c \sqrt{\frac{\mu \Omega n}{k_B T}}, \quad (\text{S52})$$

with χ a dimensionless geometric range parameter and the second term an optional small- n floor that keeps R_{int} finite as $d_n^{(c)} \rightarrow 0$; $\xi = 0$ throughout this work, so $R_{\text{int}} = \chi d_n^{(c)}$. A smooth threshold gates the transfer on geometric proximity,

$$P_{\ell d}(n) = \frac{1}{2} \left[1 + \tanh \frac{R_{\text{int}}(n) - s_{\ell d}(n)}{b_c} \right], \quad (\text{S53})$$

so $P_{\ell d} \rightarrow 0$ in the dilute limit $L_{\ell d} \gg R_{\text{int}}$ and $P_{\ell d} \rightarrow 1$ once the elastic zone of the loop reaches the mean network spacing.

The transfer frequency is set by the network, not by the loop:

$$\Lambda_n^{\text{net}} = \nu^{\text{net}} P_{\ell d}(n), \quad \nu^{\text{net}} = v_{\text{net}} \rho_{\text{net}} w_c, \quad (\text{S54})$$

$$v_{\text{net}} = \frac{a^2}{b} \left| Z_i \omega_i^{\text{eff}} c_1^i - Z_v \omega_v^{\text{eff}} c_1^v \right|.$$

in which $v_{\text{net}} \rho_{\text{net}}$ is the line area swept per unit volume per unit time and $w_c = \mathcal{O}(b_c)$ converts it into a capture probability per loop. Equation (S54) carries no factor of ω_n^{eff} , so the channel stays active for sessile loops — which are precisely the ones that dominate incorporation — and v_{net} is the Bullough–Newman climb velocity evaluated from the segment-frozen monomer concentrations, so no new transport physics is introduced. The loss term is

$$\left. \frac{dc_n}{dt} \right|_{\text{net}} = -\Lambda_n^{\text{net}} c_n, \quad n \geq n_{\text{inc}}, \quad (\text{S55})$$

applied to both characters and folded additively into the fixed-sink diagonal of Eq. (S15) rather than carried as a separate edge; loops below the incorporation onset n_{inc} are too small to register on the network and are excluded.

Each absorbed loop delivers n interstitials to the network and adds about one loop circumference of line, so the network reservoir and the fixed-sink

ledger obey

$$\frac{d\rho_{\text{net}}}{dt} = \frac{\pi}{\Omega} \sum_c \sum_{n \geq n_{\text{inc}}} d_n^{(c)} \Lambda_n^{\text{net}} c_n^{(c)} - K_{\text{rec}} \rho_{\text{net}}^{3/2}, \quad (\text{S56})$$

$$\frac{dJ_I^{\text{fix}}}{dt} += \sum_c \sum_{n \geq n_{\text{inc}}} n \Lambda_n^{\text{net}} c_n^{(c)}. \quad (\text{S57})$$

The recovery sink $K_{\text{rec}} \rho_{\text{net}}^{3/2}$ is mandatory: with the gain term alone ρ_{net} grows without bound and the model merely relocates the non-saturation from the loops to the network. Routing the absorbed interstitials into the same fixed-sink integral J_I^{fix} that the Frenkel-pair diagnostic already subtracts leaves that diagnostic unchanged in form, since every interstitial leaving the explicit inventory reappears in J_I^{fix} .

The same construction is applied to the cavity population, with one change: a sweeping line traverses the volume, so $v_{\text{net}} \rho_{\text{net}} w$ is already the encounter rate and the geometric gate would count it twice. The elastic range parameter there widens the capture width instead, $w = \chi_{\text{void}} d_m$ with $d_m = 2(3m\Omega/4\pi)^{1/3}$, and the channel starts above v_{mobile} , where mobile vacancy clusters are already charged to the diffusive dislocation sink. Table S4 lists the values used here.

Table S4: Parameters of the loop- and cavity-network loss channels for the run reported in Section 5.

Quantity	Symbol	Value	Units
Network line density	ρ_{net}	5×10^{14}	m^{-2}
Network density ceiling	$\rho_{\text{net}}^{\text{max}}$	5×10^{14}	m^{-2}
Geometric range parameter	χ	50	—
Small- n floor coefficient	ξ	0	—
Capture width	w_c	1.241	nm
Incorporation onset	n_{inc}	$i_{\text{mobile}} = 5$	SIAs
Recovery prefactor	K_{rec}	1×10^{-6}	m s^{-1}
Cavity range parameter	χ_{void}	1165	—
Cavity onset	m_{inc}	$v_{\text{mobile}} = 5$	vacancies

S3. Dislocation Loop Reaction Kernels

Three reactions transfer interstitial content between mobile $\frac{1}{2}a\langle 111 \rangle$ loops and sessile $a\langle 100 \rangle$ loops,

$$111_n \longrightarrow 100_n, \quad \text{unary rotation,} \quad (\text{S58})$$

$$111_n + 111_{n'} \longrightarrow 100_{n+n'}, \quad \text{junction,} \quad (\text{S59})$$

$$100_m + 111_n \longrightarrow 100_{m+n}, \quad \text{absorption,} \quad (\text{S60})$$

where 111_n denotes a $\frac{1}{2}a\langle 111 \rangle$ loop of n interstitials and 100_m an $a\langle 100 \rangle$ loop of m . The $\langle 100 \rangle$ population is defined only for $m \geq n_{\text{loop}}^{\min}$ and is sessile, $D_m^{100} \equiv 0$; the $\frac{1}{2}\langle 111 \rangle$ population is mobile only for $n \leq i_{\text{mobile}}$. Concentrations are atomic fractions, so unary and binary kernels both carry units of s^{-1} through $\dot{c}_a = -\Gamma_a c_a$ and $\dot{c}_a = -K_{ab} c_a c_b$; the corresponding volumetric coefficient is $k_{ab}^{(V)} = \Omega K_{ab}$.

Inserting n atomic volumes produces a platelet of volume $A_X b_X = n\Omega$, so for character $X \in \{111, 100\}$

$$A_X(n) = \frac{n\Omega}{b_X}, \quad b_{111} = \frac{\sqrt{3}}{2}a, \quad b_{100} = a. \quad (\text{S61})$$

With the hexagonal $\frac{1}{2}\langle 111 \rangle$ and square $\langle 100 \rangle$ habits,

$$P_{111}(n) = 6\sqrt{\frac{2A_{111}(n)}{3\sqrt{3}}}, \quad P_{100}(n) = 4\sqrt{A_{100}(n)}, \quad (\text{S62})$$

$$\frac{P_{111}(n)}{b_{111}} = C_P \sqrt{n}, \quad C_P = \frac{6}{b_{111}} \sqrt{\frac{2\Omega}{3\sqrt{3}b_{111}}}.$$

Every perimeter-proportional energy therefore scales as $\sqrt{n}$. For $a = 2.867 \text{ \AA}$ and $\Omega = 11.80 \text{ \AA}^3$ this gives $b_{111} = 2.483 \text{ \AA}$ and $C_P = 3.268$.

The loop formation free energy is that of Dudarev *et al.* [94],

$$E_l^X(n, T) = P_X(n) \left[\hat{F}_X(T) \ln \frac{4R_X^*(n)}{e\delta} + F_\delta^X + F_c^X \right], \quad R_X^*(n) = \sqrt{A_X(n)/\pi}, \quad (\text{S63})$$

in which δ is the core cutoff, $\hat{F}_X$ the anisotropic prelogarithmic line-energy factor, and $F_\delta^X + F_c^X$ the core traction and nonlinear core terms. The driving force for conversion at fixed content is $\Delta F(n, T) = E_l^{111}(n, T) - E_l^{100}(n, T)$,

positive when $\langle 100 \rangle$ is the lower-energy character. For a pure-edge $\langle 100 \rangle$ loop, anisotropic elasticity gives

$$\hat{F}_{100} = \frac{a^2}{4\pi}(c_{11} + c_{12}) \left[\frac{c_{44}(c_{11} - c_{12})}{c_{11}(c_{11} + c_{12} + 2c_{44})} \right]^{1/2}, \quad (\text{S64})$$

and since the tetragonal shear modulus softens as $c' = \frac{1}{2}(c_{11} - c_{12}) \simeq c'_0(1 - T/T_c)^{1/2}$ toward the bcc–fcc transformation, Eq. (S64) implies

$$\hat{F}_{100}(T) = \hat{F}_{100}^0 Q(T), \quad Q(T) = (1 - T/T_c)^{1/4}, \quad (\text{S65})$$

while $\hat{F}_{111}$ is held temperature independent. This softening is the physical origin of the temperature dependence of the whole conversion problem. The coefficient $\hat{F}_{100}^0$ is not an independent input: writing $L_X(n) = \ln[4R_X^*(n)/e\delta]$ and imposing the single crossover condition $\Delta F(n_{\text{ref}}, T^*) = 0$ gives

$$\hat{F}_{100}^0 = \frac{\frac{P_{111}}{P_{100}} [\hat{F}_{111} L_{111} + F_\delta^{111} + F_c^{111}] - (F_\delta^{100} + F_c^{100})}{Q(T^*) L_{100}} \bigg|_{n=n_{\text{ref}}}, \quad (\text{S66})$$

so $\hat{F}_{100}^0$ and T^* must not be counted as two independent parameters.

For the unary channel, a $\frac{1}{2}\langle 111 \rangle$ loop sits in a local free-energy minimum and transition-state theory gives a forward attempt rate $k_+ = \nu_0 \exp[-E_a(n)/k_B T]$. Atomistics motivates a thermally activated reorientation path but does not fix a barrier for every loop size, so we assume that the activated portion of the loop line grows with the perimeter,

$$E_a(n) = E_{a0} + \gamma_a \frac{P_{111}(n)}{b_{111}} = E_{a0} + \gamma_a C_P \sqrt{n}, \quad (\text{S67})$$

where P_{111}/b_{111} counts line segments of one Burgers vector and γ_a is an effective barrier per segment. With the calibrated values of Table S6, $E_a(n) = 2.10 + 0.0654\sqrt{n}$ eV. Local detailed balance between the two characters requires $k_+/k_- = \exp[\Delta F/k_B T]$, so the forward-minus-reverse coefficient is $k_+ - k_- = k_+[1 - \exp(-\Delta F/k_B T)]$. Retaining this net rate only where the forward reaction is downhill gives

$$\Gamma_{\text{uni}}(n, T) = \nu_0 e^{-E_a(n)/k_B T} [1 - e^{-\Delta F(n, T)/k_B T}]_+ \Theta(n - n_{\text{loop}}^{\min}), \quad [x]_+ \equiv \max(0, x), \quad (\text{S68})$$

whose bracket vanishes for $\Delta F \leq 0$, reduces to $\Delta F/k_B T$ for $0 < \Delta F \ll k_B T$, and saturates at unity for $\Delta F \gg k_B T$. This is a one-way net-rate closure consistent with the sign of the Dudarev driving force rather than a fully reversible model, since the return term $+k_- c_n^{100}$ is not retained. The population terms are $\dot{c}_n^{111}|_{\text{uni}} = -\Gamma_{\text{uni}} c_n^{111}$ and $\dot{c}_n^{100}|_{\text{uni}} = +\Gamma_{\text{uni}} c_n^{111}$, so this channel conserves both interstitial content and loop number.

The two binary channels start from the same diffusion-limited encounter problem. In the relative coordinate, with relative diffusivity $D_{\text{rel}} = D_n + D_{n'}$ and an absorbing surface of radius $R_c = R_n^c + R_{n'}^c$, the steady pair concentration satisfies $\nabla^2 C = 0$ with $C(R_c) = 0$ and $C(\infty) = C_\infty$, whose solution $C(r) = C_\infty(1 - R_c/r)$ integrates to the Smoluchowski coefficient $k_{nn'}^{(V)} = 4\pi R_c(D_n + D_{n'})$. Adopting the capture radius $R_n^c = 2\Omega^{1/3}\xi_n$ with $\xi_n = (3n/8\pi)^{1/3}$ and dividing by Ω to return to atomic fractions gives the same-character kernel

$$\mathcal{K}_{n,n'}^{ii} = \frac{8\pi}{\Omega^{2/3}} (\xi_n + \xi_{n'}) (D_n + D_{n'}). \quad (\text{S69})$$

This convention places the capture radius a factor $4^{1/3}$ above the equal-volume sphere radius $(3n\Omega/4\pi)^{1/3}$ that underlies the spherical prefactor A_{sph} of Section S2.

The Smoluchowski result assumes three-dimensional migration, which $\frac{1}{2}\langle 111 \rangle$ loops do not perform: they glide one-dimensionally along their Burgers vector and sample new directions only when that vector is reoriented. Section S2 treats the crossover through the master curve (S22). In its near-three-dimensional branch, $y \simeq x$, and with the spherical-sink strengths (S17) and (S18) the interpolation collapses to a single correlation factor on Eq. (S69),

$$\mathcal{K}_{n,n'}^{ii} = \frac{8\pi}{\Omega^{2/3}} (\xi_n + \xi_{n'}) \frac{D_n^{1D} + D_{n'}^{1D}}{[1 + B_{\text{rot}} (\ell_{nn'}/R_{nn'})^2]^{1/2}}, \quad B_{\text{rot}} = \frac{2}{9}, \quad (\text{S70})$$

where $R_{nn'} = 2\Omega^{1/3}(\xi_n + \xi_{n'})$ is the pair capture radius already used in Eq. (S69) and $\ell_{nn'}$ is the straight-line glide length of the relative motion, with $\ell_{nn'}^2 = \ell_n^2 + \ell_{n'}^2$ when both partners glide and $\ell_{nn'}^2 = \ell_n^2$ when only one does. The coefficient is not fitted. Sink density and sink radius cancel between $k_{(1)}^2 = 6\pi^2 R^4 N^2$ and $k_{(3)}^2 = 4\pi R N$, leaving $B_{\text{rot}} = R^2 k_{(3)}^4 / 12 k_{(1)}^2 = 2/9$ exactly, so the correction is a pure function of ℓ/R . It disappears when rotations are frequent, recovering Eq. (S69), and decays as ℓ^{-1} once the straight segment exceeds the capture radius, beyond which the correlated

one-dimensional kernels of Section S2 are the more accurate description [105, 124, 125].

Which regime applies is a property of the host rather than of the model. In high-purity iron the loops glide over distances far exceeding the obstacle spacing and their motion is genuinely one-dimensional [96], so the correlated kernels of Section S2 are the appropriate closure there. In the alloys of interest the picture is different: carbon and nitrogen trap a gliding cluster strongly, and the substitutional solutes act as weaker obstacles, so the Burgers vector is reoriented after a much shorter flight [97–99, 123, 126]. Treating the glider as sweeping a tube of cross section $\sigma_t = \pi b^2$ and taking one reorientation per effective obstacle encountered,

$$\ell \simeq \frac{\Omega}{\sigma_t C_{\text{eff}}}, \quad (\text{S71})$$

with C_{eff} the atomic fraction of effective obstacles. For EUROFER-97 the interstitial impurities alone, $c_{\text{C}} + c_{\text{N}} \approx 7 \times 10^{-4}$, give $\ell \approx 90$ nm, whereas counting every substitutional solute as an equally effective obstacle, $\approx 9 \times 10^{-3}$ with Mn and W included, gives $\ell \approx 7$ nm. We adopt $C_{\text{eff}} \approx 2.8 \times 10^{-3}$ — the interstitial impurities together with roughly one third of the substitutional solutes — and hence

$$\ell = 22 \text{ nm}, \quad \hat{L} = \ell/a = 77, \quad (\text{S72})$$

which is the value used throughout and tabulated in Table S12. With Eq. (S72) the correlation factor in Eq. (S70) is 19–33 for the junction channel, where both partners are small and mobile, and falls from 15 to 2 as the $\langle 100 \rangle$ host grows from $m = 10$ to $m = 10^4$ in the absorption channel. The suppression is therefore moderate and strongly size dependent rather than a uniform prefactor, and it is the size dependence that matters: large sessile loops are far less affected by one-dimensional correlation than the small mobile clusters they absorb.

The Burgers-vector reaction behind the junction channel is $\frac{1}{2}[111] + \frac{1}{2}[\bar{1}\bar{1}\bar{1}] \rightarrow [100]$ [95]. Not every encounter carries the required combination of Burgers vectors, geometry, impact parameter, and relative size, so the encounter budget of Eq. (S69) is split rather than augmented,

$$K_{nn'}^{111 \rightarrow 100} = \varphi_{\text{junc}} \mathcal{K}_{n,n'}^{ii}, \quad K_{nn'}^{111 \rightarrow 111} = [1 - \varphi_{\text{junc}}] \mathcal{K}_{n,n'}^{ii}, \quad (\text{S73})$$

so that the two branches sum to $\mathcal{K}_{n,n'}^{ii}$ and no encounter is counted twice. The Marian path passes through a metastable $\frac{1}{2}\langle 110 \rangle$ configuration from which

completion and reversal are independent activated events, $k_2 = \nu_2 e^{-\Delta H_2/k_B T}$ and $k_{\text{rev}} = \nu_r e^{-\Delta H_{\text{rev}}/k_B T}$. For two competing exponential waiting times the probability that completion occurs first is

$$P_{\text{succ}}(T) = \int_0^\infty k_2 e^{-k_2 t} e^{-k_{\text{rev}} t} dt = \frac{k_2}{k_2 + k_{\text{rev}}} = \left[1 + \frac{\nu_r}{\nu_2} e^{(\Delta H_2 - \Delta H_{\text{rev}})/k_B T}\right]^{-1}, \quad (\text{S74})$$

which for equal attempt frequencies reduces to the implemented gate $P_{\text{succ}} = [1 + e^{(\Delta H_2 - \Delta H_{\text{rev}})/k_B T}]^{-1}$. Only the barrier difference is identifiable from a scalar gate of this form; the individual barriers would become identifiable only if an absolute residence time in the intermediate were also used. The unresolved geometric compatibility is carried by a log-size Gaussian, symmetric under $n \leftrightarrow n'$, peaking at equal size and falling to $e^{-1/2}$ at $n/n' = e^{\pm\sigma_s}$, so that

$$\varphi_{\text{junc}}(n, n', T) = P_{\text{succ}}(T) \varphi_{\text{max}} e^{-\ln^2(n/n')/2\sigma_s^2} \Theta[\min(n, n') - n_j^{\text{min,eff}}], \quad (\text{S75})$$

$$n_j^{\text{min,eff}} = \max\left\{2, \min[n_j^{\text{min}}, \lceil f_j i_{\text{mobile}} \rceil]\right\}. \quad (\text{S76})$$

Equation (S76) is a numerical admissibility rule, not a result of the Marian atomistics: both junction partners must be mobile, so a fixed threshold above i_{mobile} would render the channel identically inaccessible. In mass-action form the product gain and reactant loss are

$$\dot{c}_p^{100}|_{\text{junc}} = \frac{1}{2} \sum_{n=1}^{p-1} K_{n,p-n}^{111 \rightarrow 100} c_n^{111} c_{p-n}^{111}, \quad \dot{c}_n^{111}|_{\text{junc}} = -c_n^{111} \sum_{n'} K_{nn'}^{111 \rightarrow 100} c_{n'}^{111}, \quad (\text{S77})$$

the factor $\frac{1}{2}$ preventing the double count of $(n, p-n)$ and $(p-n, n)$. Relabeling the double sums shows that interstitial content is conserved; loop number is not, since each junction replaces two loops with one.

The absorption channel uses the same encounter problem across characters,

$$K_{mn}^{\text{abs}} = P_{\text{succ}}^{\text{abs}}(T) \frac{8\pi}{\Omega^{2/3}} (\xi_m + \xi_n) (D_m^{100} + D_n^{1\text{D}}) = P_{\text{succ}}^{\text{abs}}(T) \frac{8\pi}{\Omega^{2/3}} \frac{(\xi_m + \xi_n) D_n^{1\text{D}}}{[1 + B_{\text{rot}}(\ell/R_{mn})^2]^{1/2}}, \quad (\text{S78})$$

the second form following because the host is sessile. The asymmetry is therefore one of mobility and product identity, not of size: the host size m

remains in the capture radius. The captured cluster must be incorporated into the character of a much larger host, and the same competing-hazard argument with an absorption-specific forward barrier and prefactor gives

$$P_{\text{succ}}^{\text{abs}}(T) = A_{\text{abs}} \frac{e^{-\Delta H_2^{\text{abs}}/k_B T}}{e^{-\Delta H_2^{\text{abs}}/k_B T} + e^{-\Delta H_{\text{rev}}/k_B T}} = \frac{A_{\text{abs}}}{1 + e^{(\Delta H_2^{\text{abs}} - \Delta H_{\text{rev}})/k_B T}}. \quad (\text{S79})$$

This remains a probability only if $0 \leq P_{\text{succ}}^{\text{abs}} \leq 1$. Since $\Delta H_2^{\text{abs}} > \Delta H_{\text{rev}}$ the fraction stays below one half at all temperatures, so $A_{\text{abs}} \leq 2$ preserves the interpretation; outside that range Eq. (S79) is an effective success multiplier and must be clipped at unity. Allowing $\Delta H_2^{\text{abs}} \neq \Delta H_2$ is a modeling hypothesis motivated by the Marian mechanism rather than a barrier reported for this coarse-grained reaction: junction formation nucleates a new $\langle 100 \rangle$ segment from two comparable loops, whereas absorption is templated by an existing $\langle 100 \rangle$ host, for which reversion should be less competitive. The channel is admissible for $m \geq n_{\text{loop}}^{\text{min}}$, $n \leq i_{\text{mobile}}$ and $m + n \leq I$, and the corresponding population terms are

$$\dot{c}_n^{111}|_{\text{abs}} = -c_n^{111} \sum_m K_{mn}^{\text{abs}} c_m^{100}, \quad \dot{c}_p^{100}|_{\text{abs}} = \sum_n K_{p-n,n}^{\text{abs}} c_{p-n}^{100} c_n^{111} - c_p^{100} \sum_n K_{pn}^{\text{abs}} c_n^{111}, \quad (\text{S80})$$

which conserve interstitial content. The bottom-boundary rule, by which a $\langle 100 \rangle$ loop shrinking below $n_{\text{loop}}^{\text{min}}$ returns to the $\frac{1}{2}\langle 111 \rangle$ population, is what preserves that conservation once the growth, shrinkage and emission edges also operate.

With the calibrated barriers of Table S6 the two gates take the values of Table S5. The absorption multiplier exceeds the junction completion probability by factors of 8.2, 7.6 and 6.4 at 300, 330 and 400 °C: both gates strengthen with temperature, but the junction gains faster, so the advantage of the templated route narrows. The realized junction yield is smaller still by φ_{max} and the size-compatibility factor.

Table S5: Completion factors from Eqs. (S74) and (S79) at the calibrated barriers.

T (°C)	T (K)	P_{succ}	$P_{\text{succ}}^{\text{abs}}$	ratio
300	573.15	0.0555	0.4577	8.25
330	603.15	0.0633	0.4794	7.57
400	673.15	0.0821	0.5245	6.38

Table S6: Inputs to Eqs. (S68), (S73) and (S78). “Calibration” marks a value not supplied directly by the cited elastic or atomistic theory.

Quantity	Symbol	Value	Basis
Attempt frequency	ν_0	10^{13} s^{-1}	phonon-scale prefactor
Unary barrier intercept	E_{a0}	2.10 eV	calibration
Barrier per line segment	γ_a	0.020 eV	calibration
Loop onset	$n_{\text{loop}}^{\min}$	4	population definition
Peak junction efficiency	$\varphi_{\max}$	0.10	calibration
Log-size width	σ_s	0.35	calibration
Nominal junction floor	$n_j^{\min}$	30	calibration
Mobility fraction	f_j	0.60	numerical admissibility
Junction completion barrier	ΔH_2	0.44 eV	calibration
Reverse barrier	ΔH_{rev}	0.30 eV	calibration
Absorption completion barrier	ΔH_2^{abs}	0.36 eV	calibration
Absorption prefactor	A_{abs}	2.0	calibration
Lattice parameter	a	2.867 Å	α -Fe data
Atomic volume	Ω	11.80 Å ³	α -Fe data, $\Omega = a^3/2$
Glide flight length	ℓ	22 nm	obstacle estimate, Eq. (S71)
Rotational-correlation coefficient	B_{rot}	2/9	derived, Eq. (S70)
Core cutoff	δ	4.0 Å	[94]
Softening temperature	T_c	1185 K	[94]
$\frac{1}{2}\langle 111 \rangle$ prelog factor	$\hat{F}_{111}$	0.28 eV/Å	anisotropic elasticity
$\langle 100 \rangle$ zero- T factor	$\hat{F}_{100}^0$	0.5182 eV/Å	derived, Eq. (S66)
$\frac{1}{2}\langle 111 \rangle$ core terms	$F_{\delta}^{111}, F_c^{111}$	0.345, 0.46 eV/Å	[94]
$\langle 100 \rangle$ core terms	$F_{\delta}^{100}, F_c^{100}$	0.387, 0.33 eV/Å	[94]
Crossover temperature	T^*	450 °C	calibration, Eq. (S66)
Reference size	n_{ref}	30	calibration, Eq. (S66)

The three kernels play distinct roles. Equation (S68) relabels an existing loop: the Dudarev theory supplies ΔF , while the barrier law and the one-way net-rate closure are mesoscale assumptions. Equation (S73) creates a new $\langle 100 \rangle$ loop from an encounter between two mobile $\frac{1}{2}\langle 111 \rangle$ loops; the Marian mechanism supplies the reaction path, while the log-size compatibility factor and the effective barriers coarse-grain the unresolved collision variables. Equation (S78) grows an existing sessile $\langle 100 \rangle$ loop by capture of a mobile cluster, and its separate completion gate represents templating by the host rather than nucleation. All three conserve interstitial content by

stoichiometry; the unary channel conserves loop number and the two binary channels reduce it by one per event. Keeping the thermodynamic input, the encounter kinetics and the calibrated branching factors separate in this way is what makes mechanism-ablation and uncertainty quantification possible.

S4. Mass Conservation Laws and the Swelling Identity

The master equations (26)–(28) satisfy two exact conservation identities. Their derivation rests on three facts that are built into the RADCLUSTER code:

- (i) cascade production deposits SIA and vacancy atoms in a balanced Frenkel-pair ratio, $\sum_{n=1}^{N_I} n P_n^{(i)} = \sum_{m=1}^{N_V} m P_m^{(v)} = \eta G$
- (ii) every mutual reaction $I_n + V_m$ destroys $\min(n, m)$ SIA atoms *and* $\min(n, m)$ vacancy atoms; the residual fragment (V_{m-n} if $m > n$, I_{n-m} if $n > m$) survives in its corresponding population. The two destruction sums therefore cancel identically in $d(S_I - S)/dt$;
- (iii) only mobile clusters with $n \leq n_{\max}^i$ (SIAs) or $m \leq m_{\max}^v$ (vacancies) reach fixed sinks. The corresponding rate-equation losses are summed over the full mobile range, not the monomer alone.

Cumulative loss to fixed sinks. The total fixed-sink rate for species $\alpha \in \{i, v, h\}$ collects all three sink channels, $\mathcal{D}_\alpha(n) = \mathcal{D}_\alpha^d(n) + \mathcal{D}_\alpha^{gb}(n) + \mathcal{D}_\alpha^p(n)$, and vanishes outside the mobile range. Define the cumulative atomic content delivered to fixed sinks:

$$\Delta J_i^{\text{fix}}(t) \equiv \int_0^t \sum_{n=1}^{n_{\max}^i} n \mathcal{D}_i(n) c_n(t') dt' \geq 0, \quad (\text{S81})$$

$$\Delta J_v^{\text{fix}}(t) \equiv \int_0^t \sum_{m=1}^{m_{\max}^v} m \mathcal{D}_v(m) \bar{c}_m(t') dt' \geq 0, \quad (\text{S82})$$

$$\Delta J_h^{\text{fix}}(t) \equiv \int_0^t \left[\mathcal{D}_h c_h + \mathcal{D}_v \sum_{m=1}^{m_{\max}^v} \sum_{\ell} \ell c_{m,\ell} \right] dt' \geq 0. \quad (\text{S83})$$

These three integrals are appended to the state vector and evolved as auxiliary ODEs, so that they share the integrator's local error control with the physical concentrations.

Frenkel-pair balance. With total inventories

$$S_I(t) \equiv \sum_{n=1}^{N_I} n c_n(t), \quad S(t) \equiv \sum_{m=1}^{N_V} m \bar{c}_m(t), \quad (\text{S84})$$

facts (i)–(iii) combine into the exact Frenkel-pair conservation law

$$S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t) = S(0) - S_I(0) = \text{const.} \quad (\text{S85})$$

For a defect-free initial condition, this is the *swelling identity*

$$S(t) = S_I(t) + \Delta J_i^{\text{fix}}(t) - \Delta J_v^{\text{fix}}(t), \quad (\text{S86})$$

i.e. the total vacancy content of all cavities equals the SIA inventory in loops plus the *net* atomic flux delivered to fixed sinks. Crucially, the SIA flux to fixed sinks [ΔJ_i^{fix} , Eq. (S81)] sums the contributions of *all* mobile SIA cluster sizes $n = 1, \dots, n_{\text{max}}^i$, and the vacancy flux [ΔJ_v^{fix} , Eq. (S82)] sums the contributions of *all* mobile vacancy cluster sizes $m = 1, \dots, m_{\text{max}}^v$, not just the monomer terms. Differentiating gives the rate form

$$\frac{dS}{dt} = \frac{dS_I}{dt} + \sum_{n=1}^{n_{\text{max}}^i} n \mathcal{D}_i(n) c_n - \sum_{m=1}^{m_{\text{max}}^v} m \mathcal{D}_v \bar{c}_m. \quad (\text{S87})$$

Once loops saturate ($dS_I/dt \approx 0$), swelling grows at the rate at which the biased dislocation channel ($Z_i^d \neq Z_v^d$) drains the surplus SIA inventory faster than the vacancy inventory; the unbiased grain-boundary and precipitate channels contribute equally to both sums and cancel in $dS/dt - dS_I/dt$ only insofar as the mobile SIA and vacancy populations are equal.

Helium conservation. The total helium inventory is the sum of free and trapped helium,

$$S_{\text{He}}(t) \equiv c_h(t) + \sum_{m=1}^{N_V} \sum_{\ell=1}^{\ell_{\text{max}}(m)} \ell c_{m,\ell}(t). \quad (\text{S88})$$

Trap mutation (P7), radiation re-solution (P8), and thermal He emission (P5h) only redistribute helium between c_h and the trapped pool, so their contributions to dS_{He}/dt cancel pairwise. The only net helium sinks are free

helium captured at fixed sinks ($\mathcal{D}_h c_h$) and helium carried to fixed sinks by mobile cavities ($m \leq m_{\max}^v$); immobile bubbles retain their helium. Hence

$$\frac{dS_{\text{He}}}{dt} = G_{\text{He}} - \mathcal{D}_h c_h - \mathcal{D}_v \sum_{m=1}^{m_{\max}^v} \sum_{\ell} \ell c_{m,\ell}, \quad (\text{S89})$$

i.e. $dS_{\text{He}}/dt = G_{\text{He}} - d\Delta J_h^{\text{fix}}/dt$. In particular, S_{He} is *not* conserved against $\int G_{\text{He}} dt'$ alone — the fixed-sink loss must be tracked explicitly. Both loss terms are typically small (mobile clusters carry little He) but they are non-zero and must appear in the diagnostic below for it to converge to zero.

Numerical accuracy diagnostics. The swelling identity (S86) and the helium balance (S89) provide two inexpensive a-posteriori checks, neither of which requires post-processing because the auxiliary integrals (S81)–(S83) are already state variables:

$$\delta_{\text{FP}}(t) = \frac{|S(t) - S_I(t) - \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)|}{S(t) + S_I(t) + \Delta J_i^{\text{fix}}(t) + \Delta J_v^{\text{fix}}(t)}, \quad (\text{S90})$$

$$\delta_{\text{He}}(t) = \frac{\left| S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) - \int_0^t G_{\text{He}} dt' - S_{\text{He}}(0) \right|}{S_{\text{He}}(t) + \Delta J_h^{\text{fix}}(t) + \int_0^t G_{\text{He}} dt'}. \quad (\text{S91})$$

Both should remain below $\sim 10^{-6}$ in a correctly implemented code; values above 10^{-3} almost always signal a coding error, missing or incorrect reactions, or an overly loose integrator tolerance. Despite the importance of these conservation identities, they have seldom been discussed or tested in the cluster dynamics literature.

S5. Conditions for Cavity and Loop Growth

Equations (26) and (27) decide at every node whether a cluster advances or retreats along its ladder; in a ferritic–martensitic steel that decision collapses to one number against two thresholds. Collect the mobile inventories into

$$\Phi_I = \sum_{n=1}^{i_{\text{mobile}}} n D_n c_n, \quad \Phi_V = \sum_{m=1}^{v_{\text{mobile}}} m D_m c_{m,0}, \quad x \equiv \Phi_V / \Phi_I. \quad (\text{S92})$$

Bcc iron is diffusionally cubic, so every D_α is a scalar and the capture efficiencies introduced above carry no orientation factor: the entire species asymmetry of the model is elastic, and it sits at two sinks only, $Z_i^d > 1 = Z_v^d$ and $Z_i^{\text{loop}} > 1 = Z_v^{\text{loop}}$, cavities, lath boundaries and precipitates being unbiased. Requiring the net capture rate, less thermal emission, to be positive then gives one threshold per family,

$$\underbrace{x < Z_i^{\text{loop}}}_{\text{both loop characters grow}} \qquad \underbrace{x > 1 + \varepsilon_{m,\ell}}_{\text{cavity } (m,\ell) \text{ grows}} \qquad (\text{S93})$$

with $\varepsilon_{m,\ell} = D_v c_v^{\text{th}}(m, \ell) / \Phi_I$ the emission flux of that cavity reduced by Φ_I . Loops and cavities are of opposite character and so do not share a threshold: Eq. (S93) holds simultaneously throughout $1 + \varepsilon_{m,\ell} < x < Z_i^{\text{loop}}$, a window the elastic bias alone opens and emission closes from below — the critical-radius statement in flux-ratio form — and that helium widens by lowering $\varepsilon_{m,\ell}$.

Whether x lies inside it is decided by the production bias. Only the unclustered part of each species migrates freely, so with f_i^{cl} and f_v^{cl} the incascade clustered fractions, k_L^2 the loop sink strength, and $k_I^2/k_V^2 = 1 + [(Z_i^d - 1)\rho_d + (Z_i^{\text{loop}} - 1)k_L^2]/k_V^2$ the sink-strength ratio, the quasi-steady balance gives

$$x \simeq \eta \frac{k_I^2}{k_V^2}, \quad \eta = \frac{1 - f_v^{\text{cl}}}{1 - f_i^{\text{cl}}},$$

$$\text{so } x < Z_i^{\text{loop}} \iff f_i^{\text{cl}} - f_v^{\text{cl}} < \left(Z_i^{\text{loop}} k_V^2 / k_I^2 - 1 \right) (1 - f_i^{\text{cl}}) \simeq 0.04, \quad (\text{S94})$$

so the window's upper edge is a condition on cascade statistics alone, which iron fails by an order of magnitude: Table S11 gives $f_i^{\text{cl}} - f_v^{\text{cl}} \simeq 0.43$ under a fission spectrum ($1 - f_i^{\text{cl}} = 0.42$ against $1 - f_v^{\text{cl}} = 0.85$) and 0.45 under a fusion one, hence $\eta \simeq 2.0$ –2.3, while the sink ratio exceeds unity by only a few percent at the ρ_d and biases of Table S13. Note where Z_i^d sits: opposite Z_i^{loop} , raising x and so driving the system out of the loop window and above the cavity threshold — it is the classical swelling drive — while Z_i^{loop} raises that window's ceiling. Being one elastic interaction evaluated at two sinks, they cannot be calibrated independently, and neither offsets η .

Were absorption the only channel, no loop could then grow. But Eq. (S93) is not the loop budget: the clustered fraction f_i^{cl} is deposited directly as

loops by G_n , and loops coarsen by coalescence, neither depending on x . One asymmetry thus does both jobs: the excess $f_i^{\text{cl}} - f_v^{\text{cl}}$ that drives x out of the window supplies the loops as nascent clusters and leaves the vacancy surplus that grows the cavities and produces the swelling identity above. Co-growth here is a production-bias phenomenon, not a shared-window one.

S6. Reduced Helium-Occupancy Models

The two reductions of the cavity state space (m, ℓ) summarized in the main text are derived below: fast helium equilibration under fusion conditions and a decoupled helium inventory under fission conditions. Throughout, the notation and rate constants of Sections S2 and 3 are used.

In the fusion regime, at temperatures where vacancies are mobile ($T \gtrsim 200^\circ\text{C}$), the He reaction frequency greatly exceeds the vacancy reaction frequency $\omega_h^{\text{eff}} \gg \omega_v^{\text{eff}}$. Helium equilibrates on each cavity faster than the cavity grows or shrinks, so the full two-dimensional (m, ℓ) distribution can be collapsed into two scalar quantities per vacancy size: the total cavity concentration $\bar{c}_m$ and the mean He loading

$$\bar{\ell}(m, t) \equiv \frac{\sum_{\ell} \ell c_{m, \ell}}{\bar{c}_m}. \quad (\text{S95})$$

Summing Eq. (27) over all ℓ at fixed m :

$$\frac{d\bar{c}_m}{dt} = J_{m-1 \rightarrow m} - J_{m \rightarrow m+1} - \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{ic} c_n \bar{c}_m + \Gamma_{\text{TM}}(m-1, \bar{\ell}) \bar{c}_{m-1} - \Gamma_{\text{TM}}(m, \bar{\ell}) \bar{c}_m, \quad (\text{S96})$$

where $J_{m \rightarrow m+1}$ is the net vacancy flux from cavity size m to $m+1$. The SIA-shrinkage sum $\sum_n \mathcal{K}_{n,m}^{ic} c_n$ spans all mobile SIA clusters $n = 1, \dots, i_{\text{mobile}}$. The vacancy capture rate $A_{\text{sph}} m^{1/3} \omega_v^{\text{eff}}$ is independent of ℓ , so the ℓ -summation is exact for the capture terms. The emission rate ε_v depends on ℓ through the He gas-pressure correction to $E_b^v(m, \ell)$ and is approximated by evaluation at the mean loading $\bar{\ell}(m)$.

Define the He inventory at vacancy size m as $N_{\text{He}}(m) \equiv \bar{\ell}(m) \bar{c}_m$. Multi-

plying Eq. (27) by ℓ , summing over ℓ , and collecting terms:

$$\begin{aligned}
\frac{dN_{\text{He}}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h \bar{c}_m - \varepsilon_h(m, \bar{\ell}) \bar{c}_m - b_0 \dot{\phi} N_{\text{He}}(m) \\
& + \bar{\ell}(m-1) J_{m-1 \rightarrow m}^+ - \bar{\ell}(m) J_{m \rightarrow m+1}^+ \\
& + \bar{\ell}(m+1) J_{m+1 \rightarrow m}^- - \bar{\ell}(m) J_{m \rightarrow m-1}^- \\
& - \bar{\ell}(m) \sum_{n=1}^{i_{\text{mobile}}} \mathcal{K}_{n,m}^{ic} c_n \bar{c}_m - \mathcal{D}_v N_{\text{He}}(m) \delta_{m \leq v_{\text{mobile}}}.
\end{aligned} \tag{S97}$$

The five groups of terms are: (i) He captured from the free pool minus thermal He emission (P5) minus radiation re-solution (P8); (ii–iii) He is advected in and out as clusters grow or shrink along the m -axis, where J^+ is the forward (capture) component and J^- is the backward (emission) component of the vacancy flux; (iv) He is carried away when SIA clusters annihilate vacancies; and (v) He is lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks.

Applying the quotient rule $d\bar{\ell}/dt = (\dot{N}_{\text{He}} - \bar{\ell} \dot{\bar{c}}_m)/\bar{c}_m$ and substituting Eq. (S96):

$$\begin{aligned}
\frac{d\bar{\ell}(m)}{dt} = & A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}} c_h - \varepsilon_h(m, \bar{\ell}(m)) - b_0 \dot{\phi} \bar{\ell}(m) \\
& + \frac{J_{m-1 \rightarrow m}^+}{\bar{c}_m} [\bar{\ell}(m-1) - \bar{\ell}(m)] + \frac{J_{m+1 \rightarrow m}^-}{\bar{c}_m} [\bar{\ell}(m+1) - \bar{\ell}(m)].
\end{aligned} \tag{S98}$$

Line 1 contains the local He kinetics: trapping from the free pool (proportional to c_h), thermal emission ε_h (P5), and athermal re-solution (P8), which drives $\bar{\ell}$ toward zero. Line 2 contains advection corrections: cavities flowing into size class m carry their own He loading, which may differ from $\bar{\ell}(m)$; the difference $[\bar{\ell}(m \pm 1) - \bar{\ell}(m)]$ acts as a drift source.

The second reduction applies at the opposite extreme. Under fission irradiation, the He generation rate is low (~ 0.3 – 1 appm/dpa vs. ~ 10 – 15 appm/dpa for fusion), so most cavities carry $\ell \ll m$ and the He/V ratio $\bar{\ell}/m \ll 1$. Helium acts as a weak perturbation on the void kinetics rather than controlling nucleation. Rather than tracking $\bar{\ell}(m)$ as a separate variable at each size, the total trapped He inventory is followed as a single scalar and distributed algebraically across cavity sizes. The $\bar{c}_m$ equation is identical to Case 1, Eq. (S96), with the vacancy emission rate evaluated using a He-corrected

binding energy. In the low-loading limit, the He gas pressure provides an effective binding energy.

Define $\ell_{\text{tot}} \equiv \sum_{m=1}^M \ell(m) \bar{c}_m$. Summing the He content over all cavities and using Eq. (28) with thermal emission dropped:

$$\frac{d\ell_{\text{tot}}}{dt} = A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m - b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_v \sum_{m=1}^{v_{\text{mobile}}} \ell(m) \bar{c}_m. \quad (\text{S99})$$

The first term is the total He trapping rate by all cavities (the sum is the weighted capture cross-section $\sum_m m^{1/3} \bar{c}_m$), the second is the athermal re-resolution loss (P8), and the third is He lost when mobile vacancy clusters ($m \leq v_{\text{mobile}}$) are absorbed at fixed sinks. Helium is then distributed across cavity sizes in proportion to the capture rate $A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$, as that determines the steady-state He loading when trapping and re-resolution balance.

$$\ell(m) = \ell_{\text{tot}} \frac{m^{1/3} \bar{c}_m}{\sum_{m'=1}^M m'^{1/3} \bar{c}_{m'}}. \quad (\text{S100})$$

This algebraic relation is recomputed at each time step. The $m^{1/3}$ weighting arises from the spherical-cavity capture radius $\xi(m) \propto m^{1/3}$, consistent with $\mathcal{K}_{h,m} = A_{\text{sph}} m^{1/3} \omega_h^{\text{eff}}$.

The free-He equation retains all terms from Eq. (28), including the fixed-sink loss $\mathcal{D}_h c_h$, but with thermal He emission dropped:

$$\frac{dc_h}{dt} = G_{\text{He}} - A_{\text{sph}} \omega_h^{\text{eff}} c_h \sum_{m=1}^M m^{1/3} \bar{c}_m + b_0 \dot{\phi} \ell_{\text{tot}} - \mathcal{D}_h c_h. \quad (\text{S101})$$

The trapping and re-resolution terms are exactly the negatives of those in Eq. (S99), so the He conservation law $d(c_h + \ell_{\text{tot}})/dt = G_{\text{He}} - \mathcal{D}_h c_h$ holds.

S7. Closure of the Bin-Moment Equations

The reconstruction step **R** of the bin-moment scheme requires a deterministic map from the tracked moments $(\mu_k^{(0)}, \mu_k^{(1)})$ back to per-size concentrations c_n for $n \in \mathcal{B}_k$. One closure is used throughout this work and is the default in **RadCluster**: the linear (hat-function) dual basis with $P = 2$ moments per bin, quoted in the main text as Eqs. (42)–(44). This section gives its construction, places it in the Galerkin framework from which it follows, and establishes its conservation and truncation-error properties.

Construction of the linear dual basis. With two tracked moments per bin, the reconstruction is expanded in two basis functions $(\phi_{k,0}(n), \phi_{k,1}(n))$,

$$c_n \approx \phi_{k,0}(n) \mu_k^{(0)} + \phi_{k,1}(n) \mu_k^{(1)}, \quad n \in \mathcal{B}_k. \quad (\text{S102})$$

The pair is the linear *dual basis* on the discrete index set $\mathcal{B}_k$, i.e. the unique pair satisfying

$$\sum_{n \in \mathcal{B}_k} \phi_{k,0}(n) = 1, \quad \sum_{n \in \mathcal{B}_k} \phi_{k,1}(n) = 0, \quad \sum_{n \in \mathcal{B}_k} n \phi_{k,0}(n) = 0, \quad \sum_{n \in \mathcal{B}_k} n \phi_{k,1}(n) = 1. \quad (\text{S103})$$

Solving these constraints for the linear ansatz $\phi_{k,p}(n) = \alpha_p + \beta_p n$ gives the closed form used in the code:

$$\phi_{k,0}(n) = \frac{S_2 - S_1 n}{\mathcal{N}_k}, \quad \phi_{k,1}(n) = \frac{\Delta_k n - S_1}{\mathcal{N}_k}, \quad (\text{S104})$$

where the geometry integrals are

$$S_1 \equiv \sum_{n \in \mathcal{B}_k} n, \quad S_2 \equiv \sum_{n \in \mathcal{B}_k} n^2, \quad \mathcal{N}_k \equiv \Delta_k S_2 - S_1^2 > 0, \quad (\text{S105})$$

and are computed once at setup. Equations (S104) and (S105) are evaluated on the *integer* grid, so $(\phi_{k,0}, \phi_{k,1})$ reduces to the piecewise-linear FEM hat functions only in the continuum limit $\Delta_k \rightarrow \infty$. Negative reconstructed values are clamped to zero before use in step **E**; this clamp breaks exact moment preservation only for strongly-skewed bins. A bin of unit width, $\Delta_k = 1$, makes $\mathcal{N}_k$ vanish and the reconstruction degenerates, by construction, to the single-size assignment $c_{s_k} = \mu_k^{(0)}$.

Galerkin formulation. The closure is the $P = 2$ member of the Galerkin projection of the master equation (26) onto basis functions $\{\phi_p\}_{p=0}^{P-1}$ defined on $\mathcal{B}_k$:

$$\sum_{n \in \mathcal{B}_k} \phi_p(n) \frac{dc_n}{dt} = \sum_{n \in \mathcal{B}_k} \phi_p(n) f_n(\mathbf{c}), \quad p = 0, 1, \dots, P-1, \quad (\text{S106})$$

with f_n the right-hand side of (26). Taking $P = 2$ and choosing $\phi_p(n) = n^p$, $p = 0, 1$, as the test functions yields the moment projection (45) used in the code; the trial functions enter only through the reconstruction step **R**.

Mass conservation. Because the projection (45) sums the *exact* per-size rates, the bin-moment system inherits the Frenkel-pair and helium-balance identities of the full per-size system without modification. In particular, the swelling identity (Sec. S4) holds exactly at the level of the moment ODE system, with S_I and S assembled as the discrete + first-moment sums introduced after Eq. (41).

Truncation errors. Closure error enters only through the reconstructed c_n in step **R**. For a smooth, slowly varying distribution $f(n) = c_n$ inside a bin and any reaction kernel $\mathcal{K}(n)$ that is smooth on the same scale, the linear dual basis gives

$$\sum_{n \in \mathcal{B}_k} \mathcal{K}(n) c_n - \sum_{n \in \mathcal{B}_k} \mathcal{K}(n) c_n^{\text{rec}} = O((r-1)^3), \quad (\text{S107})$$

so that the per-bin rate-constant error is $\sim 12\%$ at $r = 2$ and $\sim 1.5\%$ at $r = 1.5$. Together with the linear growth of the ODE count with I_{bin} , this makes $r \approx 1.5$ – 2 the best accuracy/cost compromise, and it is the default.

S8. Numerical Validation

The bin-moment reduction of Section S7 replaces a discrete master equation over cluster sizes of 10^4 – 10^5 with a hybrid representation in which sizes up to i_{discrete} are carried individually and the remainder is partitioned into logarithmically spaced bins I_{bin} that carry only the zeroth and first moments, the intra-bin distribution being reconstructed from the linear dual-basis closure; the vacancy axis is treated identically with v_{discrete} and V_{bin} . The economy is substantial, but it is only useful if the reduced solution reproduces the discrete one. Two distinct questions follow. The first is internal consistency: do the observables cease to move as the numerical parameters of the reduction are refined? The second and more demanding one is the accuracy: does the reduced solution agree with the exact discrete solution of the same equations? This section answers them in turn and then isolates the single parameter that governs the one residual discrepancy the first two leave behind.

Unless stated otherwise, every calculation reported below shares the EUROFER97 host declaration of the main text, a fission cascade spectrum, a displacement rate $G = 10^{-7}$ dpa s $^{-1}$, an irradiation temperature of 330 °C, GMRES with the Woodbury preconditioner, an absolute integration tolerance of 10^{-20} and mobility cutoffs $i_{\text{mobile}} = v_{\text{mobile}} = 5$. Six observables are

scored in every configuration: the number densities N_{111} , N_{100} and N_{cav} and the mean diameters d_{111} , d_{100} and d_{cav} of the two loop families and of the cavity population. The Frenkel-pair residual δ_{FP} is reported along with them as a diagnostic of integration rather than as a measure of closure error; the two are distinct quantities and are not conflated here.

Where a comparison with measurement is meaningful it is made against the EUROFER97 neutron irradiation data at 300–350 °C, which span 0.30–34.0 and $2.24\text{--}12.0 \times 10^{21} \text{ m}^{-3}$ in $\frac{1}{2}\langle 111 \rangle$ and $\langle 100 \rangle$ loop density, 3.3–35 and 2.8–48 nm in the corresponding mean diameters, and $0.32\text{--}6.30 \times 10^{21} \text{ m}^{-3}$ and 1.6–16 nm for cavities [87, 91–93, 127–129]. The dose coverage of that compilation governs where the comparison can be drawn at all: the loop measurements begin at 14.6 dpa, so the calculations carried to the production horizon are scored at 15.72 dpa, while the verification calculations of Section S8, which are integrated only to 2 dpa, cannot be set against loop data. One $\langle 100 \rangle$ row reports a mean diameter of 48 nm at 17.4 dpa and 350 °C against 2.8–8.0 nm for the other nine rows, with an unremarkable density; the full range is retained in the text but that single diameter is excluded from the figures, where it would set the ordinate on its own.

Sensitivity to the parameters of the reduction. The first requirement of a converged reduction is that refining its own resolution parameter changes nothing. Refining the discrete core at the production domain, $I = 80\,000$ and $V = 20\,000$, with the bin count adjusted to hold the bin ratio approximately constant as the core contracts, satisfies it: as i_{discrete} falls from 1600 to 400 to 100 the system shrinks by an order of magnitude, from 2048 to 562 to 189 equations, while d_{100} spans only 5.868–6.001 nm and N_{111} only $3.449\text{--}3.512 \times 10^{21} \text{ m}^{-3}$. The cavity population is the least settled of the three: at the coarsest core N_{cav} and d_{cav} stand at $3.609 \times 10^{20} \text{ m}^{-3}$ and 2.890 nm against 2.233 and 2.232 at the two finer cores, a larger excursion than any loop observable shows.

Taking the 189-equation configuration as the reference and varying one parameter at a time separates the remaining numerical degrees of freedom (Table S7). Two orders of magnitude in relative integration tolerance, from 10^{-4} to 10^{-6} , move no observable in any reported digit, so the tolerance is not a free parameter concealed within the results. Reducing the bin count from 18 to 10 displaces the observables by a few percent at most, consistent with the insensitivity to the discrete core established above.

A reduction in physics rather than numerics can be examined in the same

Table S7: One parameter varied at a time from the reference configuration ($I = 80\,000$, $V = 20\,000$, $i_{\text{discrete}} = 100$, $I_{\text{bin}} = 18$, linear closure), scored at 15.72 dpa. Densities are in units of 10^{21} m^{-3} for the loops and 10^{20} m^{-3} for the cavities, diameters in nm.

Configuration	N_{eq}	N_{111}	d_{111}	N_{100}	d_{100}	N_{cav}	d_{cav}	δ_{FP}
Reference	189	3.498	4.637	4.715	5.939	3.609	2.890	0.060
$I_{\text{bin}} = 10$	171	3.543	4.282	4.021	5.861	2.900	2.340	0.070
$\text{rtol} = 10^{-4}$	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
$\text{rtol} = 10^{-6}$	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
<i>Measured range</i>	—	0.30–34.0	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

way. Replacing the dynamic integration of the helium sub-system by its quasi-steady-state form at otherwise identical settings changes 3.533, 4.354, 4.153, 5.919, 3.253 and 2.434 into 3.533, 4.354, 4.153, 5.919, 3.253 and 2.434: the two treatments agree in every digit, and the reduction may be used under fission without consequence for any of the six observables. The result should be read as it is. Under fission, the reference calculation holds a total helium concentration of $1.18 \times 10^{13} \text{ m}^{-3}$, so there is almost nothing to integrate in time, and the agreement is equally consistent with a helium treatment that does not matter and with a comparison never sensitive to it. The safety of the reduction is established at fission helium levels only; the corresponding statement under fusion generation rates would require a separate calculation and is not made here.

Verification against the exact discrete solution. The calculations above compare reduced solutions with each other. Verification proper requires the exact solution of the same equations, which is obtainable only on a domain small enough for the fully discrete system to be integrated; $I = V = 4\,000$ is such a domain, and Table S8 reports deviations from the discrete arm on it at 2 dpa.

Two restrictions define what the comparison can support. Both arms are run with the $\frac{1}{2}\langle 111 \rangle$ loop–loop coarsening channel disabled, so that the two solvers integrate identically the same equations; the closure error measured here is therefore that of the no-coarsening variant rather than of the production configuration, in which the channel is active and costs $O(N_C^2)$ in the populated sessile sizes. With the channel disabled the $\frac{1}{2}\langle 111 \rangle$ mean size pins at the mobility cutoff and its density exceeds the measurements by one to two orders of magnitude, so no entry in the table may be read against ex-

periment. Neither restriction bears on the verification itself, which asks only whether the reduction reproduces the exact solution of whatever equations both arms solve.

Table S8: Verification against the exact discrete solution at $I = V = 4000$, coarsening channel disabled in both arms, scored at 2.0000 dpa. Parenthesised numbers are percentage deviations from the discrete run D. N_{eq} is the main block; *total* adds the appended $\langle 100 \rangle$ loop block that the integrator also carries. Densities as in Table S7.

Rung	N_{eq}	total	N_{111}	N_{100}	d_{100}	N_{cav}	d_{cav}
D (exact)	8006	12006	12.568	1.065	5.741	2.948	2.544
C1 (1/10)	830	1242	12.335 (−1.9)	1.044 (−2.0)	5.690 (−0.9)	2.555 (−13.3)	2.273 (−10.6)
C2 (1/40)	244	364	12.144 (−3.4)	1.044 (−2.0)	5.681 (−1.0)	2.574 (−12.7)	2.279 (−10.4)
C3 (1/160)	108	161	7.027 (−44.1)	1.018 (−4.4)	5.349 (−6.8)	3.219 (+9.2)	2.451 (−3.6)

The linear closure holds d_{100} to within 1 percent at the two finer rungs, from 830 equations down to 244 — a thirtyfold reduction in cost for a one-percent error — and N_{100} to 2 percent there, degrading only to 4.4 percent at the coarsest rung. The exceptions are informative rather than incidental. N_{111} degrades sharply below C2, by 44 percent at C3: the $\frac{1}{2}\langle 111 \rangle$ population is peaked at small sizes, and once i_{discrete} falls to 25 that peak lies inside the binned region, where the closure represents it rather than resolving it. d_{100} follows the same trend more mildly, losing 6.8 percent at that rung. The cavity observables behave in the opposite sense: N_{cav} is low by some 13 percent at C1 and C2 and crosses to +9.2 percent at C3, while d_{cav} recovers from −10.6 percent to −3.6 percent as the grid coarsens. The pattern is consistent across the ladder — the coarse grids buy their speed at real cost in the loop density while the mean loop size remains accurate.

Convergence along this ladder is not monotonic. C1 and C2 are nearly identical, C3 is markedly worse than C2 in both N_{111} and d_{100} , yet C3 is the only rung that improves on the others in d_{cav} . A ladder that does not improve monotonically is not by itself evidence of a converged closure, and the likeliest reason is that the rungs vary two parameters at once: they hold the bin ratio fixed, but the boundary between the discrete and binned regions moves relative to the peak of the distribution as i_{discrete} contracts. Section S8 separates the two axes and confirms this.

Dose dependence along the verification ladder. Figures S1 to S5 carry the same ladder through dose, with the measurements and the band spanning

them overlaid. The trajectories terminate at two different doses by construction: the coarse grids were integrated to 40 dpa in order to reach the measured range, while the exact arm and its nearest reduced neighbour stop at 2 dpa, beyond which the discrete system could not be carried at any affordable cost. The consequence is worth stating plainly, because it governs how the figures should be read. Below 2 dpa the coarse grids may be judged against the exact solution; above it they may not, and where the extended trajectories cross the experimental band that agreement speaks to the model rather than to the reduction. The $\frac{1}{2}\langle 111 \rangle$ mean diameter is omitted from the figures: with the coarsening channel disabled it is pinned at the mobility cutoff and carries no information about the closure.

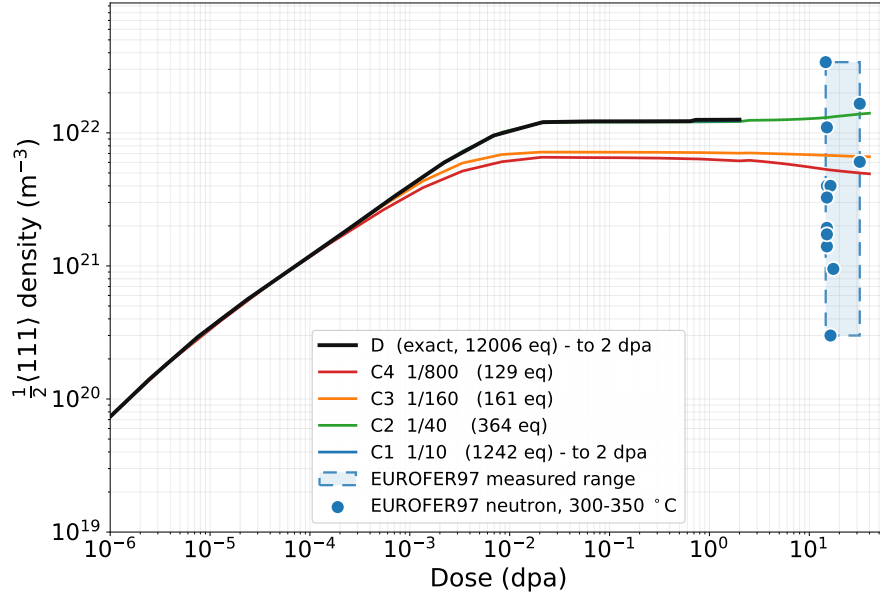


Figure S1: $\frac{1}{2}\langle 111 \rangle$ loop number density against dose. The exact solution separates from the coarse grids from 0.1 dpa onwards, the two coarsest losing approximately half of the population.

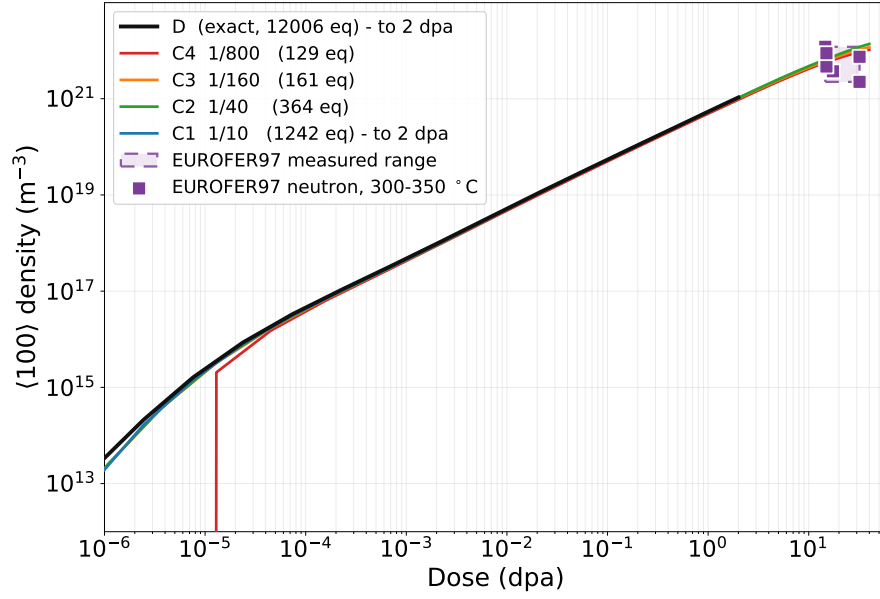


Figure S2: $\langle 100 \rangle$ loop number density against dose. All grids track the exact solution closely; this is the density observable the reduction represents best.

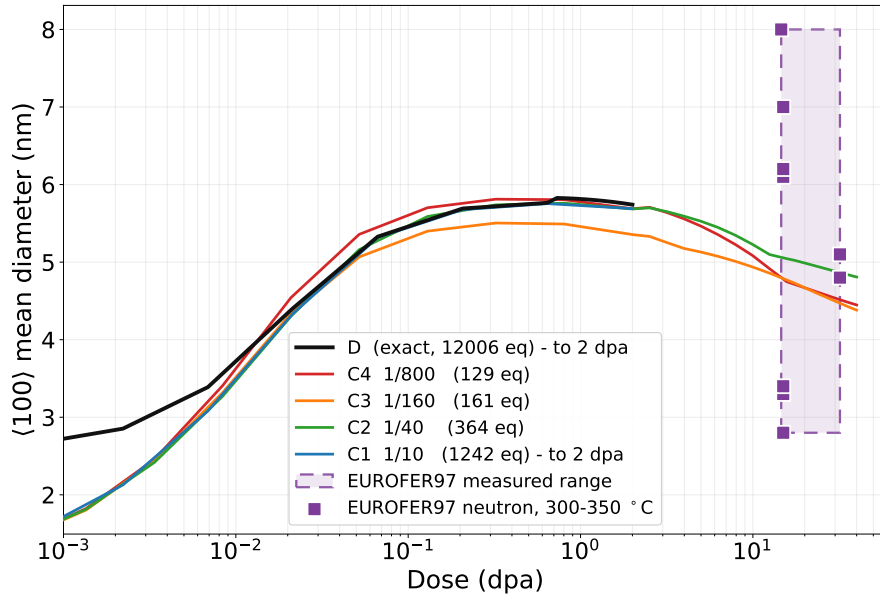


Figure S3: $\langle 100 \rangle$ mean diameter, held to about one percent across the whole ladder. The single 48 nm measurement discussed above is excluded.

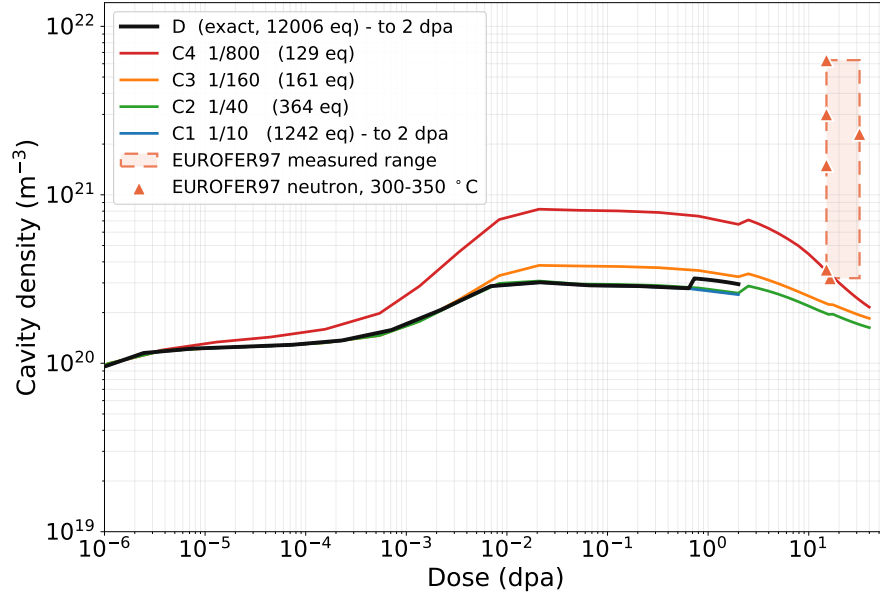


Figure S4: Cavity number density, the observable most sensitive to the extent of the discrete core, with the coarsest grid lying more than a factor of two above the exact solution.

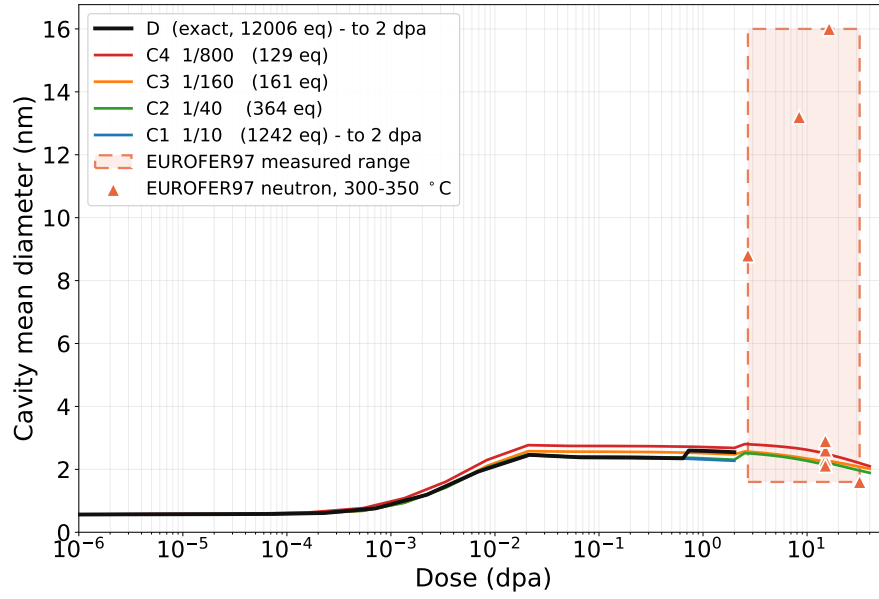


Figure S5: Cavity mean diameter. The cavity measurements extend down to 2.7 dpa, so this is the one panel in which the measured band and the calculated trajectories nearly meet.

Separating the discrete core from the bin count. The verification ladder moved i_{discrete} and I_{bin} together, so a deviation along it cannot be attributed to either. A second group of calculations separates them, varying the domain with no closure at all, then the bin count at a fixed discrete core, and then both independently, so that every configuration has a neighbor differing in one axis alone. All run in production mobility, under a fission cascade spectrum, on one shared 45-point logarithmic grid to 20 dpa.

The reference domain is necessarily small, and the cost of the exact arm is the reason. At $I = V = 1000$ the fully discrete system of 2006 equations completes 20 dpa in 5.2 minutes; at $I = V = 2000$ it reaches 0.038 dpa of the same target in 61 minutes; at $I = V = 10\,000$ it diverges outright, the per-step cost growing roughly four times per output step at 104 per CPU and , therefore, entirely within the serial region. That region is the sum of pairs $O(N_C^2)$ of the coarsening channel, whose participation gate admits a sessile population ten times larger at $I = 10\,000$ than at $I = 1000$, so the pair count is a hundred times higher. The exact reference is confined to small domains for this and no other reason. One consequence is that at $V = 1000$ a cavity is capped at about 2.8 nm and the exact arm returns $d_{\text{cav}} = 1.74$ nm at 20 dpa, 62 percent of that ceiling; both arms sit on the same truncated domain, so the truncation cancels from the measured closure error , but nothing in this group can be compared with experiment.

Varying only the bin count at a fixed discrete core of 50 gives Table S9. Five of the six observables converge, and they converge quickly: by thirteen bins, where 158 equations stand against 2006 for the exact arm, N_{100} , d_{100} , N_{cav} , and d_{cav} all lie within 0.3 percent of the exact solution and N_{111} has plateaued at a residual +1.0 percent. Figure S6 shows the approach for the $\langle 100 \rangle$ and cavity observables directly.

The sixth observable does not converge. The $\frac{1}{2}\langle 111 \rangle$ mean diameter remains between -5 and -6.6 percent throughout and does not improve with refinement; the finest of the nine rungs is the worst of them. Sweeping both parameters independently identifies the reason (Table S10): the error falls with the extent of the discrete core and is nearly flat in the bin count. Read down the eight-bin column the deviations are -6.3 , -3.4 , -2.0 , -0.1 and $+0.7$ percent; read across the row at $i_{\text{discrete}} = 100$ they are -3.4 , -4.2 and -5.1 , worsening with every bin added. At $i_{\text{discrete}} = 800$ the deviation is better than one percent at every bin count. The axis that governs this error is therefore not the one the bin sweep varied, which is why nine rungs of refinement at a fixed core of 50 could not remove it.

Table S9: Deviation from the exact arm (percent) at 20 dpa, $I = V = 1000$, with $i_{\text{discrete}} = v_{\text{discrete}} = 50$ held fixed throughout, so that only the bin count and the resulting bin ratio r vary.

I_{bin}	r	N_{eq}	N_{111}	d_{111}	N_{100}	d_{100}	d_{cav}
2	4.47	114	+3.7	-4.4	+1.2	-0.6	-2.7
3	2.71	118	+1.8	-5.4	-2.1	+1.0	-2.1
4	2.11	122	+1.4	-5.3	-1.2	-5.4	-2.4
6	1.65	130	+1.2	-5.8	-2.1	-2.3	-1.6
8	1.45	138	+1.1	-6.3	-1.2	-0.8	-0.8
13	1.28	158	+1.0	-4.8	-0.2	+0.1	-0.1
17	1.21	174	+0.9	-5.4	-0.1	+0.3	+0.1
25	1.13	206	+1.0	-6.4	-0.2	+0.1	+0.1
33	1.10	238	+1.0	-6.6	-0.2	+0.1	+0.1

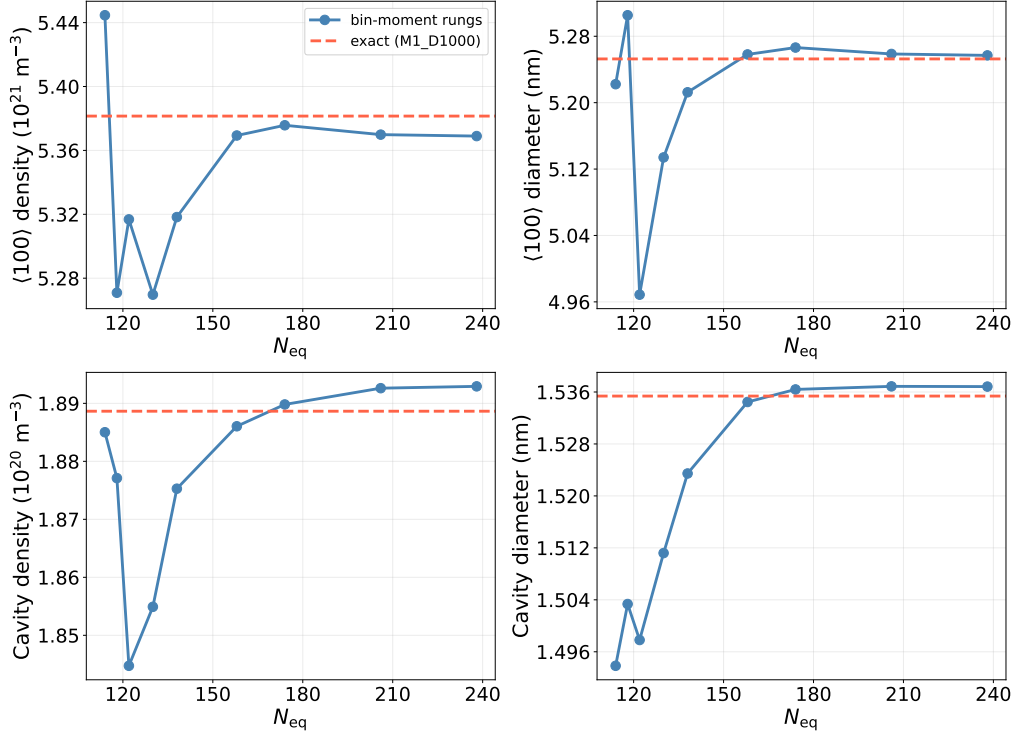


Figure S6: The $\langle 100 \rangle$ and cavity observables against N_{eq} at fixed $i_{\text{discrete}} = 50$, the exact arm shown dashed. Each closes on the reference by thirteen bins, at 158 equations. The $\frac{1}{2}\langle 111 \rangle$ panels are omitted; that population is given quantitatively in Table S9.

Table S10: Deviation of d_{111} from the exact arm (percent) at 20 dpa, with the discrete core and the bin count varied independently. The error falls down a column and is nearly flat across a row. *d.n.r.* marks a configuration that reached the one-hour limit short of 20 dpa, a cost result rather than a failure of the method; *n/r* marks a bin layout that cannot be realised, since at $i_{\text{discrete}} = 800$ only 200 sizes remain to be binned.

i_{discrete}	$I_{\text{bin}} = 4$	$I_{\text{bin}} = 8$	$I_{\text{bin}} = 16$	$I_{\text{bin}} = 32$
25	−7.2	<i>d.n.r.</i>	−6.7	−6.3
50	−5.3	−6.3	−5.4	−6.6
100	<i>d.n.r.</i>	−3.4	−4.2	−5.1
200	−6.1	−2.0	−2.5	−3.4
400	<i>d.n.r.</i>	−0.1	<i>d.n.r.</i>	−1.1
800	+1.0	+0.7	+0.6	<i>n/r</i>

The residual is the ordinary closure error and tracks the fraction of $\frac{1}{2}\langle 111 \rangle$ content that lies above the discrete core, the fraction that the closure must represent rather than resolve. 0.05 dpa, against an exact $d_{111} = 2.9787$ nm, raising i_{discrete} through 25, 50, 100, 200, 400 and 800 leaves 96.6, 91.2, 83.4, 70.6, 47.7 and 8.0 percent of binned content and returns 2.7232, 2.7744, 2.8128, 2.8692, 2.9403 and 2.9813 nm, a monotone approach from −8.6 to +0.1 percent. Conversely, at a fixed core of 400, raising the bin count from 4 to 8 to 33 gives 2.9615, 2.9503 and 2.9245 nm — slightly worse each time. Nothing special about the $\frac{1}{2}\langle 111 \rangle$ population in this: its mean size is about 147 interstitials, about three times the discrete core, so the peak of its distribution lies in the binned region and is carried by the closure, whereas d_{100} and d_{cav} improve monotonically along the same sweep with about a tenth the amplitude.

Three alternative explanations were excluded by direct measurement. A difference between the two code paths is ruled out by running the bin-moment path without closure $I_{\text{bin}} = 0$ and $i_{\text{discrete}} = I$, which reproduces the discrete arm at 20 dpa to within rounding. A domain-edge artifact is ruled out because the exact spectrum decays smoothly to the top of the grid, of size 1000, holding 0.01 per cent of the content. Slow accumulation is ruled out because the deficit already stands at −8.4 per cent at 0.005 dpa and drifts only to −6.6 per cent by 20 dpa.

S9. Input Parameters

Every quantity a RADCLUSTER run requires for EUROFER-97 is collected here, in the order in which it enters the master equations: the primary-damage source that fills the cluster ladders, the lattice geometry that fixes the diffusion-limited prefactors of the reaction kernels, and the defect energetics, solute traps, and fixed sinks that set the reaction and emission frequencies. These are the only quantities supplied externally; the kernels, sink strengths, and emission rates of Section S2 are computed from them once at setup. The values are those of the production run reported in Section 5, as recorded in its run provenance.

Primary damage. The source term is fixed by the defect production characteristics of the displacement cascade: the number of Frenkel pairs that survive intracascade recombination, and the fractions of the surviving vacancies and self-interstitials that emerge already bound in clusters rather than as monomers. These fractions differ between a fission and a fusion neutron spectrum, and the difference propagates directly into the production bias that decides whether cavities and loops can grow together (Section S5). Table S11 reconstructs them from published molecular-dynamics cascade simulations.

Table S11: Comparison of defect production characteristics under fission and fusion neutron spectra in α -Fe [14, 23, 25, 130–141].

Quantity	Fission	Fusion
Mono-SIA fraction, $1 - f_i^{\text{cl}}$	0.42	0.35
Mono-vacancy fraction, $1 - f_v^{\text{cl}}$	0.85	0.80
Mean SIA cluster size $\langle m \rangle_{\text{cl}}$	4.4	6.3
Mean vacancy cluster size $\langle n \rangle_{\text{cl}}$	2.6	3.1
Largest SIA cluster i_{cascade}	20	50
Largest vacancy cluster v_{cascade}	10	20
He co-production (appm He/dpa)	0.5–1	~ 10

Lattice geometry and rate-constant prefactors. The diffusion-limited kernels of Table 2 carry numerical prefactors that follow from the bcc lattice alone once the atomic-volume convention $\Omega = a^3/2$ is fixed. Table S12 lists them together with the capture radii and jump distances from which they are built, so that no prefactor entering a kernel is left implicit.

Table S12: Lattice geometry and diffusion-limited rate-constant prefactors for the EUROFER-97 reaction kernels. All entries follow from the bcc geometry with $\Omega = a^3/2$ and are derived in Section S2 of the Supplementary Material.

Parameter	Symbol	Value	Units
Lattice parameter (bcc Fe)	a	0.2867	nm
Atomic volume	$\Omega = a^3/2$	1.18×10^{-2}	nm ³
Burgers vector, $\frac{1}{2}\langle 111 \rangle$	$b = \sqrt{3}a/2$	0.248	nm
Loop core radius	r_c	$0.333 b$	—
Spherical-sink prefactor	$A_{\text{sph}} = (48\pi^2)^{1/3}$	7.82	—
Loop prefactor, $\frac{1}{2}\langle 111 \rangle$	$A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}}$	10.77	—
Loop prefactor, $\langle 100 \rangle$	$A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}}$	10.77	—
Recombination prefactor	$4\sqrt{3}\pi$	21.8	—
Interstitial bias at loops	Z_i^{loop}	1.181	—
Vacancy bias at loops	Z_v^{loop}	1	—
1D/3D rotation coefficient	$B_{\text{rot}} = 2/9$	0.222	—
Glide flight length	ℓ	22	nm
Direction-change free path	$\hat{L} = \ell/a$	77	—

Defect energetics, solute traps and sinks. The remaining inputs are material data: formation, migration and binding energies for vacancies, self-interstitials and helium; the solute concentrations and trap binding energies through which Cr, W and the minor alloying elements retard defect migration; and the fixed-sink microstructure — dislocation density, grain size and precipitate dispersion — that terminates mobile clusters. Table S13 collects them with their sources, which are largely first-principles.

S10. Experimental Data Base

The measurements used for comparison with the calculations of Section 5 are collected here exactly as the source papers report them. Neutron and ion irradiations are tabulated separately throughout, because the damage rate, the alloy set, and the dose range differ systematically between them. Loops, cavities, and helium bubbles are given in separate tables, and the $\langle 100 \rangle$ loop fraction, which measures loop character rather than loop size, is tabulated on its own. Each row carries the temperature and dose at which it was measured and the source from which it came from; the short source keys are expanded in Table S21.

Table S13: Reaction rate input parameters for EUROFER cluster dynamics [28, 80, 142–153].

Parameter	Symbol	Value	Units
<i>Dislocation sink:</i>			
Network dislocation density	ρ_{net}	5×10^{14}	m^{-2}
Interstitial bias factor	Z_i^d	1.35	—
<i>Grain-boundary sink:</i>			
Grain diameter	d_g	1	μm
<i>Precipitate sink:</i>			
Mean precipitate radius	r_p	20	nm
Precipitate number density	N_p	1×10^{20}	m^{-3}
<i>Mobility cutoffs:</i>			
Max mobile SIA cluster	i_{mobile}	5	SIAs
Max mobile vacancy cluster	v_{mobile}	5	vacancies
<i>Vacancy energetics:</i>			
Formation energy	E_f^v	2.0	eV
Migration energy	E_m^v	0.80	eV
Divacancy binding energy (2NN)	E_B^{2v}	0.22	eV
Divacancy migration energy	E_m^{2v}	0.70	eV
<i>SIA energetics:</i>			
Formation energy ($\langle 110 \rangle$ dumbbell)	E_f^i	3.64	eV
Migration energy ($\langle 110 \rangle$ T-R)	E_m^i	0.402	eV
One-dimensional migration energy	E_m^{1D}	0.34	eV
Di-SIA binding energy	E_B^{2i}	0.80	eV
Di-SIA migration energy	E_m^{2i}	0.42	eV
<i>Helium energetics:</i>			
He migration energy (tetrahedral)	E_m^h	0.074	eV
He–vacancy binding (1st He)	E_b^{h-V}	2.21	eV
He–vacancy binding (2nd He)	$E_b^{h-V_2}$	2.00	eV
<i>Attempt frequencies:</i>			
Vacancy	ν_v	1.0×10^{13}	s^{-1}
SIA ($\langle 110 \rangle$)	ν_i	1.0×10^{13}	s^{-1}
He interstitial	ν_h	3.0×10^{12}	s^{-1}
<i>He equation of state (virial coefficients, 600–1000 K):</i>			
Second virial coefficient	B_2	1.67×10^{-29}	m^3/atom
Third virial coefficient	B_3	1.84×10^{-58}	m^6/atom^2

Entries are reproduced verbatim from the database, including ranges, approximate values, and bounds, so that no interpretation is imposed here: a cell reading $\sim 10\text{--}15$ is a range the authors quoted as such, and a cell reading <5 is an upper bound. A dash means that the quantity was not reported. Densities follow the unit stated in the column head.

Table S14: Dislocation loops under neutron irradiation: mean loop diameter $\bar{d}$, loop number density N and dominant Burgers vector, with the irradiation temperature and dose.

Material	T ($^{\circ}\text{C}$)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Loop character	Source
EUROFER97	330	15	3.4	1.4	$\frac{1}{2}\langle 111 \rangle$ dominant at $\leq 300^{\circ}\text{C}$; $\langle 100 \rangle$ at $\geq 350^{\circ}\text{C}$	Klimenkov 2012
EUROFER97	335	32	4.8	1.7	$\langle 100 \rangle$ dominant	Klimenkov 2012
EUROFER97	300	15	<5 (black dots)	High (>1.4)	$\frac{1}{2}\langle 111 \rangle$	Gaganidze 2016
EUROFER97	250	16.3	<5 (black dots dominant)	~ 1.0	$\frac{1}{2}\langle 111 \rangle$	Gaganidze 2011
EUROFER97	350	16.3	$\sim 10\text{--}15$	~ 0.8	$\langle 100 \rangle$ dominant	Gaganidze 2011
F82H	345	20	$\sim 3\text{--}8$ (small clusters)	High	$\frac{1}{2}\langle 111 \rangle$	Wan 2010
T91	376	15.4	$\sim 5\text{--}15$	~ 1.0	$a\langle 100 \rangle$	Gao 2018
T91	376	35.1	$\sim 10\text{--}20$	~ 0.7	$a\langle 100 \rangle$ dominant	Gao 2018
T91	469	4.3	$\sim 8\text{--}12$	~ 0.5	$a\langle 100 \rangle + \frac{a}{2}\langle 111 \rangle$	Zhong 2021
HT9	377	17.1	$\sim 8\text{--}15$	~ 0.9	$a\langle 100 \rangle$ dominant	Gao 2019
HT9	377	35.1	$\sim 12\text{--}22$	~ 0.6	$a\langle 100 \rangle$ dominant	Gao 2019
HT9	300	4.02	18.9	~ 0.3	$a\langle 100 \rangle + \frac{a}{2}\langle 111 \rangle$	Yan 2021
Fe-9Cr (model)	300	2	5	0.01	$\frac{1}{2}\langle 111 \rangle$ dominant; some $\langle 100 \rangle$	Lambrecht 2014
Pure Fe	60	0.79	$\sim 5\text{--}15$ (rafts >0.3 dpa)	$\sim 1.0\text{--}3.0$	$\frac{1}{2}\langle 111 \rangle \rightarrow \langle 100 \rangle$	Eldrup 1999
Pure Fe	300	0.19	$\sim 8\text{--}12$	~ 0.5	$\langle 100 \rangle$	Lambrecht 2009
9Cr-NbMo (model)	400	—	—	—	$a\langle 100 \rangle + \frac{a}{2}\langle 111 \rangle$	Zhong 2024
EUROFER97	300	15	4.7	0.14	$\frac{1}{2}\langle 111 \rangle$ (ZA $[\bar{1}\bar{3}5]$, $g=\{\bar{3}10\}$)	Dethloff 2016

Table S14 (continued)

Material	T (°C)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Loop character	Source
EUROFER97	300	15	3.4	0.49	a $\langle 100 \rangle$ (ZA [011], g= $\{\bar{2}11\}$)	Dethloff 2016
EUROFER97	300	15	4.8	0.14	$\frac{1}{2}\langle 111 \rangle$ (ZA [011], g= $\{200\}$)	Dethloff 2016
EUROFER- ODS	350	16.3	17.1	$\sim 0.4\text{--}0.7$ (total)	a $\langle 100 \rangle$ (45–60% at 350°C)	Klimenkov 2017
EUROFER- ODS	350	16.3	26.4	$\sim 0.4\text{--}0.7$ (total)	$\frac{1}{2}\langle 111 \rangle$ (40–55% at 350°C)	Klimenkov 2017

Table S15: Dislocation loops under ion irradiation.

Material	Ion	T (°C)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Loop character	Source
HT9	Kr ⁺	420	4	$\sim 5\text{--}8$	~ 5	$\frac{a}{2}\langle 111 \rangle$ dom. (<4 dpa); a $\langle 100 \rangle$	Zheng 2018
HT9	Kr ⁺	440	4	$\sim 6\text{--}9$	~ 4	dom. (>4 dpa) mixed $\frac{a}{2}\langle 111 \rangle$ + a $\langle 100 \rangle$	Zheng 2018
HT9	Kr ⁺	470	4	$\sim 8\text{--}12$	~ 2	a $\langle 100 \rangle$ dominant	Zheng 2018
HT9	Fe ⁺⁺	440	140	$\sim 12\text{--}20$	$\sim 1\text{--}3$	a $\langle 100 \rangle$ dominant at high dose	Toloczko 2015
F82H	Fe ³⁺	300	30	~ 6	~ 3	$\frac{a}{2}\langle 111 \rangle$ + a $\langle 100 \rangle$	Ando 2018
Fe-9Cr ODS	Fe ²⁺	500	3	$\sim 4\text{--}6$	~ 5	$\frac{a}{2}\langle 111 \rangle$	Swenson & Wharry 2018
HCM12A	Fe ²⁺	500	3	$\sim 4\text{--}5$	~ 5	$\frac{a}{2}\langle 111 \rangle$	Swenson & Wharry 2018
T91	Fe ³⁺ + He ²⁺ (dual)	369–520	16.6–72	$\sim 7\text{--}14$	$\sim 1\text{--}5$	$\frac{a}{2}\langle 111 \rangle$ + a $\langle 100 \rangle$	Toloczko 2024
T91 (Round Robin)	Fe ²⁺ (various)	376 (target)	~ 47	Wide scatter	Wide scatter	$\frac{a}{2}\langle 111 \rangle$ + a $\langle 100 \rangle$	Was 2025 (IAEA CRP)
Fe-9Cr	Fe ¹³⁺	450	3	~ 4	Higher vs Fe- 9Cr-C	$\frac{a}{2}\langle 111 \rangle$	Jiang 2024
Fe-9Cr-C	Fe ¹³⁺	450	3	~ 6 (larger)	Lower vs Fe-9Cr	$\frac{a}{2}\langle 111 \rangle$	Jiang 2024

Table S15 (continued)

Material	Ion	T (°C)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Loop character	Source
RUSFER-EK-181	Fe ion	250–400	~6	~2–4	High	Disloc. loops + clusters	Rogozhkin 2020
CLAM 14YWT (NFA)	Fe + He (dual) Fe ⁹⁺	350–550	11	~3–8	~1–10	$\frac{a}{2}\langle 111 \rangle + a\langle 100 \rangle$	Liu 2023
		450	21	~2–5 (small)	–	—	Auger 2019
EUROFER97	Kr ²⁺	350	3	9.9	–	$\frac{1}{2}\langle 111 \rangle + \langle 100 \rangle$	Boulanger & Serruys 2009

Table S16: Cavities under neutron irradiation.

Material	T (°C)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Swelling (%)	Source
EUROFER97	330	15	2.6	0.14	<0.01	Klimenkov 2012
EUROFER97	335	32	1.6	0.11	<0.01	Klimenkov 2012
EUROFER97	300	15	~2–3	High (>HFR)	<0.1	Gaganidze 2016
T91	376	15.4	<2	High density (small)	~0.01	Gao 2018
T91	376	35.1	Bimodal: <2 + >2	~0.5 (small); <0.1 (large)	~0.02	Gao 2018
HT9	377	17.1	<2 (small only)	~0.3	0.02	Gao 2019
HT9	377	35.1	Bimodal: small + large	~0.2 (small); <0.05 (large)	0.07	Gao 2019
HT9	420	200	–	–	0.09–1.02 (heat-treat. dependent)	Gelles 1990s (Garner review)
Pure Fe	60	0.79	~1	>100	<0.01	Eldrup 1999

Table S16 (continued)

Material	T ($^{\circ}\text{C}$)	Dose (dpa)	$\bar{d}$ (nm)	N (10^{22} m^{-3})	Swelling (%)	Source
Pure Fe	400	1	~ 5 (max at 400°C)	~ 0.1	0.07 (max at 400°C)	English 1982
EUROFER97	400	26	< 2	—	< 0.1	Rouffié 2014 (HAL)

Table S17: Voids under ion irradiation.

Material	Ion	T ($^{\circ}\text{C}$)	Dose (dpa)	$\bar{d}$ (nm)	N (m^{-3})	Swelling (%)	Source
HT9	Fe^{++}	460	75–650	$\sim 15\text{--}30$ (650 dpa)	$\sim 5 \times 10^{20}$	$\sim 2\text{--}20$ (at 650 dpa)	Getto 2015 (Gao/Was group)
T91	Fe^{++}	460	75–650	$\sim 10\text{--}20$	$\sim 1 \times 10^{21}$	< 2 (to 650 dpa)	Getto 2015
T92	Fe^{++}	460	75–650	$\sim 8\text{--}15$	$\sim 1 \times 10^{20}$	$< 1\%$	Getto 2015
HT9 (raster)	Fe^{++}	440	140	~ 12	$\sim 3 \times 10^{20}$	~ 0.8	Toloczko 2015
HT9 (de- focused)	Fe^{++}	440	140	~ 18	$\sim 8 \times 10^{20}$	~ 2.5	Toloczko 2015
ODS-FM (Y-Ti-O)	Fe^{2+}	550	500	$\sim 10\text{--}20$ (at high dose)	Low (at oxide interface)	Low (suppressed)	Liu 2023
F82H (IEA heat)	Fe + He + H (triple)	$\sim 400\text{--}500$	$\sim 20\text{--}50$	$\sim 2\text{--}5$	$\sim 10^{21}$	< 1	Wakai 2002
FC92-B	Fe^{++}	475	480	Reported	Reported	Reported	Lee 2021
Gr.92	Fe^{++}	475	480	Reported	Reported	Reported	Lee 2021
17Cr ferritic (F17)	Dual beam (ion)	< 500	—	$\sim 5\text{--}10$	Observable	Non-negligible	Gilbon 1988

Table S18: Helium bubbles under ion irradiation (dual-beam and helium-implantation experiments).

Material	Mode	T ($^{\circ}\text{C}$)	Dose (dpa)	He (appm)	$\bar{d}$ (nm)	N (m^{-3})	Source
Y-Ti- ODS	Dual Fe+He	550	30	~ 3500	2.8	1.1×10^{23}	Song 2018
Y-Al- ODS	Dual Fe+He	550	30	~ 3500	6.6	2.7×10^{22}	Song 2018
Y-Al-Zr- ODS	Dual Fe+He	550	30	~ 3500	4.5	3.6×10^{22}	Song 2018
CLAM	Dual Fe+He	350–550	11	Variable	~ 3 –8	$\sim 10^{22}$ – 10^{23}	Liu 2023
ODS- F/M (CLAM base)	Dual Fe+He	350–550	11	Variable	~ 2 –5	Higher than CLAM	Liu 2023
14.5Cr- ODS	Single He $^{+}$	300–700	0.2	3500	~ 2 –3	$\sim 10^{23}$	Yutani 2007
14.5Cr- 3.5Al(Zr)- ODS	Single He $^{+}$	300–700	0.2	3500	~ 3 –5	$\sim 10^{22}$ – 10^{23}	Yutani 2007 / Kimura group
F82H (SINQ)	Spallation (n+p+He)	165–305	10.7–19.6	850– 1740	~ 1	$\sim 10^{23}$ – 10^{24}	Dai 2004
T91 (SINQ)	Spallation (n+p+He)	165–305	up to 11.8	~ 550	~ 1	Slightly F82H	> Jia 2005
ODS- EUROFER	He $^{+}$ single + Au dual	RT	–	~ 8500 peak	~ 1 –3 (single); larger (dual)	High	Konstantinovic 2021
Fe-12Cr (model)	Single He $^{+}$	250–400	<1	<10 (nucle- ation)	~ 1 –2	Linear in- crease with He conc.	Schroeder & Trinkaus 1986

Table S19: Loop character under neutron irradiation: the measured $\langle 100 \rangle$ fraction of the loop population, with the diameter and density of each loop family where these are resolved separately.

Material	T ($^{\circ}\text{C}$)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m^{-3})	Source
Eurofer97	300	15	78	$\langle 100 \rangle$	2.8	4.9×10^{21}	Dethloff 2016
Eurofer97	300	15	78	$\frac{1}{2}\langle 111 \rangle$	4.2	1.4×10^{21}	Dethloff 2016
Eurofer97	300	15	77	$\langle 100 \rangle$	3.3	4.697×10^{21}	Dethloff 2018
Eurofer97	300	15	77	$\frac{1}{2}\langle 111 \rangle$	3.3	1.403×10^{21}	Dethloff 2018
Eurofer97	300	15	0	$\frac{1}{2}\langle 111 \rangle$	14	4×10^{21}	Dethloff 2018
Eurofer97	330	15	72	$\langle 100 \rangle$	6.1	4.968×10^{21}	Dethloff 2018
Eurofer97	330	15	72	$\frac{1}{2}\langle 111 \rangle$	6.1	1.932×10^{21}	Dethloff 2018
Eurofer97	330	15	45	$\langle 100 \rangle$	3.4	9×10^{21}	Dethloff 2018
Eurofer97	330	15	45	$\frac{1}{2}\langle 111 \rangle$	3.4	1.1×10^{22}	Dethloff 2018
Eurofer97	330	32	27	$\langle 100 \rangle$	5.1	2.241×10^{21}	Dethloff 2018
Eurofer97	330	32	27	$\frac{1}{2}\langle 111 \rangle$	5.1	6.059×10^{21}	Dethloff 2018
Eurofer97	330	32	31	$\langle 100 \rangle$	4.8	7.44×10^{21}	Dethloff 2018
Eurofer97	330	32	31	$\frac{1}{2}\langle 111 \rangle$	4.8	1.656×10^{22}	Dethloff 2018
Eurofer97	250	16.3	–	$\frac{1}{2}\langle 111 \rangle$	7	2×10^{21}	Klimenkov 2011
Eurofer97	300	16.3	0	$\frac{1}{2}\langle 111 \rangle$	14	4×10^{21}	Klimenkov 2011
Eurofer97	350	16.3	0	$\frac{1}{2}\langle 111 \rangle$	35	3×10^{20}	Klimenkov 2011
Eurofer97	400	16.3	–	–	12	5×10^{19}	Klimenkov 2011
Eurofer97	450	16.3	–	–	10	1×10^{19}	Klimenkov 2011

Table S19 (continued)

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
EUROFER-ODS with addition of 0.5% Y2O3	250	16.2	0	$\langle 100 \rangle$	–	–	Klimenkov 2017
EUROFER-ODS with addition of 0.5% Y2O3	350	16.2	45	$\frac{1}{2} \langle 111 \rangle$	26	2.2×10^{21}	Klimenkov 2017
EUROFER-ODS with addition of 0.5% Y2O3	350	16.2	45	$\langle 100 \rangle$	15	1.8×10^{21}	Klimenkov 2017
EUROFER-ODS with addition of 0.5% Y2O3	350	16.2	60	$\langle 100 \rangle$	15	2.8×10^{21}	Klimenkov 2017
EUROFER-ODS with addition of 0.5% Y2O3	350	16.2	60	$\frac{1}{2} \langle 111 \rangle$	26	4.2×10^{21}	Klimenkov 2017
EUROFER-ODS with addition of 0.5% Y2O3	450	16.2	100	$\langle 100 \rangle$	60	–	Klimenkov 2017
Eurofer97	250	13.4	10	$\frac{1}{2} \langle 111 \rangle$	5	1.5×10^{22}	Klimenkov 2020
Eurofer97	250	13.4	10	$\langle 100 \rangle$	5	1.5×10^{21}	Klimenkov 2020
Eurofer97	300	14.6	27	$\frac{1}{2} \langle 111 \rangle$	6	3.4×10^{22}	Klimenkov 2020
Eurofer97	300	14.6	27	$\langle 100 \rangle$	8	1.2×10^{22}	Klimenkov 2020
Eurofer97	350	17.4	79	$\frac{1}{2} \langle 111 \rangle$	20	9.5×10^{20}	Klimenkov 2020
Eurofer97	350	17.4	79	$\langle 100 \rangle$	48	3.72×10^{21}	Klimenkov 2020

Table S19 (continued)

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
Eurofer97	400	17.2	87	$\frac{1}{2}\langle 111 \rangle$	25	2.5×10^{19}	Klimenkov 2020
Eurofer97	400	17.2	87	$\langle 100 \rangle$	110	2.25×10^{20}	Klimenkov 2020
Eurofer97	415	18.1	80	$\frac{1}{2}\langle 111 \rangle$	100	1×10^{19}	Klimenkov 2020
Eurofer97	415	18.1	80	$\langle 100 \rangle$	180	7×10^{19}	Klimenkov 2020
Eurofer97	330	15	–	–	3.4	1.4×10^{22}	Weiß 2012
Eurofer97	330	32	–	–	4.8	1.7×10^{22}	Weiß 2012
Eurofer97	330	15	73	$\langle 100 \rangle$	6.2	4.672×10^{21}	Chauhan 2021
Eurofer97	330	15	73	$\frac{1}{2}\langle 111 \rangle$	6.2	1.728×10^{21}	Chauhan 2021
Eurofer97	330	15	73	$\langle 100 \rangle$	7	8.833×10^{21}	Chauhan 2021
Eurofer97	330	15	73	$\frac{1}{2}\langle 111 \rangle$	7	3.267×10^{21}	Chauhan 2021
Eurofer97	330	15	–	–	8	2×10^{21}	Chauhan 2021
Eurofer97	330	15	–	–	13	4.1×10^{21}	Chauhan 2021
F82H	250	0.5	–	$\frac{1}{2}\langle 111 \rangle$	2.4	–	Schaublin 2002
F82H	310	8.8	–	$\langle 100 \rangle$	5.4	–	Schaublin 2002
HT-9	400	7.4	–	$\langle 100 \rangle$	14	1×10^{22}	Chen 2013
HT-9	400	7.4	–	$\frac{1}{2}\langle 111 \rangle$	10	3×10^{21}	Chen 2013
HT-9	420	35	–	–	100	5×10^{20}	Chen 2013
EUROFER-ODS with addition of 0.5% Y2O3	300	16.2	0	–	7	1×10^{21}	Klimenkov 2021
EUROFER-ODS with addition of 0.5% Y2O3	300	16.2	5	$\frac{1}{2}\langle 111 \rangle$	13	4.75×10^{21}	Klimenkov 2021

Table S19 (continued)

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
EUROFER-ODS with addition of 0.5% Y2O3	300	16.2	5	$\langle 100 \rangle$	13	2.5×10^{20}	Klimenkov 2021
EUROFER-ODS with addition of 0.5% Y2O3	400	16.2	10	$\frac{1}{2}\langle 111 \rangle$	71	4.05×10^{20}	Klimenkov 2021
EUROFER-ODS with addition of 0.5% Y2O3	400	16.2	10	$\langle 100 \rangle$	71	4.5×10^{19}	Klimenkov 2021
EUROFER-ODS with addition of 0.5% Y2O3	400	16.2	15	$\frac{1}{2}\langle 111 \rangle$	94	8.5×10^{20}	Klimenkov 2021
EUROFER-ODS with addition of 0.5% Y2O3	400	16.2	15	$\langle 100 \rangle$	94	1.5×10^{20}	Klimenkov 2021

Table S20: Loop character under ion irradiation.

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
Eurofer97	350	3	—	—	7	3×10^{21}	Boulanger 2009
Eurofer97	550	3	—	$\langle 100 \rangle$	150	1×10^{20}	Boulanger 2009
Eurofer97	250	3	—	—	5	5×10^{21}	Rogozhkin 2019
Eurofer97	350	3	—	—	9	4×10^{21}	Rogozhkin 2019
Eurofer97	400	3	—	—	20	7×10^{20}	Rogozhkin 2019

Table S20 (continued)

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
Eurofer97	250	5	–	–	7	6.3×10^{21}	Rogozhkin 2019
Eurofer97	350	5	–	–	10	6.7×10^{21}	Rogozhkin 2019
Eurofer97	400	5	–	–	19	8×10^{20}	Rogozhkin 2019
Eurofer97	250	10	–	–	6	9.8×10^{21}	Rogozhkin 2019
Eurofer97	350	10	–	–	10	7.7×10^{21}	Rogozhkin 2019
Eurofer97	400	10	–	–	15	3.8×10^{21}	Rogozhkin 2019
Eurofer97	400	26	87	$\frac{1}{2}\langle 111 \rangle$	7.9	2.99×10^{21}	Brimbal 2015
Eurofer97	400	26	87	$\langle 100 \rangle$	7.9	2×10^{22}	Brimbal 2015
EuroferODS(tempered martensite gains)	400	26	76	$\frac{1}{2}\langle 111 \rangle$	7.1	4.56×10^{21}	Brimbal 2015
EuroferODS(tempered martensite gains)	400	26	76	$\langle 100 \rangle$	7.1	1.44×10^{22}	Brimbal 2015
EuroferODS(ferro grains)	400	26	32	$\frac{1}{2}\langle 111 \rangle$	11	6.528×10^{20}	Brimbal 2015
EuroferODS(ferro grains)	400	26	32	$\langle 100 \rangle$	11	3.072×10^{21}	Brimbal 2015
Eurofer97	148	9.5	–	–	3.32	3.4×10^{21}	Zhang 2014
Eurofer97	266	17.7	–	–	11.21	1.48×10^{22}	Zhang 2014
Eurofer97	322	18.5	–	–	9.45	1.26×10^{22}	Zhang 2014
Eurofer97	279	18.5	–	–	8.68	2.23×10^{22}	Zhang 2014
Eurofer97	309	20.4	–	–	9.25	1.45×10^{22}	Zhang 2014
Eurofer97	357	20.4	–	–	11.5	1.3×10^{22}	Zhang 2014
Eurofer97	250	10	–	–	7.5	1×10^{22}	Rogozhkin 2019
Eurofer97	300	10	–	–	10	7.5×10^{21}	Rogozhkin 2019
Eurofer97	400	10	–	–	12	3.5×10^{21}	Rogozhkin 2019
Eurofer97	250	6	–	–	–	6×10^{21}	Rogozhkin 2019
Eurofer97	300	6	–	–	–	5.8×10^{21}	Rogozhkin 2019

Table S20 (continued)

Material	T (°C)	Dose (dpa)	$f_{\langle 100 \rangle}$ (%)	Loop type	$\bar{d}$ (nm)	N (m ⁻³)	Source
Eurofer97	400	6	–	–	–	8×10^{20}	Rogozhkin 2019
Eurofer97	250	3	–	–	–	5×10^{21}	Rogozhkin 2019
Eurofer97	300	3	–	–	–	4.3×10^{21}	Rogozhkin 2019
Eurofer97	400	3	–	–	–	5×10^{20}	Rogozhkin 2019
Eurofer97	330	16	29	$\frac{1}{2}\langle 111 \rangle$	4.2	6.035×10^{21}	Kaiser 2018
Eurofer97	330	16	29	$\langle 100 \rangle$	4.2	2.465×10^{21}	Kaiser 2018
Eurofer97	400	16	22	$\frac{1}{2}\langle 111 \rangle$	6.1	5.694×10^{21}	Kaiser 2018
Eurofer97	400	16	22	$\langle 100 \rangle$	6.1	1.606×10^{21}	Kaiser 2018

Table S21: Sources of the tabulated measurements. The key is the short form used in the Source column of Tables S14–S20; the second column gives the DOI or the full title as recorded in the database.

Key	Reference / DOI
Ando 2018	10.1016/j.jnucmat.2018.xx
Auger 2019	10.1016/j.jnucmat.2019.xx
Boulanger & Serruys 2009	https://www.sciencedirect.com/science/article/abs/pii/S0022311508008982
Boulanger 2009	Boulanger and Serruys - 2009 - Dislocation loops in Eurofer and a Fe–Cr alloy irradiated by ions at 350 and 550°C at 3dpa Effect of dose rate
Brimbal 2015	Brimbal et al. - 2015 - Microstructural characterization of Eurofer-97 and Eurofer-ODS steels before and after multi-beam ions irradiations
Chauhan 2021	Chauhan et al. - 2021 - Post-irradiation annealing of neutron-irradiated EUROFER97
Chen 2013	Chen2013-IRRADIATION EFFECTS OF HT-9 MARTENSITIC STEEL
Dai 2004	10.1016/j.jnucmat.2004.325001
Dethloff 2016	https://www.sciencedirect.com/science/article/pii/S2352179115300508
Dethloff 2018	Dethloff et al. - 2018 - Review and critical assessment of dislocation loop analyses on EUROFER 97
Eldrup 1999	10.1016/S0022-3115(98)00767-3
English 1982	10.1016/0022-3115(82)90490-1
Gaganidze 2011	10.1016/j.jnucmat.2010.12.261

Table S21 (continued)

Key	Reference / DOI
Gaganidze 2016	10.1016/j.nme.2015.12.004
Gao 2018	10.1016/j.actamat.2018.01.016
Gao 2019	10.1016/j.jnucmat.2019.03.026
Gelles 1990s (Garner review)	10.1016/j.jnucmat.2019.03.026
Getto 2015	–
Getto 2015 (Gao/Was group)	–
Gilbon 1988	–
Jia 2005	10.1016/j.jnucmat.2005.343212
Jiang 2024	ResearchGate (2024)
Kaiser 2018	Kaiser et al. - 2018 - Investigation of microstructure defects in EUROFER97 under He+Fe3+ dual ion beam irradiation
Klimenkov 2011	Klimenkov et al. - 2011 - Characterization of radiation induced defects in EUROFER 97 after neutron irradiation
Klimenkov 2012	10.1016/j.jnucmat.2012.01.022
Klimenkov 2017	J. Nucl. Mater. 493 (2017) 426 (ADS 2017JNuM..493..426K)
Klimenkov 2020	Klimenkov et al. - 2020 - Correlation of microstructural and mechanical properties of neutron irradiated EUROFER97 steel
Klimenkov 2021	Klimenkov et al. - 2021 - Post-irradiation microstructural examination of EUROFER-ODS steel irradiated at 300°C and 400°C
Konstantinovic 2021	10.1016/j.jnucmat.2021.152828
Lambrecht 2009	10.1016/j.jnucmat.2009.09.015
Lambrecht 2014	10.1016/j.jnucmat.2014.04.011
Lee 2021	–
Liu 2023	10.1016/j.jnucmat.2023.154378
Rogozhkin 2019	Rogozhkin et al. - 2019 - Study of Microscopic Origins of Radiation Hardening of Eurofer 97 in Simulation Experiment with Ion
Rogozhkin 2020	10.1134/S207511332002032X
Rouffié 2014 (HAL)	cea-02383837
Schaublin 2002	Schaublin2002-Microstructure of irradiated ferritic-martensitic steels
Schroeder & Trinkaus 1986	10.1016/0022-3115(86)90080-2
Song 2018	10.1016/j.jnucmat.2018.503078
Swenson & Wharry 2018	10.1016/j.jnucmat.2018.503047
Toloczko 2015	10.1016/j.jnucmat.2015.300064
Toloczko 2024	10.1016/j.jnucmat.2024.155147
Wakai 2002	–

Table S21 (continued)

Key	Reference / DOI
Wan 2010	ResearchGate 245051589
Was 2025 (IAEA CRP)	10.1016/j.jnucmat.2025.004593
Weiß 2012	Weiß et al. - 2012 - Quantitative characterization of microstructural defects in up to 32dpa neutron irradiated EUROFER97
Yan 2021	10.1016/j.jnucmat.2021.153161
Yutani 2007	10.1016/j.jnucmat.2007.367423
Yutani 2007 / Kimura group	10.1016/j.jnucmat.2007
Zhang 2014	Zhang et al. - 2014 - Irradiation-induced evolution of mechanical properties and microstructure of Eurofer 97
Zheng 2018	10.1080/14786435.2018.1464065
Zhong 2021	10.1016/j.jnucmat.2021.152971
Zhong 2024	10.1016/j.jnucmat.2024.154990

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